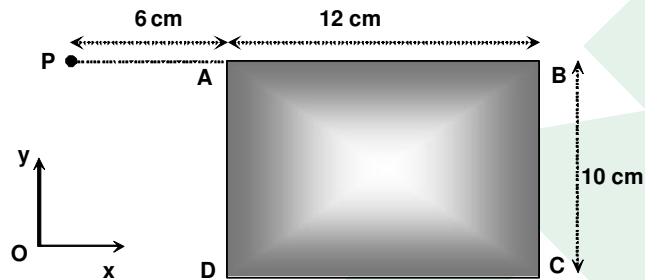


ADVANCE REVISION ASSIGNMENT-01

PHYSICS

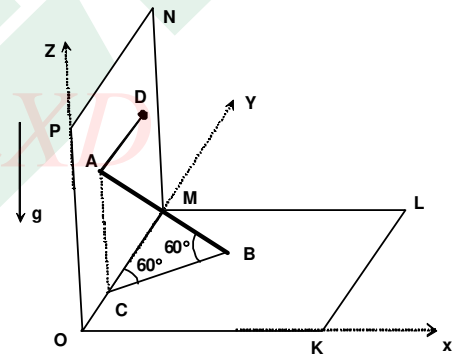
Single Correct Answer Type

1. At an instant, the point P in the figure represents the instantaneous centre of rotation for a uniform rectangular plate ABCD ($12 \text{ cm} \times 10 \text{ cm}$). At the given instant the point A (corner) has velocity $24 \text{ cm/sec} (-\hat{j})$. The magnitude of the velocity of the centre of mass of the plate is



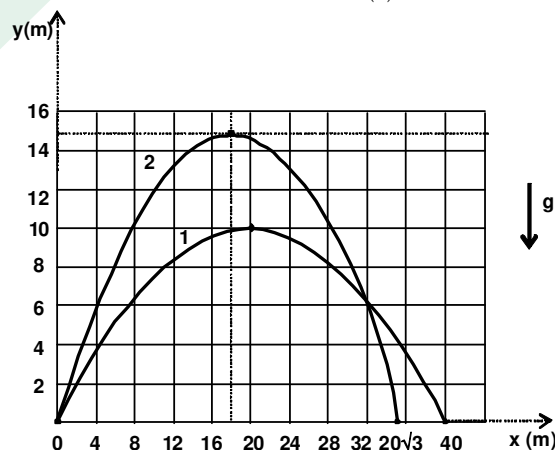
- (A) 13 cm/s (B) 26 cm/s (C) 52 cm/s (D) 104 cm/s

2. Two horizontal cords AD and BC hold a rod AB of mass m and length L in an inclined position between smooth horizontal surface OKLMO and vertical smooth wall OMNPO as shown in the figure. At point A, the rod leans against a vertical wall on which point D is located. At point B, the rod rests on the horizontal floor. The points A and C are on the same vertical line. The rod is in equilibrium. T_1 and T_2 are the magnitudes of the tensions in the cords AD and BC respectively. N_A and N_B are the normal forces exerted by the vertical wall and the horizontal surface on the rod respectively. $\angle ABC = \angle MCB = 60^\circ$. Choose the correct option.



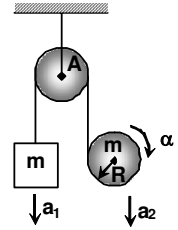
- (A) $2N_A = N_B, 2T_1 = T_2$
(B) $4N_A = N_B, 4T_1 = T_2$
(C) $2T_1 = T_2, 4N_A = N_B$
(D) $4T_1 = T_2, 2N_A = N_B$

3. Trajectory of two particles projected from origin with speed v_1 and v_2 and angles θ_1 and θ_2 with positive x-axis respectively as shown in the figure. Given that $\vec{g} = -10 \text{ m/s}^2 (\hat{j})$. Choose the correct option related to diagram.



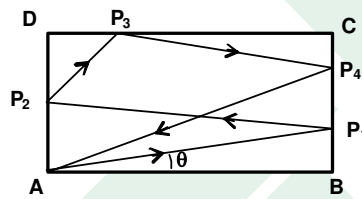
- (A) $v_1 - v_2 = 2v_1$ (B) $\theta_2 - \theta_1 = 2\theta_1$ (C) $3(v_1 - v_2) = v_1$ (D) $3(\theta_2 - \theta_1) = \theta_1$

4. An ideal inextensible string is wrapped over the disc of mass m and radius R . The other end of the string is connected to mass m . The string is passing over an ideal pulley A as shown in the figure. At any time t , mass m and disc are moving downward with acceleration of magnitudes a_1 and a_2 respectively. The disc is rotating clockwise with angular acceleration of magnitude α . There is no slipping between string and disc. Choose the INCORRECT option.



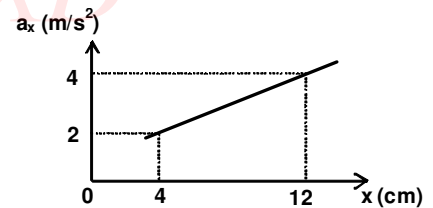
- (A) $a_1 = a_2$ (B) $\alpha R > a_1$ (C) $\alpha R > a_2$ (D) $\alpha R < a_2$

5. From one corner A of a rectangular smooth billiard table ABCD (length a and width b) placed on a horizontal surface, a ball of mass m and negligible dimension is projected in the direction making θ with side AB. It strikes in order sides BC, AD, DC and BC and then returns to same points A. Then the value of coefficient of restitution is (ignore friction anywhere)



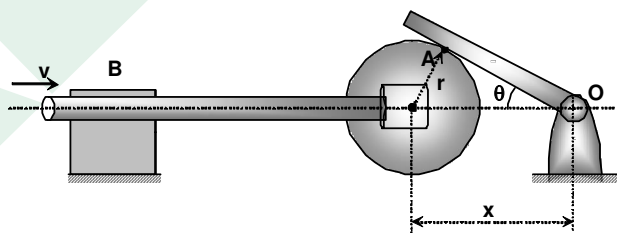
- (A) $\sqrt{\frac{a \sin \theta}{b \cos \theta - a \sin \theta}}$ (B) $\sqrt{\frac{a \sin \theta}{b \cos \theta + a \sin \theta}}$ (C) $\sqrt{\frac{a \sin \theta}{b \sin \theta - a \cos \theta}}$ (D) $\sqrt{\frac{a \sin \theta}{b \sin \theta + a \cos \theta}}$

6. A particle moves along the positive x -axis with an acceleration a_x in meters per second squared which increases linearly with x expressed in centimetres, as shown on the graph for an interval of its motion. If the velocity of the particle at $x = 4$ cm is 0.40 m/s, determine the velocity at $x = 12$ cm.



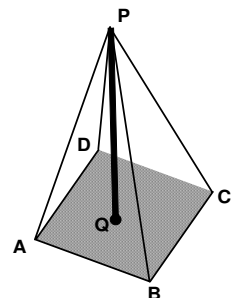
- (A) 0.2 m/s (B) 0.4 m/s
(C) 0.8 m/s (D) 1.6 m/s

7. Rotation of the lever OA is controlled by the motion of the contacting circular disk whose centre is given a horizontal velocity v . Determine the expression of the angular velocity of the lever OA in terms of x .



- (A) $\frac{vr}{3x\sqrt{x^2 - r^2}}$ (B) $\frac{2vr}{x\sqrt{x^2 - r^2}}$ (C) $\frac{vr}{2x\sqrt{x^2 - r^2}}$ (D) $\frac{vr}{x\sqrt{x^2 - r^2}}$

8. Four identical symmetrically located ideal inextensible strings (PA, PB, PC and PD) support a rod PQ of mass m in vertical position as shown in the figure. The angle between two consecutive strings is equal to 60° . Determine the normal force on the rod PQ given by the ground, assuming that the tension in each string is $mg/\sqrt{2}$ Newton (ABCD is a square on the ground)

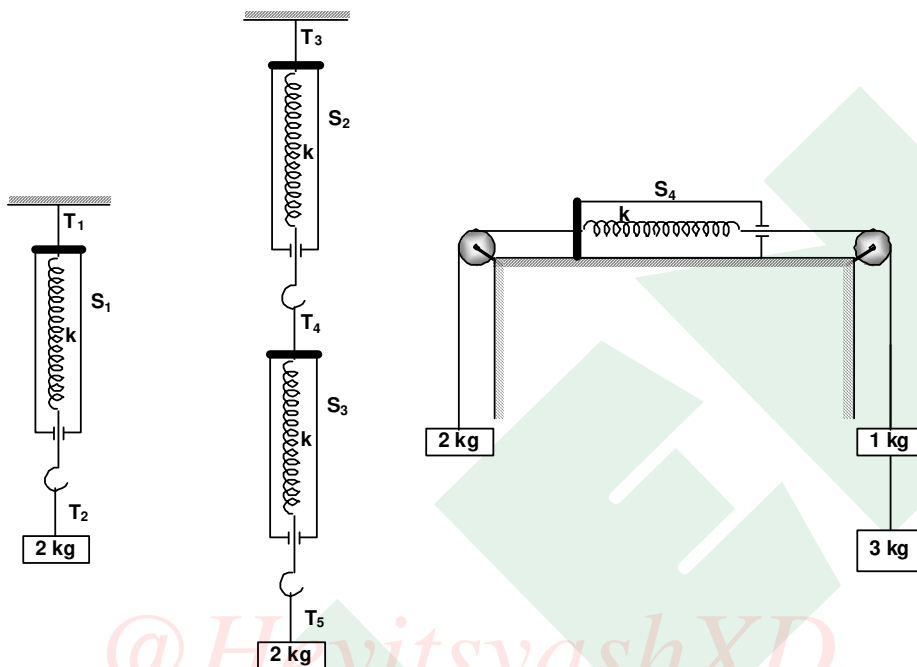


- (A) mg (B) $2mg$
(C) $3mg$ (D) none

COMPREHENSION TYPE

Paragraph for Questions 9 and 10

In each of the experimental arrangements illustrated below, the spring-balances, marked S, are having mass 1 kg each and springs are ideal. The each spring balance reads directly in kilogram

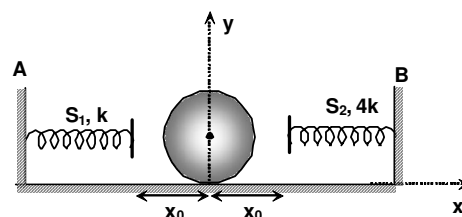


In each figure, each mass in equilibrium. The coefficient of friction between horizontal table and spring balance-4 is μ . The stiffness of each spring of spring balance is 1000 N/m. Now answer the following questions on the basis of above passage.

9. If readings of spring balances S_1 , S_2 , S_3 and S_4 are a kg, b kg, c kg and d kg respectively, then choose the correct option.
 (A) $a < b < c < d$ (B) $b < c < d < a$ (C) $c = a < b < d$ (D) $d = a < b < c$
10. The coefficient of static friction between the horizontal and spring balance-4 can be
 (A) 0.5 (B) 0.8 (C) 1.6 (D) 2.5

Paragraph for Questions 11 and 12

In the adjacent figure, S_1 and S_2 are two light springs of stiffness k and $4k$ respectively. The springs are horizontally oriented and they are rigidly attached to the wall A and B. When springs are in relaxed state, separation between their free ends is $2x_0$. A uniform solid cylinder of mass M and radius $R (< x_0)$ is placed symmetrically exactly midway between the free ends of the springs with its axis horizontal and perpendicular to the springs. Now consider the two cases separately.



Case I: The cylinder is imparted speed v_0 towards left. When the cylinder comes to rest momentarily the co-ordinates of its centre are either $(-x_1, R)$ or (x_2, R) .

Case II: The cylinder is imparted speed v_0 toward left and angular speed ω_0 in clockwise sense simultaneously. At the time of maximum compression in the springs the co-ordinates of centre of the cylinder are either $(-x_3, R)$ or (x_4, R) . Now answer the following questions assuming that friction is absent every where.

11. Choose the correct option.
 (A) $x_1 = x_2$ (B) $x_1 = x_3$ (C) $x_1 = x_4$ (D) $x_3 = x_4$
12. The value of x_3 is
 (A) $x_0 + v_0 \sqrt{\frac{m}{k}}$ (B) $x_0 + 2v_0 \sqrt{\frac{m}{k}}$ (C) $x_0 + \frac{v_0}{2} \sqrt{\frac{m}{k}}$ (D) $x_0 + v_0 \sqrt{\frac{m}{2k}}$

Paragraph for Questions 13 and 14

The top is particularly important example of the conservation of angular momentum, because it exemplifies the behaviour of any spinning object in a force field. As such it has important application to the discussion of rotating atomic and nuclear particles. Consider a top of mass m which, when viewed, is spinning in a counter clockwise direction with angular momentum $\vec{L}$. By convention, this is indicated by a vector length $\vec{L}$ that points outward along the axis OA of the top as shown in the figure-1.

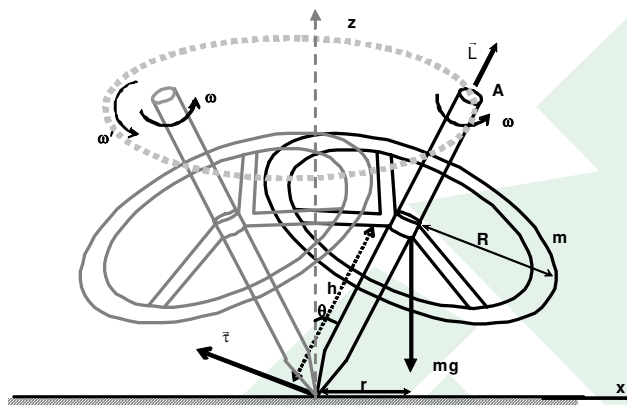


Figure -1

If the top is not upright, then the weight force mg vertically down will provide a torque $\vec{\tau}$ about point O in the base of the top, which must tend to produce an increase in angular momentum in the direction of that torque. This direction will always be perpendicular to the direction of $\vec{L}$; hence this torque can never change the magnitude of $\vec{L}$, but only its direction. It is seen that L , and hence the axis of the top, will precess around in a circle about the vertical line z -axis.

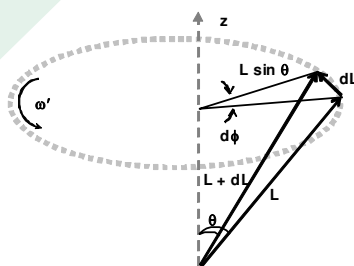
- (a) the centre of mass of the top is seen to be at a distance r from the vertical. Consequently the torque is of magnitude: $\tau = mgr$

This is, then, the magnitude of the time rate of change of angular momentum dL/dt .

- (b) the angular momentum is depicted as precessing about the vertical with angular velocity

$$\omega' = \frac{d\phi}{dt} = \frac{dL / L \sin \theta}{dt} = \frac{dL / dt}{L \sin \theta} = \frac{mgr}{L \sin \theta} = \frac{mgh}{L} = \frac{mgh}{I\omega}$$

where h is the distance from the centre of mass to the base of the top, and $r = h \sin \theta$.



For the top shown, is made of ring of mass m radius R with three identical rods each of mass m and length R as shown in figure-3. The dimension and mass of the central rod which behaves as axis of top is negligible.

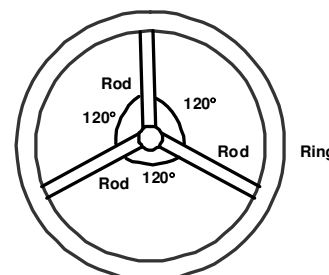


Figure -3: Top view of top when spinning in vertical position

Now answer the following questions:

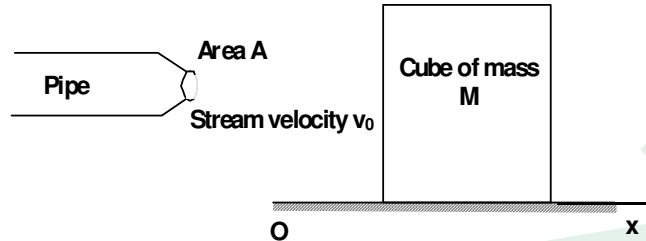
13. If m is 100 g, R is 4 cm, h is 10 cm and ω is 10 rad/s, then find the value of ω' .
(A) 10 rad/s (B) 62.5 rad/s (C) 31.25 rad/s (D) none
14. Choose the correct statements regarding motion of the top.
(A) the magnitude of angular momentum of the top is continuously changing during the motion.
(B) the magnitude of angular momentum is not changing during the motion.

(C) $\vec{\tau} \times \vec{L} = \vec{0}$

(D) $\vec{\tau} \neq \frac{d\vec{L}}{dt}$ because axis of the top is spinning.

Paragraph for Questions 15 and 16

A horizontal stream of water, of uniform cross sectional area 'A' and having constant speed v_0 along positive x-axis, is continuously hitting a vertical face of a cube which is lying on a smooth horizontal surface as shown in the figure. The mass of the cube is M and the area of its face is $a(> A)$. After hitting the cube the water molecules slide down the vertical



face of the cube. In the beginning the cube is at rest. Assume that the speed acquired by the cube, v , is much less than v_0 . At the time $t = 0$, the rear end of the cube is at the origin. Now answer the following questions.

15. The value of the x co-ordinate of the rear end at time t will be

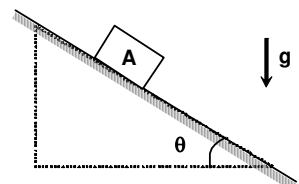
(A) $\frac{\rho A v_0^2 t^2}{2M}$ (B) $\frac{\rho A v_0^2 t^2}{M}$ (C) $\frac{2\rho A v_0^2 t^2}{M}$ (D) $\frac{\rho A v_0^2 t^2}{4M}$

16. The value of v at time t will be

(A) $\frac{\rho A v_0^2 t}{2M}$ (B) $\frac{\rho A v_0^2 t}{3M}$ (C) $\frac{\rho A v_0^2 t}{M}$ (D) $\frac{2\rho A v_0^2 t}{M}$

Matching list Type

17. A block of mass m kg is sliding from rest on an inclined rough surface of inclination θ as shown in the figure under the action of three forces, weight, normal and frictional force only. The magnitude of forces are 20 N, 40 N and 50 N not necessarily in the same order. The value of g is 10 m/s^2 . Now match the physical quantity in List - I with their magnitude in List - II in SI unit and select the correct answer using the codes given below the lists: (assume that the coefficient of kinetic friction is less than 1).



List - I

- (P) Mass of the block (A)
(Q) Acceleration of block (A)
(R) Net force acting on the block during motion
(S) Coefficient of friction between inclined surface and block A is μ , then value of 8μ will be

List - II

- (1) 2
(2) 4
(3) 5
(4) 10

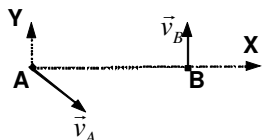
Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	3	1	4	2
(D)	2	3	4	1

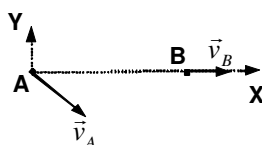
18. There are two point masses A and B, situated at origin and point (5m, 0m) respectively, at a certain time t. $\vec{v}_A$ and $\vec{v}_B$ are respectively the velocities of point masses A and B at this instant t. Match the situations under List - I with their correct option under List - II and select the correct answer using the codes given below the lists:

List - I

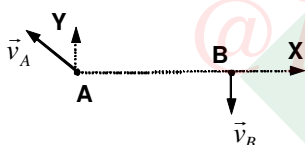
(P) $\vec{v}_A = 5\hat{i} - 5\hat{j}$, and $\vec{v}_B = 20\hat{j}$



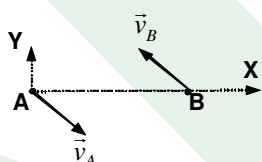
(Q) $\vec{v}_A = 5\hat{i} - 5\hat{j}$, and $\vec{v}_B = 20\hat{i}$



(R) $\vec{v}_A = -5\hat{i} + 5\hat{j}$, and $\vec{v}_B = -20\hat{j}$



(S) $\vec{v}_A = 5\hat{i} - 5\hat{j}$, and $\vec{v}_B = -10\hat{i} + 10\hat{j}$



List - II

- (1) The speed of approach between point masses A and B at the given time t is 5 m/sec

- (2) The speed of separation between point masses A and B at the given time t is 5 m/sec

- (3) At the given time t = t sec, the magnitude of angular velocity of point mass A with respect to B is 3 rad/sec.

- (4) At the given time t = t sec, the magnitude of angular velocity of point mass A with respect to B is 1 rad/sec.

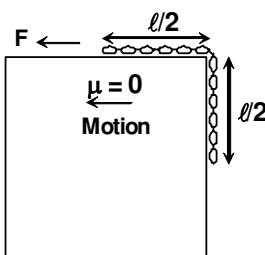
Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	4	1	2	3
(D)	2	3	4	1

19. In section A, some diagrams are shown and in section B, related work done by weight or by F in different cases are given. Match them correctly. Match List - I with List - II and select the correct answer using the codes given the lists:

List - I

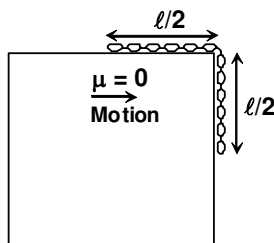
Chain of weight W is pulled slowly by a variable force F so that complete chain comes to table.



List - II

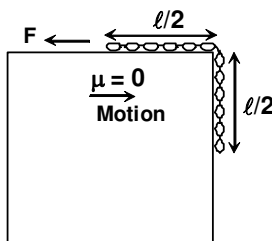
- (1) Work done by weight of chain = $\frac{3W\ell}{4}$

- (Q) Chain slip from the table due to weight $2W$ and becomes vertical



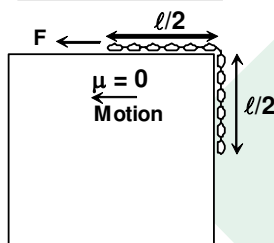
(2) Work done by $F = \frac{W\ell}{4}$

- (R) Chain slips from the table slowly due to weight W of the chain and variable force F and it becomes vertical



(3) Work done by $F = \frac{W\ell}{8}$

- (S) Chain of weight W is pulled by constant force F slightly more than F so that complete chain comes to table. Initially chain is in equilibrium due to F .



(4) Work done by $F = -\frac{3W\ell}{8}$

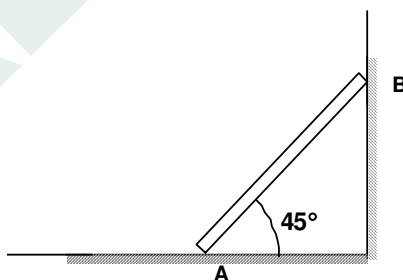
Codes:

	P	Q	R	S
(A)	3	1	4	2
(B)	1	4	2	3
(C)	4	1	2	3
(D)	2	3	4	1

Integer Type

20. The uniform rod AB of length ℓ and mass ' m ' kg is supported on a horizontal smooth surface as shown in figure. If the coefficient of kinetic friction between ends and vertical wall is $2/3$. The rod is released from rest in the shown position.

The angular acceleration α just after released is $\frac{g}{\ell\sqrt{k}}$. Find the value of k .



21. A heavy load is suspended on a light spring. The spring is slowly pulled down at the midpoint (a certain work A is done thereby) and then released.

The maximum kinetic energy W_k of the load in the subsequent motion is $\frac{A}{n}$. Find the value of n .

ADVANCE REVISION ASSIGNMENT-01

PHYSICS

ANSWER KEY

1. (C)	2. (C)	3. (D)	4. (D)	5. (A)	6. (C)	7. (D)	8. (C)
9. (C)	10. (D)	11. (B)	12. (A)	13. (C)	14. (B)	15. (A)	16. (C)
17. (C)	18. (B)	19. (A)	20. (2)	21. (2)			

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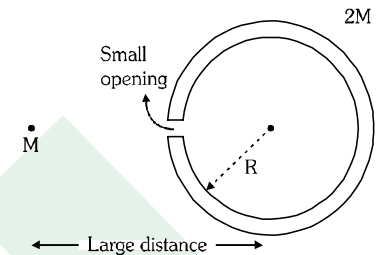
ADVANCE REVISION ASSIGNMENT-02

PHYSICS

Single Correct Answer Type

1. The shown set up is released at rest obtain the speed of mass M when it just enters the shell. Assume all collisions to be elastic.

- (A) $\sqrt{\frac{2GM}{3R}}$ (B) $\frac{2}{3}\sqrt{\frac{GM}{R}}$
(C) $\sqrt{\frac{8GM}{3R}}$ (D) $\sqrt{\frac{2GM}{R}}$



2. A hypothetical spherical planet of radius R, and its density varies as $\rho = Kr$, where K is constant and r is the distance from the center. The pressure caused by gravitational pull inside ($r < R$) the planet at a distance r measured from its center given by [Assume pressure on the surface of planet is P_0]

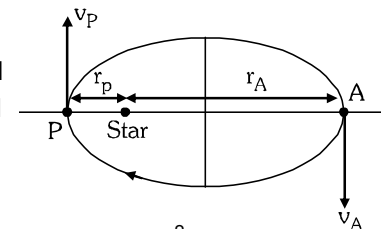
- (A) $P_0 + \frac{1}{2}K^2G\pi(R^4 - r^4)$ (B) $P_0 + \frac{2}{27}K^2G\pi(R^4 - r^4)$
(C) $P_0 + \frac{K^2G\pi}{4}(r^4)$ (D) None of these

3. A test particle of mass m is released from a point far away from a system of two particles, each of mass M ($> m$), kept 2m apart from each other. The speed of the test particle when it is at a point on perpendicular bisector of line joining two masses at which it experiences maximum force due to gravity will be given by (approx)

- (A) $u = \sqrt{3.3GM}$ (B) $u = 0.707 GM$ (C) $u = 1.73\sqrt{GM}$ (D) $u = \sqrt{2.2GM}$

Multiple Correct Type

4. A planet orbits in an elliptical path of eccentricity e around a massive star considered fixed at one of the foci. The point in space where it is closest to the star is denoted by P and the point where it is farthest is denoted by A. Let v_P and v_A be the respective speeds at P and A and



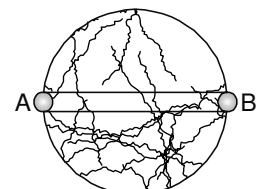
- (A) $\frac{v_P}{v_A} = \frac{1+e}{1-e}$ (B) $\frac{r_P}{r_A} = \frac{(1-e^2)}{(1+e^2)}$ (C) $\frac{r_P}{r_A} = \frac{(1-e)}{(1+e)}$ (D) $\frac{v_P}{v_A} = \frac{1+e^2}{1-e^2}$

5. Mass density of a planet varies as $\rho(r) = \frac{\rho_0 r}{R}$; where R is radius of planet and r is distance from center of planet:

- (A) A particle of mass m is to be projected from surface so as to escape the planet, its velocity of projection relative to surface is $\sqrt{2\pi G\rho_0 R^2}$
(B) A particle of mass m is to be projected from surface so as to escape the planet, its velocity of projection relative to surface is $2\sqrt{2\pi G\rho_0 R^2}$
(C) Gravitational intensity on surface of planet is $\pi G\rho_0 R$
(D) Time period of a satellite in circular orbit very close to planet is $2\pi\sqrt{\frac{R}{\pi G\rho_0}}$

6. Suppose that a narrow and straight tunnel is made passing through the centre of the earth. Two small balls A and B are released simultaneously at the two ends of tunnel. After collision. A comes to rest and B escapes earth's gravitational field. If mass of B is $(\sqrt{3} - 1)kg$.

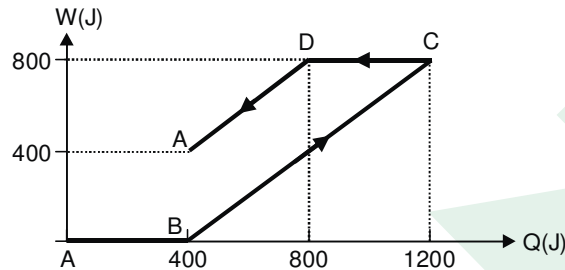
- (A) $m_A = 2 kg$ (B) $m_A = \sqrt{3} + 1 kg$
(C) $e = \frac{1}{\sqrt{2}}$ (D) $e = \frac{\sqrt{3}}{2}$



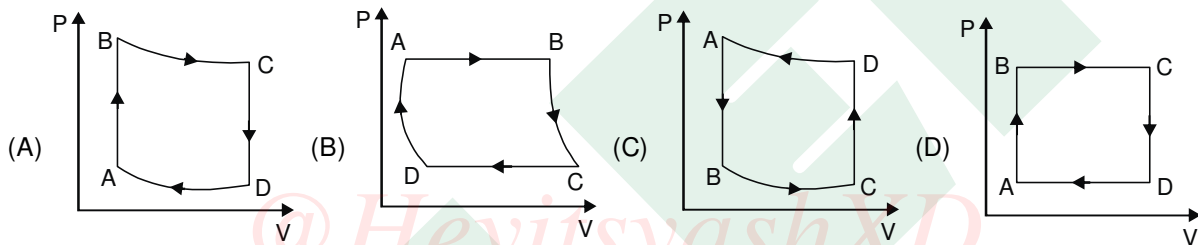
COMPREHENSION TYPE

Paragraph for Ques. no. (7 to 8)

A sample of mono-atomic ideal gas is taken through a cyclic process ABCDA. The cyclic process is such that each of the subprocesses have constant molar specific heat capacity. A graph is plotted in which heat exchanged by the gas upto an instant is shown on x-axis and work done by the gas upto that instant is shown on y-axis.



7. The given cyclic process can be represented on a pressure-volume graph as :-



8. Choose INCORRECT statement :
- (A) Magnitudes of temperature change in the process AB and CD are same
 - (B) Process AB and CD are isochoric
 - (C) Process BC and DA are isothermal
 - (D) Process BC is adiabatic but CD is isothermal

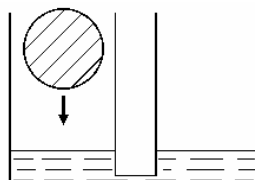
Paragraph for Ques. no. (9 to 11)

An ideal monoatomic gas is confined in a thermally insulated container of volume $2.4 \times 10^{-3} \text{ m}^3$ by a massless piston of area of cross-section of $8 \times 10^{-3} \text{ m}^2$, attached to a light spring. The other end of the spring carries a 10 kg block placed on a rough horizontal surface of coefficient of friction $\mu = 0.8$. Initially, the gas is at 300 K and the spring is in normal state. Now, the gas is heated by an electric heater, till the block just starts moving and till then the piston slowly moves out a distance of 0.1 m without friction. [Take $g = 10 \text{ ms}^{-2}$ and pressure $1 \text{ atm} = 1 \times 10^5 \text{ Nm}^{-2}$]

9. The final temperature of the gas will be
(A) 620 K (B) 440 K (C) 580 K (D) 730 K
10. The work done by the gas till it attains final temperature is
(A) 48 J (B) 4 J (C) 84 J (D) 8 J
11. Net heat supplied by the heater will be
(A) 156 J (B) 84 J (C) 72 J (D) 252 J

Paragraph for question nos. 12 to 14 (one or more than correct)

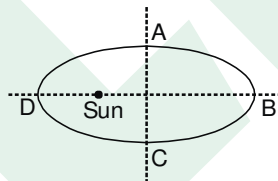
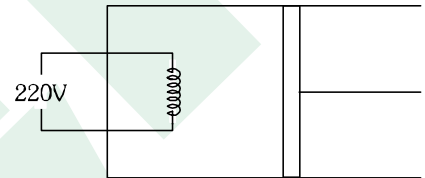
In two identical communicating vessel we poured water (see picture). In one of them we put an ice ball of volume $V = 100 \text{ cm}^3$ which gets exactly half immersed in the water. The density of water $\rho_w = 1000 \text{ kg/m}^3$, the density of ice $\rho_i = 900 \text{ kg/m}^3$.



12. Select the correct statement(s). Soon after placing the ice ball in left vessel.
 (A) The volume of water flowing to the right vessel is 25cc.
 (B) The volume of water flowing to the right vessel is 50cc..
 (C) The ice ball is resting on the bottom of vessel.
 (D) The ice ball is floating on the water surface.
13. After a long time, select the correct statement(s)
 (A) If ice melts, the level of water in both vessel will rise.
 (B) If water freezes, such that ball's radius increases the level of water in both vessel will fall.
 (C) If ice melts, the level of water in left vessel will rise and level of water in right vessel will fall.
 (D) If water freeze such that ball radius increases the level of water in left vessel will fall and level of water in right vessel will rise.
14. Which of the following actions will raise the water level.
 (A) Heating the system
 (B) Putting a coin on top of ice ball.
 (C) Putting a coin in right vessel
 (D) Accelerating the system upwards.

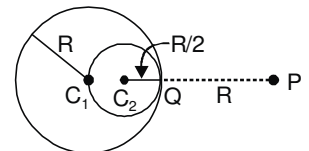
Integer Type

15. There is a certain mass of hydrogen gas in a container closed with a frictionless piston as shown in figure. The gas is heated for 25 s by an electric heater of resistance 50Ω built in the container using a power supply of 220 volt. While the gas expands at constant pressure its temperature increases by 250°C . If efficiency of electric heater is 75%, then find the mass of gas approximately (in gm) in container.
16. Consider a thin hollow spherical shell with uniform distributed charge Q . The force between upper and lower hemisphere is $\eta \frac{kQ^2}{32R^2}$. Find η
17. A planet moves around sun in elliptical orbit with eccentricity $\frac{\pi}{4}$ as shown.



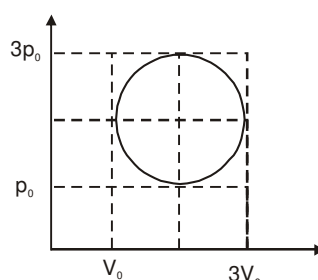
Find the ratio of time taken by planet to travel ABC with that of CDA

18. Inside a solid sphere of radius R and mass M , a spherical cavity of radius $\frac{R}{2}$ is made, such that the surface of the cavity passes through the centre C_1 of the sphere. C_2 is the centre of the cavity. A particle of mass m is released from rest at a point P , a distance R from the surface of the sphere nearer to the cavity as shown in the Figure. The



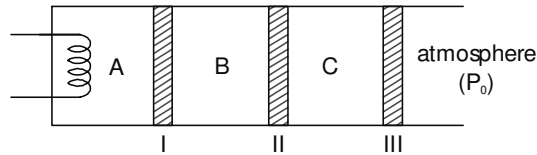
speed with which the particle strikes the point Q is given by $u = \sqrt{\frac{nGM}{3R}}$. Find the value of n .

19. An ideal gas undergoes a process that is shown on the p - V diagram by a circle. What is the ratio of the highest temperature to the lowest temperature attained by the gas during the process? Round off to nearest integer.



20. Three identical chambers have same gas each. Initially volume and temperature of each chamber is V_0 and T_0 . Piston I is fixed and diathermic, II is movable and diathermic while III is movable and adiabatic. Given adiabatic coefficient

v. The temperature rise of chamber C when we supply a heat Q is $\frac{Q(v-1)}{(1+Kv)} \frac{T_0}{P_0 V_0}$. Find K .



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ANSWER KEY

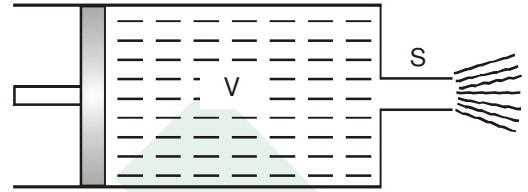
1. (C)	2. (A)	3. (A)	4. (AC)	5. (ACD)	6. (AD)	7. (A)	8. (D)
9. (B)	10. (C)	11. (D)	12. (AC)	13. (AB)	14. (AC)	15. (7)	16. (4)
17. (3)	18. (2)	19. (4)	20. (2)				

ADVANCE REVISION ASSIGNMENT-03

PHYSICS

Single Correct Type

1. Water is filled in a horizontal cylindrical tank filled with position. The volume of tank is 'V' and there is small orifice of area of cross-section 'S' at other side of tank as shown in figure. If the 'W' is total amount of work done in squeezing all volume of water due to applying a constant force on piston then (Assume time to empty vessel is constant)



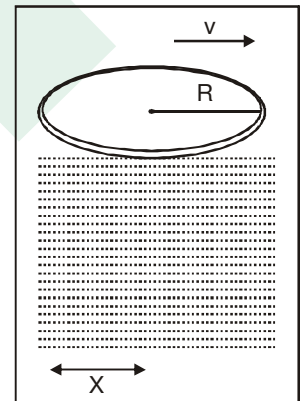
- (A) $W \propto V^{3/2}$
(B) $W \propto V^3$
(C) $W \propto S^2$
(D) $W \propto V$

Paragraph for Q.no. (2 to 4)

Consider a disk of mass m , radius R lying on a liquid layer of thickness T and coefficient of viscosity η as shown in the figure.

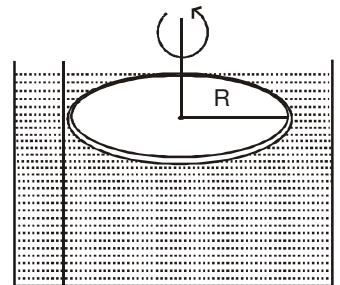
Answer the following questions.

2. The coefficient of viscosity varies as $\eta = \eta_0 x$ (x measured as shown in the figure) at the given instant the disk is floating towards right with a velocity v as shown. Find the force required to move the disk slowly at the given instant.



- (A) $\frac{2\eta_0 R^2 v}{T}$ (B) $\frac{8\eta_0 R^2 v}{T}$
(C) $\frac{\pi\eta_0 R^3 v}{T}$ (D) $\frac{16\eta_0 R^3 v}{T}$

3. The torque required to rotate the disk at a constant angular velocity ω given the viscosity is uniformly η .



- (A) $\frac{4\pi\omega\eta R^4}{T}$
(B) $\frac{\pi\omega\eta R^4}{2T}$
(C) $\frac{2\pi\omega\eta R^4}{T}$
(D) $16\pi\omega\eta R^4$

4. A disc rotating with angular velocity ω is placed on a viscous liquid of thickness T . Find the angle rotated by the disc before it comes to rest. (Viscosity = η , mass of disc = M , radius of disc = R)

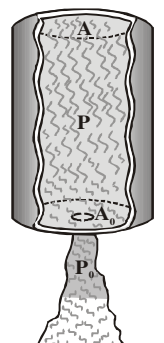
- (A) $\frac{4\omega_0 TM}{\eta\pi R^2}$ (B) $\frac{2\omega_0 TM}{\eta\pi R^2}$ (C) $\frac{\omega_0 TM}{\eta\pi R^2}$ (D) $\frac{\omega_0 TM}{2\eta\pi R^2}$

Paragraph for Q.no. (5 and 6)

Thrust on a rocket & Bernoulli's equation

Thrust is the force which moves the rocket through the air and through space. Thrust is generated by the propulsion system of the rocket through the application of Newton's third law of motion. In the propulsion system, an engine does work on a gas or liquid, called a working fluid.

Consider a chamber of cross-sectional area A filled with a gas of density ρ at a pressure P as shown in figure. Let there be a small orifice of cross-sectional area A_0 at the bottom of the chamber. The speed v_0 with which the gas escape through the orifice can be found by Bernoulli's equation as



$P - P_0 = \frac{1}{2} \rho (v_0^2 - v^2)$, where P_0 represents atmospheric pressure just outside the orifice, v is the speed of the flowing gas inside the chamber and v_0 is the speed of gas through orifice. Since the gas density is negligible, pressure variation with height in the chamber can be neglected.

5. For $A \gg A_0$, the speed of gas through the chamber will be

(A) $\sqrt{\frac{P - P_0}{\rho}}$ (B) $\sqrt{\frac{P - P_0}{2\rho}}$ (C) $\sqrt{\frac{2(P - P_0)}{\rho}}$ (D) $\sqrt{\frac{(P - P_0)}{3\rho}}$

6. The thrust force acting on the rocket due to this will be

(A) $2A_0(P - P_0)$ (B) $\frac{A_0(P - P_0)}{2}$ (C) $A_0(P - P_0)$ (D) $2A(P - P_0)$

7. A transverse sinusoidal wave moves along a string in the positive x direction. In figure (I) displacement of particle at P as a function of time is given and in figure (II) at a particular time t -the snap shot of wave is shown. The wave velocity (cm/s) and velocity of particle at P (cm/s) will be :

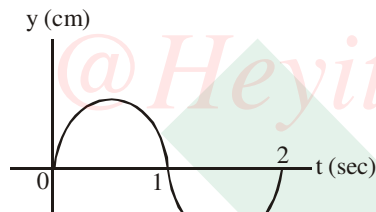


Figure-I

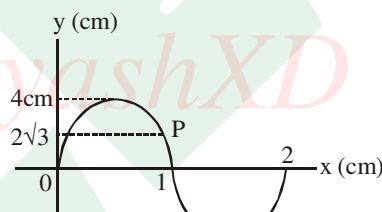
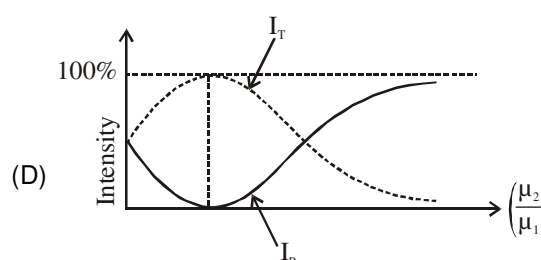
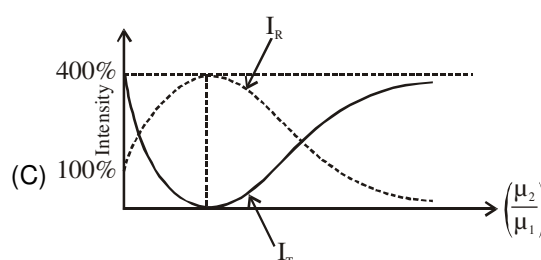
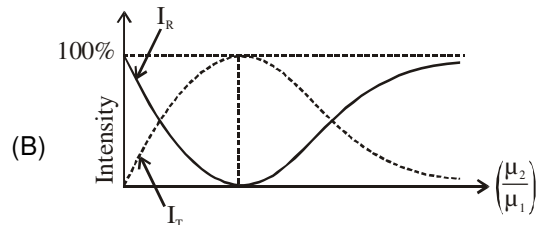
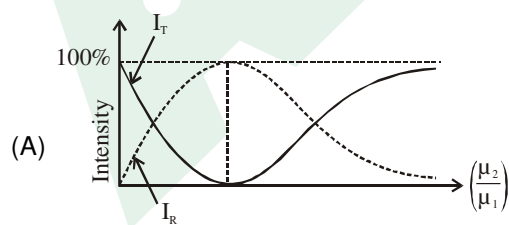
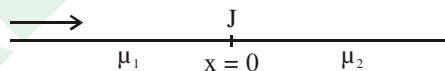


Figure-II

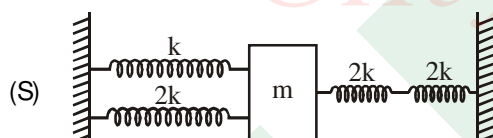
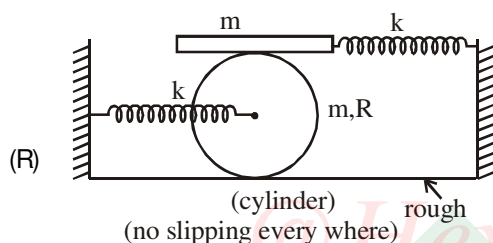
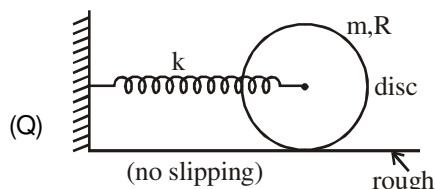
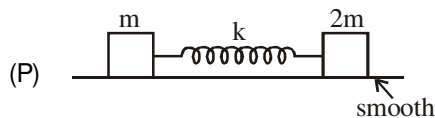
(A) $1 \hat{i}, -3\pi \hat{j}$ (B) $1 \hat{i}, 3\pi \hat{j}$ (C) $1 \hat{i}, 2\pi \hat{j}$ (D) $-1 \hat{i}, -2\pi \hat{j}$

8. A transverse wave travelling in a string of linear mass density μ_1 is incident on junction point J. The intensities of reflected and transmitted wave both as a function of $\left(\frac{\mu_2}{\mu_1}\right)$ are plotted. The correct sketch is :



9. In list-I, the systems are performing SHM and in list-II, the time period of SHM is shown then match list-I with list-II.

List-I



List-II

(1) $\left[2\pi\sqrt{\frac{11m}{10k}} \right]$

(2) $2\pi\sqrt{\frac{m}{4k}}$

(3) $2\pi\sqrt{\frac{3m}{2k}}$

(4) $2\pi\sqrt{\frac{2m}{3k}}$

Codes :

	P	Q	R	S
(A)	3	2	1	4
(B)	4	3	2	1
(C)	3	4	2	1
(D)	4	3	1	2

10. Water rises to a height of 10 cm in a capillary tube and mercury falls to a depth of 3.42 cm in the same capillary tube. If the density of mercury is 13.6 gm/cm^3 and the angle of contact is 135° . Surface tension of water and mercury are T

and $\frac{XT}{2}$ respectively. Find the value of X in nearest integer.

- (A) 13.2 (B) 9 (C) 16 (D) None of these

More than One Correct Type

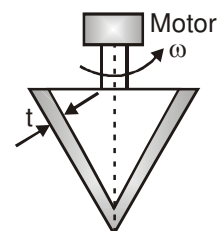
11. The shown figure, there is a conical shaft rotating on a bearing of very small clearance t . The space between the conical shaft and the bearing, is filled with a viscous fluid having coefficient of viscosity η the shaft is having radius R and height h . If the external torque applied by the motor is τ and the power delivered by the motor is P working in 100% efficiency to rotate the shaft with constant ω then

(A) $P = \frac{\pi\omega^2\eta R^3\sqrt{R^2+h^2}}{2t}$

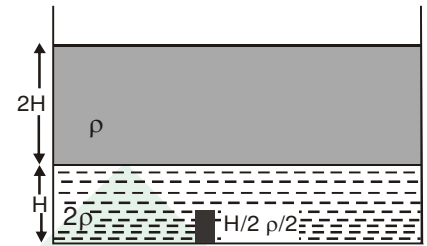
(B) $\tau = \frac{\pi\omega\eta R^3\sqrt{R^2+h^2}}{2t}$

(C) $P = \frac{\pi\omega^2\eta R^2h}{2t}$

(D) $\tau = \frac{\pi\omega\eta^2 R^3h^2}{2t\sqrt{R^2+h^2}}$



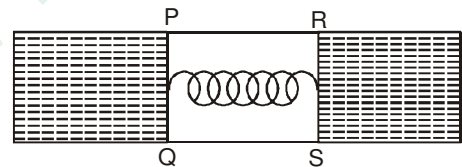
12. A container of very large cross-section filled with two immiscible liquids as shown. A small solid cylinder of height $H/2$ and but its cross-section is negligible when compared to cross-section of the container and its bottom surface just touches the bottom surface of the container and released freely from rest then which of the following statements are correct (neglect viscous forces)



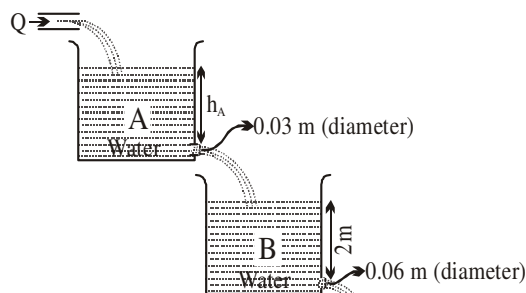
[Assume there is no toppling of the cylinder and heights of the common interface and free surface remains same, density of denser liquid 2ρ , lighter liquid ' ρ ' and the density of solid cylinder $\rho/2$ Acceleration due to gravity " g "]

- (A) Acceleration of the cylinder when it moving completely in the denser liquid of 2ρ is $3g$.
 (B) Acceleration of the cylinder when it moving completely in the lighter liquid of density ρ is $g/2$.
 (C) The velocity of the cylinder when its top surface just before reaches to the common interface of liquids is $\sqrt{3gH}$.
 (D) Maximum height reached by bottom surface of the cylinder from bottom surface of the container just before comes to rest in Air (neglect Air buoyancy) is $7H$.

13. Two identical straight wires PQ and RS each of mass m and length ℓ can move smoothly on a fixed rectangular frame. Two thin films of a liquid of surface tension T are formed between each wire and the frame. The two wires are connected by a massless spring of stiffness k and initially in a natural length position and released then choose the correct option(s).

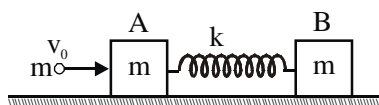


- (A) Maximum elongation of spring, $\Delta x_m = \frac{2T\ell}{k}$
 (B) Maximum elongation of spring, $\Delta x_m = \frac{4T\ell}{k}$
 (C) Each wire executes SHM with the time period, $T_0 = 2\pi\sqrt{\frac{m}{2k}}$
 (D) Each wire executes SHM with the time period, $T_0 = 2\pi\sqrt{\frac{2m}{k}}$
14. Which of the following is/are CORRECT about the following system (take $\pi = \sqrt{10}$)



- (A) h_A should be 16 m for water column to be remains steady in container B
 (B) h_A should be 32 m for water column to be remains steady in container B
 (C) Q should be 4.5 litre/sec for water column to be remains steady in both the containers
 (D) Q should be 18 litre/sec for water column to be remains steady in both the containers

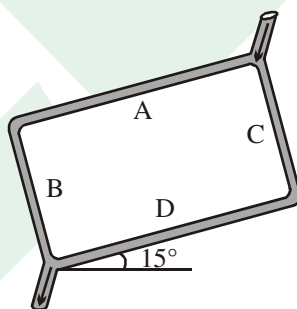
15. Two blocks A and B, each of mass m , are connected by an ideal spring of stiffness k and placed on a smooth horizontal surface. A ball of mass m moving with a velocity v_0 strikes the block A and gets embedded to it. Then,



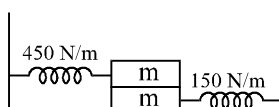
- (A) velocity of block A just after collision is $v_0/2$
 (B) velocity of block B just after collision is zero
 (C) the maximum compression produced in the spring is $v_0 \sqrt{\frac{m}{6k}}$
 (D) the kinetic energy lost during collision is $\frac{1}{4}mv_0^2$
16. Standing waves are produced on a stretched string of length L with fixed ends. When there is a node at a distance $L/3$ from one end, then
 (A) minimum and next higher number of nodes excluding the ends are 2, 5 respectively
 (B) minimum and next higher number of nodes excluding the ends are 2, 4 respectively
 (C) frequency produced may be $V/(3L)$
 (D) frequency produced may be $3V/(2L)$
 [V = velocity of waves in the string]

Integer Type

17. A smooth tube in the shape of a rectangle with rounded corners is placed in a vertical plane, as shown in figure. Two ball bearings are introduced at the upper right-hand corner. One travels by path AB and the other by path CD to arrive at the lower left-hand corner. If the time taken by them is t_{AB} and t_{CD} respectively, find the value of t_{AB}/t_{CD} .



18. A point source of sound is located somewhere along the x-axis. Experiments show that the same wave front simultaneously reaches listeners at $x = -8$ m and $x = +2.0$ m.
 A third listener is positioned along the positive y-axis. What is her y-coordinate (in m) if the same wave front reaches her at the same instant as it does the first two listeners?
19. A spherical ball of density ρ and radius 0.003 m is dropped into a tube containing a viscous fluid. Viscosity of the fluid = $1.260 \text{ N}\cdot\text{m}^{-2}$ and its density $\rho_L = \rho/2 = 1260 \text{ kg}\cdot\text{m}^{-3}$. Find its terminal speed in cm/s. (g = acceleration due to gravity = 10 ms^{-2})
20. When the system shown in the diagram is in equilibrium, the right spring is stretched by 1 cm. The coefficient of static friction between the blocks is 0.3. There is no friction between the bottom block and the supporting surface. The force constants of the springs are 150 N/m and 450 N/m (refer figure). The blocks have equal mass of 2 kg each. Find the maximum amplitude (in cm) of the oscillations of the system shown in the figure that does not allow the top block to slide on the bottom.





ADVANCE REVISION ASSIGNMENT

JEE (Main + Advanced) 2021
ENTHUSIAST COURSE

ADVANCE REVISION ASSIGNMENT-03

PHYSICS

ANSWER KEY

- | | | | | | | | | |
|---------|----------|-----------|----------|----------|------------|----------|---------|---------|
| 1. (B) | 2. (C) | 3. (B) | 4. (C) | 5. (C) | 6. (A) | 7. (C) | 8. (B) | 9. (D) |
| 10. (A) | 11. (AB) | 12. (ACD) | 13. (BC) | 14. (BD) | 15. (ABCD) | 16. (AD) | 17. (1) | 18. (4) |
| 19. (2) | 20. (3) | | | | | | | |

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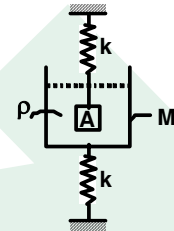
ADVANCE REVISION ASSIGNMENT-04

PHYSICS

Single Correct Type

1. The system shown in the adjacent figure is in equilibrium. The mass of the container with liquid is M , density of liquid in the container is ρ and volume of the block is V . If the container is now displaced downward through a distance x_0 and released such that the block remains well inside the liquid, then during subsequent motion

- (A) Time period of SHM of the container will be $2\pi\sqrt{\frac{M}{k}}$
 (B) Time period of SHM of the container will be $2\pi\sqrt{\frac{(M + \rho V)}{k}}$
 (C) Amplitude of SHM of the container will be greater than x_0
 (D) Container will not perform SHM

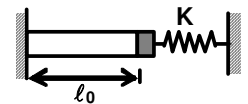


2. A plane progressive harmonic wave is given by the equation : $\phi = \phi_m \sin(2t - 3x + 4y + \pi/3)$, where x and y are in meters, and t is in seconds. Let $\hat{n}$ is the unit vector in the direction of wave propagation, and v is the speed of wave w.r.t. to the wave medium, then :

- (A) $\hat{n} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$; $v = 0.4 \text{ m/s}$ (B) $\hat{n} = \frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}$; $v = \frac{2}{3} \text{ m/s}$
 (C) $\hat{n} = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$; $v = 0.4 \text{ m/s}$ (D) $\hat{n} = -\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$; $v = 0.5 \text{ m/s}$

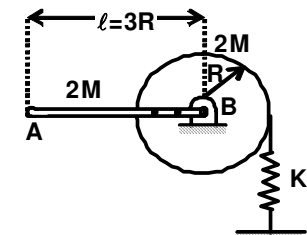
3. A light uniform rod of Young's modulus Y , cross sectional area A , coefficient of linear expansion α and length ℓ_0 is rigidly connected to support at one end and the other end of the rod is connected to the spring as shown in the figure. The temperature of the rod is increased by $\Delta\theta$ with supports remaining fixed. Initially, the spring is in natural length position. Spring force on the rod acts uniformly over the cross section during elongation of the rod. Find the net elongation of the rod. (Assume thermal strain to be small)

- (A) $\frac{K\ell_0\alpha\Delta\theta}{(K + YA/\ell_0)}$ (B) $\frac{YA\alpha\Delta\theta}{(K + YA/\ell_0)}$
 (C) $YA\alpha\Delta\theta$ (D) None of the above



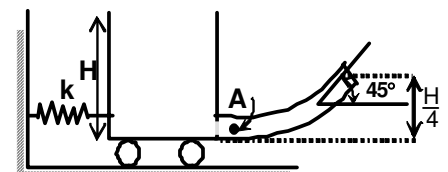
4. A 2M kg slender rod AB is riveted to a 2M kg uniform disk as shown. A belt is attached to the rim of the disk and to a spring which holds the rod at rest in the position shown. If the rod is rotated by a small angle then find the time period of oscillation. Given that the length of the rod is $\ell = 3R$.

- (A) $2\pi\sqrt{\frac{M}{5K}}$ (B) $2\pi\sqrt{\frac{7M}{K}}$
 (C) $\pi\sqrt{\frac{3M}{K}}$ (D) None of the above



Paragraph for Questions no. 5 and 7

A tank is filled with a liquid of density ' ρ ' upto height H . Water flows out from the tank through a nozzle of uniform cross sectional area " a ". The exit of the nozzle makes an angle of 45° with the horizontal. At a given instant, the tank is in equilibrium and rests against a compressed spring of spring constant k . The top of the tank is open to atmosphere. The cross sectional area of the tank is very large in comparison to area of the nozzle. The exit of nozzle is at a height $H/4$ from the base of the tank.



Now, answer the following questions.

5. The compression of the spring at the given instant is

- (A) $\frac{3\rho agH}{2\sqrt{2}k}$ (B) $\frac{\rho agH}{2\sqrt{2}k}$ (C) $\frac{\rho agH}{2k}$ (D) None of the above

6. Pressure inside the nozzle just after entry point at A is
- (A) $P_0 + \rho gH$ (B) $P_0 - \rho gH/4$ (C) $P_0 + \frac{3\rho gH}{4}$ (D) None of the above
7. The maximum height, from the base of the tank, attained by liquid coming out of the nozzle at the given instant is
- (A) H (B) $\frac{3H}{4}$ (C) $\frac{5H}{8}$ (D) $\frac{3H}{4\sqrt{2}}$

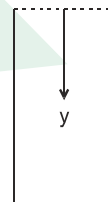
Paragraph for Questions 8 and 9

In an organ pipe (may be closed or open) of 99 cm length standing wave is setup, whose equation is given by longitudinal displacement

$$\xi = (0.1 \text{ mm}) \cos \frac{2\pi}{0.8} (y + 1 \text{ cm}) \cos 2\pi(400)t$$

where y is measured from the top of the tube in meters and t in second.

Here 1 cm is the end correction.



8. The air column is vibrating in
- (A) First overtone (B) Fifth harmonic (C) Third harmonic (D) Fundamental mode
9. Equation of the standing wave in terms of excess pressure is (Bulk modulus of air $B = 5 \times 10^5 \text{ N/m}^2$)
- (A) $P_{\text{ex}} = (125 \pi \text{ N/m}^2) \sin \frac{2\pi}{0.8} (y + 1 \text{ cm}) \cos 2\pi(400)t$
- (B) $P_{\text{ex}} = (125 \pi \text{ N/m}^2) \cos \frac{2\pi}{0.8} (y + 1 \text{ cm}) \sin 2\pi(400)t$
- (C) $P_{\text{ex}} = (225 \pi \text{ N/m}^2) \sin \frac{2\pi}{0.8} (y + 1 \text{ cm}) \cos 2\pi(200)t$
- (D) $P_{\text{ex}} = (225 \pi \text{ N/m}^2) \cos \frac{2\pi}{0.8} (y + 1 \text{ cm}) \sin 2\pi(200)t$

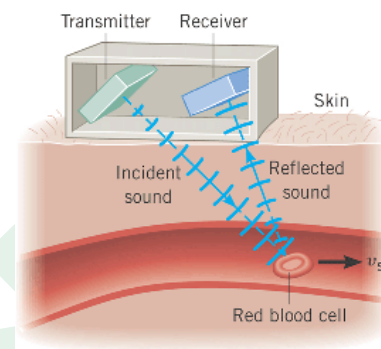
Paragraph for Question Nos. 10 to 12

A tube of length ℓ and radius R carries a steady flow of a fluid whose density is ρ and viscosity η . The velocity v of flow is given by $v = v_0[1 - (r^2/R^2)]$, where r is the distance of flowing fluid from the axis. Then

10. Volume of fluid flowing across the section of tube in unit time is
- (A) $\frac{1}{4} \pi v_0 R^2$ (B) $\frac{1}{2} \pi v_0 R^2$ (C) $\pi v_0 R^2$ (D) $\frac{1}{8} \pi v_0 R^2$
11. Kinetic energy of the fluid within the volume of the tube is
- (A) $\frac{1}{6} \pi \ell R^2 \rho v_0^2$ (B) $\frac{1}{8} \pi \ell R^2 \rho v_0^2$ (C) $\frac{1}{4} \pi \ell R^2 \rho v_0^2$ (D) $\frac{1}{3} \pi \ell R^2 \rho v_0^2$
12. The difference of pressure at the end of the tube is
- (A) $\frac{4\eta \ell v_0}{R^2}$ (B) $\frac{8\eta \ell v_0}{R^2}$ (C) $\frac{2\eta \ell v_0}{R^2}$ (D) $\frac{\eta \ell v_0}{R^2}$

Paragraph for Question no. 13 to 15

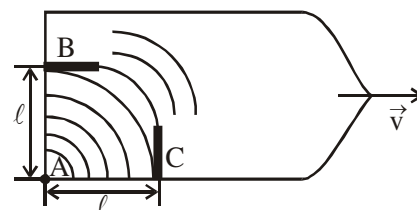
The Doppler flow meter is a particularly interesting medical application of the Doppler effect. This device measures the speed of blood flow, using transmitting and receiving elements that are placed directly on the skin, as in Figure. The transmitter emits a continuous sound whose frequency is typically about 5 MHz. When the sound is reflected from the red blood cells, its frequency is changed in a kind of Doppler effect because the cells are moving with the same velocity as the blood. The receiving element detects the reflected sound, and an electronic counter measures its frequency, which is Doppler-shifted relative to the transmitter frequency. From the change in frequency the speed of the blood flow can be determined. Typically, the change in frequency is around 600 Hz for flow speeds of about 0.1 m/s.



13. Assume that the red blood cell is directly moving away from the source and the receiver. What is the (approx) speed of the sound wave in the blood?
 (A) 1700 m/s (B) 330 m/s (C) 5000 m/s (D) 3000 m/s
14. An abnormal segment of the artery is narrowed down by an arteriosclerotic plaque to one-fourth the normal cross-sectional area. What will be the change in frequency due to reflection from the red blood cell in that region?
 (A) 150 Hz (B) 300 Hz (C) 600 Hz (D) 2400 Hz
15. At what extra rate does the heart have to work due to this narrowing down of the artery? Assume the density to be 1.5 gm/cc and the area of the normal artery to be 0.1 cm².
 (A) 1.125×10^{-4} W (B) 2.5×10^{-4} W (C) 6.25×10^{-5} W (D) 5.625×10^5 W

Paragraph for Question no. 16 and 17

A ship swims at constant velocity $\vec{v}$ on a windless ocean. A short sound pulse is emitted from a sound source located at point A of the open deck of the ship, the speed of sound is v_0 . The sound is reflected from wall B that is at distance ℓ from point A and is perpendicular to the direction of travel. The sound is also reflected from wall C that is also at distance ℓ but is parallel to the direction of travel. The sound reflected from wall B is received at A after time t_B and similarly from wall C after time t_C .



16. If $t_B = n t_C$, then the ratio of $\frac{v_0}{v}$ would be
 (A) $\frac{1}{n}$ (B) $\frac{1}{\sqrt{n}}$ (C) $\frac{1}{\sqrt{1-n^2}}$ (D) $\frac{1}{1-n^2}$
17. If rather than pulse, a continuous sound wave of frequency f is produced by source, then the beat frequency as observed by an observer at A would be
 (A) $\frac{f}{n^2}$ (B) $\frac{f}{1-n^2}$ (C) $\frac{f}{n}$ (D) None of these

More than One Correct Type

18. A closed organ pipe of length 28 cm closed at one end is found to be at resonance when a tuning fork of frequency 850 Hz is sounded near the open end. If velocity of sound in air is 340 m/s, then the
 (A) air in the pipe is vibrating in fundamental mode
 (B) air in the pipe is vibrating in first overtone
 (C) end correction of the pipe is 1 cm
 (D) end correction of the pipe is 2 cm

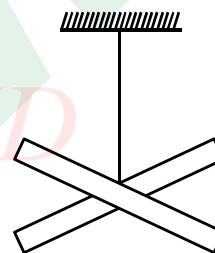
19. Two identical particles A and B are performing SHM along x and y-axis respectively with equal amplitude and frequency of 2 cm and 1 Hz respectively. Equilibrium positions of the particles A and B are at the co-ordinates (3, 0) and (0, 4) respectively. At $t = 0$, B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin. Choose the correct option(s) :
- (A) The distance between A and B is 3 cm at $t = t_0$ sec. It will again be 3 cm at $t = (t_0 + 1)$ sec.
- (B) The distance between A and B can never be 5 cm.
- (C) The maximum magnitude of force acting on each particle during the motion is same.
- (D) Kinetic energy of the system (A and B) is conserved
20. A torsional pendulum consists of two identical rods each of mass M & length ℓ joined perpendicular to each other in horizontal plane and is hanged through the wire having torsional constant C . The system performs simple harmonic motion $\theta = A \sin(\omega t + \phi)$ such that initially angular displacement of the system is equal to half of amplitude and its angular velocity is ω_0 and is decreasing, then :-

(A) Time period of pendulum is $T = 2\pi \sqrt{\frac{M\ell^2}{6C}}$

(B) Amplitude of oscillation is $\frac{\omega_0 \ell}{3} \sqrt{\frac{2M}{C}}$

(C) Initial phase of SHM is $\frac{\pi}{6}$ or $\frac{7\pi}{6}$

(D) Energy stored in wire at $t = 0$ is $\frac{M\ell^2 \omega_0^2}{36}$



* * * * *

ANSWER KEY

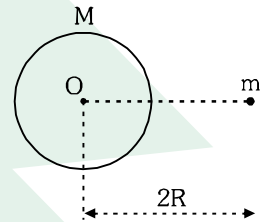
- | | | | | | | | |
|---------|----------|----------|------------|---------|---------|---------|---------|
| 1. (B) | 2. (C) | 3. (B) | 4. (B) | 5. (A) | 6. (D) | 7. (C) | 8. (B) |
| 9. (A) | 10. (B) | 11. (A) | 12. (A) | 13. (A) | 14. (D) | 15. (A) | 16. (C) |
| 17. (D) | 18. (BD) | 19. (AC) | 20. (ABCD) | | | | |

ADVANCE REVISION ASSIGNMENT-05

PHYSICS

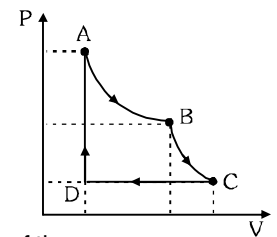
Single Correct Type

- The binding energy of a particle of mass m with a planet, when it is on the planets surface, is $\frac{1}{2}mv_0^2$. A tunnel is dug along a diameter of the planet and the particle is dropped into it from the surface, when the body reaches the centre of the planet, its speed is
 (A) v_0 (B) $\frac{v_0}{\sqrt{2}}$ (C) zero (D) $\frac{v_0}{2}$
- A particle of mass m is at a distance $2R$ from the centre of a thin shell of mass M and having radius R as shown in figure. The gravitational field at the centre of shell is :
 (A) Zero (B) $\frac{GM}{R^2}$
 (C) $\frac{G(M+m)}{4R^2}$ (D) $\frac{Gm}{4R^2}$
- An artificial satellite of mass m is moving in circular orbit at a height equal to the radius R of the earth. Suddenly due to internal explosion the satellite breaks into two parts of equal masses. One part of the satellite stops just after the explosion and then falls to the surface of the earth. The increase in the mechanical energy of the system (satellite + earth) due to explosion will be (g is the acceleration due to gravity on the surface of earth)
 (A) mgR (B) $\frac{mgR}{2}$ (C) $\frac{mgR}{4}$ (D) $\frac{3mgR}{4}$
- A particle of mass equal to half of the earth's mass is released at a distance h ($h < R$) from the surface of earth. The time after which the particle will hit earth's surface is :
 (A) $\sqrt{\frac{4h}{3g}}$ (B) $\sqrt{\frac{2h}{g}}$ (C) $\sqrt{\frac{h}{g}}$ (D) $\sqrt{\frac{2h}{3g}}$

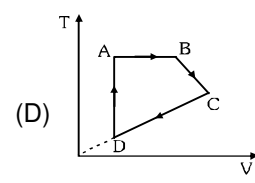
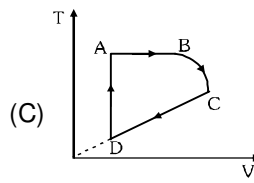
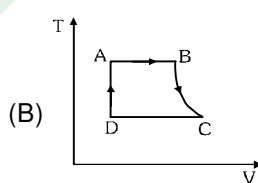
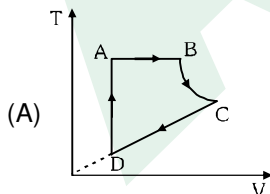


Paragraph for Q.no. (5 to 6)

Two moles of a diatomic gas are carried through the cycle ABCDA shown in the figure PV diagram as shown in figure. The segment AB represents as isothermal expansion, the segment BC an adiabatic expansion. The pressure and temperature at A are 5 atm and 600 K respectively. The volume at B is twice that at A. The pressure at D is 1 atm.

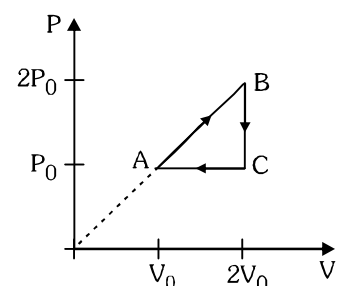


- What is the pressure at B ?
 (A) 2.5 atm (B) 3 atm (C) 3.5 atm (D) None of these
- Find the T versus V graph of the above cyclic process :



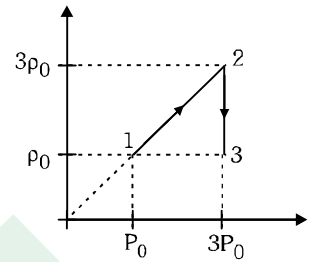
- In the process shown, which of the following option is correct ?

- $\Delta Q_{AB} = +ve, \Delta Q_{BC} = +ve, \Delta Q_{CA} = +ve$
- $\Delta Q_{AB} = +ve, \Delta Q_{BC} = -ve, \Delta Q_{CA} = -ve, \frac{dT}{dP} -ve$ at B
- $\Delta Q_{AB} = +ve, Q_{BC} = \infty, \frac{dT}{dP} = +ve$ at B
- $\Delta Q_{AB} = +ve, \Delta Q_{BC} = -ve, \Delta Q_{CA} = -ve, \frac{dT}{dP}$ not defined at B



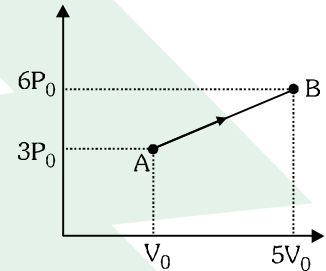
8. The density ρ vs pressure P graph of an ideal monoatomic gas undergoes a process as shown in the figure. Find the total work done during the complete process. Molecular weight of gas is M and number of moles is 1

- (A) $\frac{P_0 M}{\rho_0} (1 - \ln 3)$ (B) $\frac{P_0 M}{\rho_0} (2 - \ln 3)$
(C) $\frac{P_0 M}{\rho_0} (3 - \ln 3)$ (D) None of these



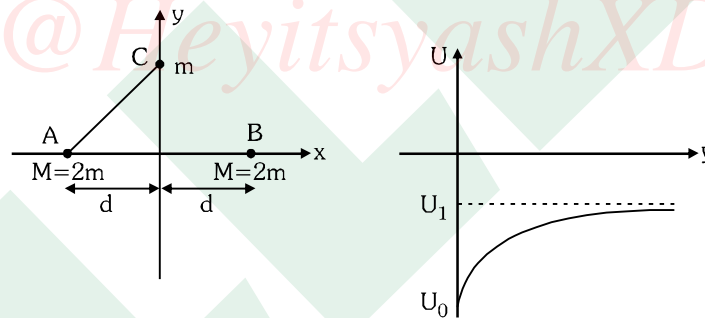
9. One mole of a monoatomic ideal gas undergoes the process $A \rightarrow B$ as indicated in the given P - V diagram. The molar heat capacity for the process is :

- (A) $3R/2$ (B) $13 R/6$
(C) $5 R/2$ (D) $2R$



Multiple Correct Type

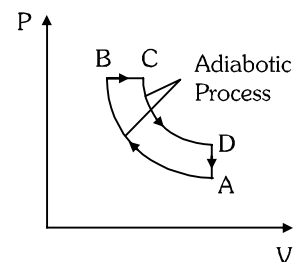
10. Two point masses, each of mass M are kept fixed at points A and B respectively. A third point m is released from infinity with a negligible speed, so that it can move along y -axis under the influence of mutual gravitational attraction on it due to point masses kept at A and B respectively as shown in the figure-1. Figure-2 represents the potential energy of system (includes m , M at A and M at B) with position of m at y -axis. (Neglect any other $Gm^2/d = 12 \text{ Joule}$ ($m = 6 \text{ kg}$). Choose the correct option(s)



- (A) Point mass m will perform periodic motion (B) $U_1 = -24 \text{ Joule}$
(C) Maximum speed of particle is 24 m/s (D) Maximum speed of particle is 4 m/s

11. An idealized diesel engine operates in a cycle known as air standard diesel cycle as shown. Fuel is sprayed into the cylinder at the point of maximum compression, B . Combustion occurs during expansion BC .

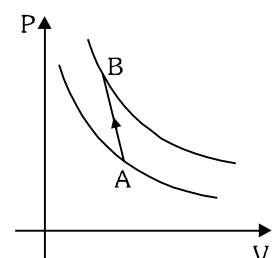
- (A) Heat is absorbed during process BC
(B) Heat is released during process DA
(C) Heat is absorbed during process CD



- (D) Efficiency of the cycle is $\left[1 - \frac{1}{\gamma} \left(\frac{T_D - T_A}{T_C - T_B} \right) \right]$

12. Two adiabatic processes involving an ideal gas are plotted on a P - V diagram. A and B are two points on these curves as shown in the diagram. Which of the following statements is true ?

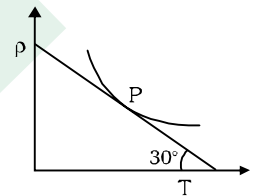
- (A) Heat is given to system in the process AB
(B) Heat is rejected by system in the process AB
(C) Whether heat is rejected or absorbed by the system during the process AB depends on the magnitude of the slope of line AB
(D) Whether heat is rejected or absorbed by the system during the process AB depends on the position of the points A and B on the P - V diagram.



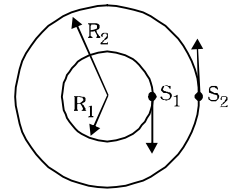
13. Two moles of O_2 ($\gamma = \frac{7}{5}$) at temperature T_0 and 3 moles of CO_2 ($\gamma = \frac{4}{3}$) at temperature $2T_0$ are allowed to mix together in a closed adiabatic vessel. The resulting mixture finally comes in thermal equilibrium. Then,
- (A) final temperature of the mixture is $\frac{23T_0}{14}$ (B) final temperature of the mixture is $\frac{31T_0}{19}$
- (C) adiabatic exponent of the mixture formed is $\frac{34}{25}$ (D) adiabatic exponent of the mixture formed is $\frac{19}{14}$

Integer Type

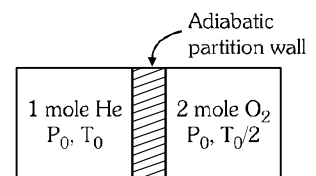
14. A particle is taken to a height K times radius of Earth, above the surface of earth and projected at an angle of 30° to the vertical. During its subsequent motion, it just grazes the surface of earth with a velocity $\frac{\sqrt{15}}{4}$ times the escape velocity from the surface of earth. Find the value of K . ($K < 9$)
15. The variation of density (ρ) with temperature (T) of an ideal gas of molecular mass (M) at constant pressure is plotted as shown in figure. The pressure of the gas at state P is $\frac{RT^2}{\sqrt{KM}}$. Find the value of K .



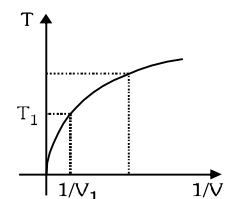
16. The density inside a solid sphere of radius R is given by $\rho = \rho_0 R/r$ where ρ_0 is the density at the surface and r is the distance from the centre. The gravitational field due to this sphere at a distance $2R$ from its centre is found to be $\frac{1}{\eta} \pi G \rho_0 R$. Find η .
17. Two satellites S_1 and S_2 revolve around a planet in coplanar circular orbits in the opposite sense. The periods of revolutions are T and ηT respectively. Find the angular speed (in radian/hr) of S_2 as observed by an astronaut in S_1 , when they are closest to each other. (Given $T = \frac{\pi}{3}$ hr, $\eta = 8$).



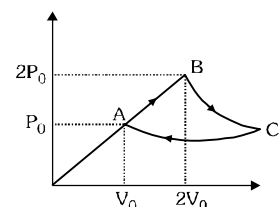
18. Consider the adjacent figure, a piston divides a cylindrical container into two parts. The left part contains 1 mole of helium gas and the right part contains two moles of oxygen gas. The initial temperatures and pressures of the gases in the left chamber are T_0 & P_0 and the right chamber are $\frac{T_0}{2}$ & P_0 respectively, as shown in the figure. The piston as well as the walls of the container are adiabatic. After removal of the piston, gases mix homogeneously and the final pressure becomes $P = \frac{4n}{13} P_0$. Find the value of n .



19. Figure shows a parabolic graph between T and $\frac{1}{V}$ (T = temperature, V = Volume) for a gas undergoing an adiabatic process. If the ratio of rms velocity of molecules and speed of sound in the gas at the same temperature is $\sqrt{n}$, then find the value of n .



20. One mole of a diatomic gas is taken around a cyclic process shown in the figure. The process BC is adiabatic expansion and the process CA is isothermal compression. If the volume of the gas at C is given by $V_C = (2)^n V_0$, then the value of n is



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ADVANCE REVISION ASSIGNMENT

JEE (Main + Advanced) 2024

ENTHUSIAST COURSE

ADVANCE REVISION ASSIGNMENT-05

PHYSICS

ANSWER KEY

- | | | | | | | | |
|---------|-----------|-----------|---------|----------|---------|---------|---------|
| 1. (B) | 2. (D) | 3. (C) | 4. (A) | 5. (A) | 6. (A) | 7. (D) | 8. (B) |
| 9. (B) | 10. (ABD) | 11. (ABD) | 12. (A) | 13. (AD) | 14. (5) | 15. (3) | 16. (2) |
| 17. (3) | 18. (3) | 19. (2) | 20. (6) | | | | |

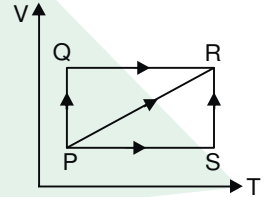
@HeyitsyashXD

ADVANCE REVISION ASSIGNMENT-06

PHYSICS

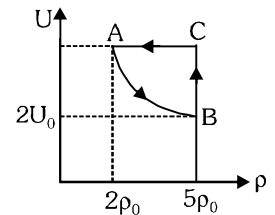
Single Correct Type

- Six stars of equal mass are moving about the centre of mass of the system such that they are always on the vertices of a regular hexagon of side length a . Their common time period will be
 (A) $4\pi\sqrt{\frac{a^3}{Gm}}$ (B) $2\pi\sqrt{\frac{4\sqrt{3}a^3}{Gm(5\sqrt{3}+4)}}$ (C) $4\pi\sqrt{\frac{3a^3}{Gm}}$ (D) None of these
- One mole of a monoatomic gas is taken from P to R, via three paths PQR, PR and PSR. If work done by the gas in PQR is W_1 , in PR work done is W_2 and in PSR work done is W_3 , then
 (A) $W_2 > W_3 > W_1$ (B) $W_1 > W_2 > W_3$
 (C) $W_2 > W_1 > W_3$ (D) $W_3 > W_2 > W_1$
- Four identical samples of same polyatomic ideal gas undergo addition of same amount of heat under four different processes namely $PV^{1/2} = C$; $PV^{3/2} = C$; $P = C$ and $PV^2 = C$. The final internal energy will be maximum for the sample which has undergone
 (A) $PV^{1/2} = C$ (B) $PV^{3/2} = C$ (C) $P = C$ (D) $PV^2 = C$



Paragraph of question nos. 4 to 6

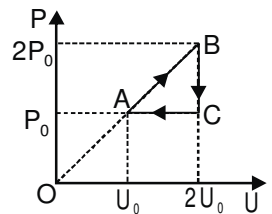
Figure shows the variation of the internal energy U with the density ρ of one mole of ideal monoatomic gas for a thermodynamics cycle ABCA. Here process AB is a part of rectangular hyperbola.



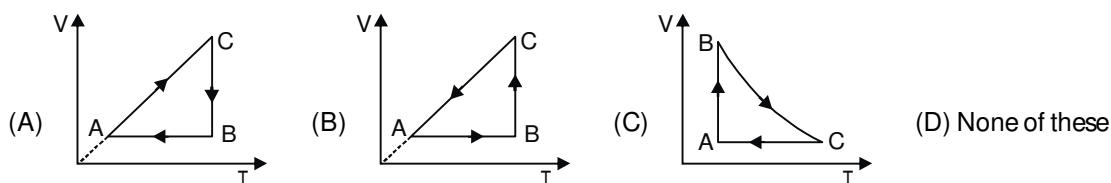
- The P-V diagram of above process
 (A) (B) (C) (D) None of these
- The total amount of heat absorbed by the system for cyclic process is-
 (A) $\left(\frac{10}{3} \ln 2.5 - 2\right) U_0$ (B) $\left(\frac{10}{5} \ln 0.4 - 2\right) U_0$ (C) $50 U_0$ (D) None of these
- The work done in process AB is
 (A) $-U_0$ (B) $-2U_0$ (C) $-5U_0$ (D) None of these

Paragraph of question nos. 7 to 9

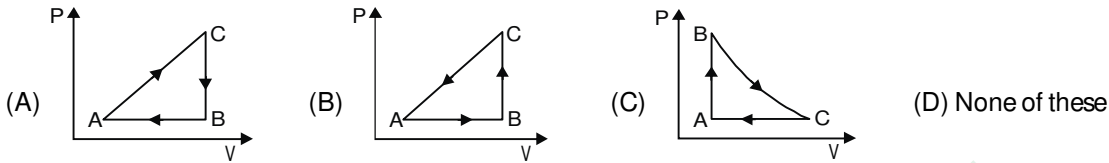
2 mole of an ideal monoatomic gas undergoes a cyclic process ABCA as shown in the figure. R is the universal gas constant. (given $\ln 2 = 0.693$)



- Percentage efficiency of cycle is (approximately)
 (A) 10.38% (B) 13.38%
 (C) 40.38% (D) None
- Graph of volume versus temperature of the cyclic process ABCA is



9. Graph of pressure verses volume of the cyclic process ABCA is



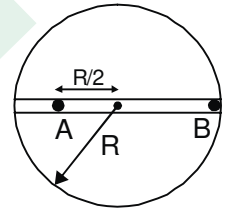
10. A certain thermodynamic cycle comprises of two isothermal and two adiabatic processes. The highest temperature obtained in the entire cycle is 600 K while the lowest temperature is 300 K. When the cycle represented on a P-V curve the sense of the cycle is clockwise. The efficiency of the cycle is

(A) 50% (B) 75% (C) 25% (D) None of these

11. The specific heat capacity of a monoatomic gas for the process $TV^2 = \text{constant}$ is (where R is gas constant)

(A) R (B) 2R (C) $\frac{R}{3}$ (D) None

12. A massive planet of radius R has a diametrical hole as shown in figure, such that hole does not effect the spherocity of the planet. The planet has uniform density. two equal mass A and B [$m \ll M$] are simultaneously released, from positions shown. If the collisions are elastic then the total distance traveled by mass A after being released at the time of 6th collision is : [M → mass of planet, m → mass of A and B neglect the gravitational interaction between mass A and mass B.]



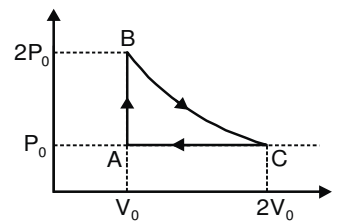
(A) 8.5 R (B) 5 R (C) 6.5 R (D) 9.5 R

13. A cubical vessel of edge 1 m and total thermal resistance (of its walls) is $\frac{1}{R}$ (where, R is universal gas constant) has a small hole in one of its walls. It is kept in a very big closed chamber whose temperature T_0 remains constant. In the chamber and vessel, a mono-atomic gas is filled at a same constant pressure P_0 . At time $t = 0$, temperature of the gas in the vessel is $T_1 (< 2T_0/3)$. When temperature of the gas in the vessel becomes $0.8 T_0$, Rate of change of moles in it will be

(A) $\frac{1}{10}$ (B) $\frac{1}{6}$ (C) $-\frac{1}{10}$ (D) $-\frac{1}{6}$

14. 1 mole of a mono atomic ideal gas undergoes a cyclic process shown in the diagram. The process BC is isothermal and the temperature at B is 800 K. The work done by the gas during the process ABC equals

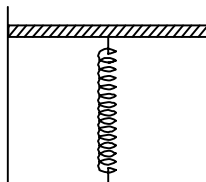
(A) $1600 R \ln 2$ (B) $3200 R \ln 2$
(C) $800 R \ln 2$ (D) Zero

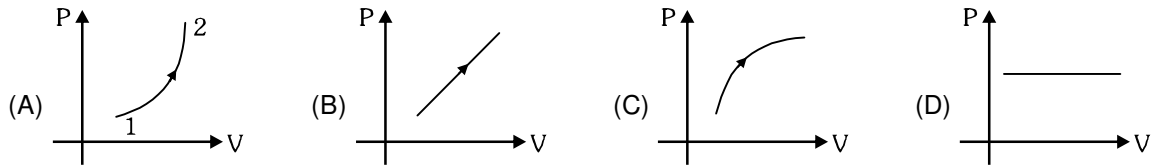


15. The gas inside a spherical bubble expands uniformly and slowly so that its radius increases from R to 2R. Let the atmospheric pressure be p_0 and surface tension be S. The work done by the gas in the process is

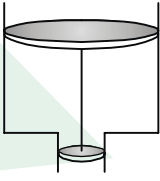
(A) $\frac{28\pi p_0 R^3}{3} + 24\pi SR^2$ (B) $\frac{25\pi p_0 R^3}{3} + 24\pi SR^2$ (C) $\frac{25\pi p_0 R^3}{3} + \frac{23\pi SR^2}{2}$ (D) None of these

16. The piston is massless and the spring is ideal and initially stretched the piston cylinder arrangement encloses an ideal gas. If the gas is heated quasistatically the PV graph





Multiple Correct Type

17. A vertical cylinder contains ideal gas with the help of two pistons whose combined mass is M . Pistons are connected by an inextensible string. The difference in the area of pistons is A . Atmospheric pressure is P_0 . Initially the system is in equilibrium as shown in the figure. Now heat is supplied to the system. Mark the correct statement(s).
- 
- (A) Volume will increase (B) Pressure will remain constant
(C) Temperature will increase (D) Internal energy of gas will remain constant.
18. The equation of a process of diatomic gas is $P^2 = \alpha^2 V$ where α is a positive constant. Then choose the INCORRECT option(s).
- (A) Work done by gas for a temperature change T is $\frac{2}{3} \alpha n R T$
(B) The change in internal energy is $\frac{5}{2} n R T$ for a temperature change T
(C) The specific heat for the process is $\frac{19}{9} R$
(D) The change in internal energy for a temperature change T is $\frac{5}{2} \alpha n R T$
19. One mole of an ideal gas is enclosed in a cylinder fitted with a frictionless piston and occupies a volume of 1.5 litre at a pressure of 1.2 atm. It is subjected to a process given by equation $T = \alpha V^2$, γ (adiabatic constant) for the gas = 1.5. Choose the correct statement.
Given $R\alpha = 80 \text{ J mol}^{-1} \text{ lit}^{-2}$ (R = gas constant and α is constant)
- (A) The $P - V$ diagram of the process is a straight line.
(B) The work done by the gas in increasing the volume of the gas to 9 litre is 3150 J.
(C) The change in the internal energy of the gas is 12600 J.
(D) The heat supplied to the gas in the process is 15750 J.
20. A planet of mass m revolves round the sun in an elliptical orbit of eccentricity e . At apogee, its speed is v_1 . At perigee. Its speed is v_2 . Semi-major axis of the elliptical orbit is a .
Choose the correct answers
- (A) sun is located at one of the foci of the ellipse
(B) $\frac{v_1}{v_2} = \frac{1-e}{1+e}$
(C) Time period of revolution of the planet is $T = \frac{m\pi a^2 \sqrt{1-e^2}}{L}$ where L is the angular momentum of the planet
(D) Angular momentum of the planet with respect to sun is conserved.



ADVANCE REVISION ASSIGNMENT

JEE (Main + Advanced) 2024

ENTHUSIAST COURSE

ADVANCE REVISION ASSIGNMENT-06

PHYSICS

ANSWER KEY

- | | | | | | | | |
|-----------|-----------|------------|-----------|---------|---------|---------|---------|
| 1. (B) | 2. (D) | 3. (B) | 4. (A) | 5. (A) | 6. (B) | 7. (B) | 8. (B) |
| 9. (C) | 10. (A) | 11. (A) | 12. (A) | 13. (D) | 14. (C) | 15. (A) | 16. (B) |
| 17. (ABC) | 18. (ACD) | 19. (ABCD) | 20. (ABD) | | | | |

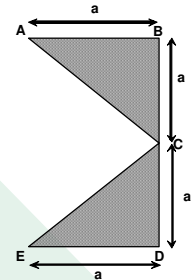
@HeyitsyashXD

ADVANCE REVISION ASSIGNMENT-07

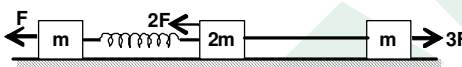
PHYSICS

Single Correct Type

1. ABCDECA is a planar body of mass m of uniform thickness and same material. The dimensions are as shown in the figure. The moment of inertia of the body about an axis passing through point A and perpendicular to planar body is I_1 and that of about an axis passing through C and perpendicular to planar body is I_2 . If I_1/I_2 is k . Find the value of k .

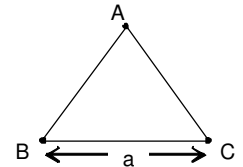


- (A) 1 (B) 2
(C) 3 (D) 4
2. Three point masses m , $2m$ and m , connected with ideal spring (of spring constant k) and ideal string as shown in the figure, are placed on a smooth horizontal surface



At $t = 0$, three constant forces F , $2F$ and $3F$ start acting on the point masses m , $2m$ and m respectively, as shown in figure. Find the maximum extension in the spring.

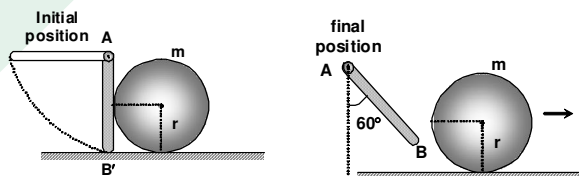
- (A) $\frac{2F}{k}$ (B) $\frac{5F}{2k}$ (C) $\frac{3F}{k}$ (D) $\frac{8F}{3k}$
3. Three particles are located at corners of an equilateral triangle of side a , as shown in the figure. The particles start moving with a constant speed v such that the particle initially at A always heads for the particle which was initially at B. Similarly the particle at B heads for the particle at C and the particle at C heads for the particle at A. The magnitude of initial acceleration of the particles is



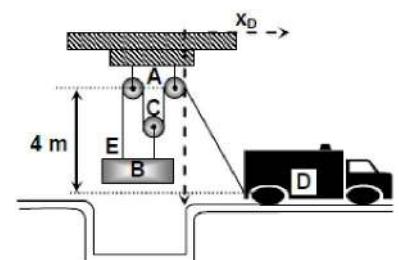
- (A) $\frac{v^2}{a}$ (B) $\frac{1}{2} \frac{v^2}{a}$ (C) $\frac{\sqrt{3}v^2}{2a}$ (D) $\frac{v^2}{\sqrt{2}a}$
4. A uniform rod AB of mass M is attached to a hinge at one end A, and released from rest from the horizontal position. The rod rotates about A, and when it reaches the vertical position the rod strikes a sphere of mass m and radius r initially at rest on the smooth horizontal surface as shown in the adjacent figure. The impact is along the horizontal direction and perfectly elastic.

If at the moment of impact the lowest end of the rod is very close to the smooth horizontal surface. After the impact, the sphere moves along the horizontal and the rod, subsequently rises to a maximum of 60° with the vertical. Choose the correct statement(s) from the following, taking into account the information given above. The

length of the rod equals $\sqrt{2}r$. ($r = \frac{6\sqrt{2}}{10} m$)

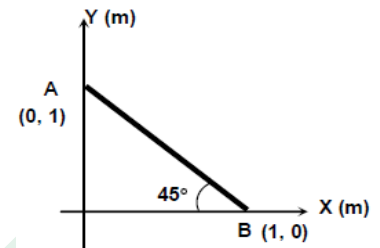


- (A) The ratio M/m is $3/2$ (B) The ratio m/M is $2/3$ (C) The ratio m/M is $2/9$ (D) The ratio m/M is $9/2$
5. A block B is suspended from a cable that is attached to the block at E, wraps around three pulleys and is tied to the back of a truck D. The truck starts from rest when x_D is zero and moves forward with a constant acceleration of $a_p = 3/2 m/s^2$, the speed of the block at the instant when $x_D = 3 m$ is

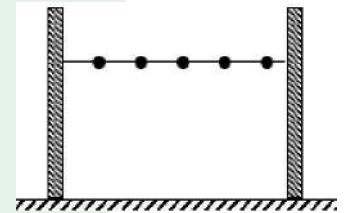


- (A) $\frac{1}{5} m/s$ (B) $\frac{2}{5} m/s$
(C) $\frac{3}{5} m/s$ (D) $1 m/s$

6. In an XY horizontal plane a force field $\vec{F} = -(40\text{N/m})(y\hat{i} + x\hat{j})$ is present where x and y are the coordinates of any point on the plane. A smooth rod AB is fixed in the plane as shown in the figure. A particle of mass 5 kg is to be released with a velocity in this force field such that it reaches to point B. Find the minimum velocity that must be imparted along the rod at A such that it reaches to B.



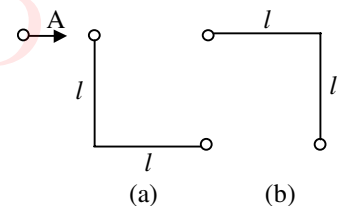
- (A) 1 m/s (B) 2 m/s
(C) 3 m/s (D) 4 m/s
7. Five identical balls each of mass m and radius r are strung like beads at random and are at rest along a smooth, rigid horizontal thin rod of length L, mounted between immovable supports as shown in the figure. Assume $10r < L$ and that the collision between balls or between balls and supports are elastic. If one ball is struck horizontally so as to acquire a speed v, the average force felt by the support is



- (A) $\frac{5mv^2}{L-5r}$ (B) $\frac{mv^2}{L-10r}$ (C) $\frac{5mv^2}{L-10r}$ (D) $\frac{mv^2}{L-5r}$

Paragraph for Q. Nos. 8 to 10

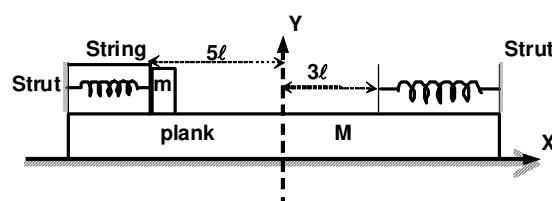
A spherical ball of mass M moving with initial velocity v collides elastically with another ball of mass M, which is fixed at one end of L shaped rigid massless frame as shown in figure (A). The L shaped frame contains another mass M connected at the other end.



8. The speed of the striking mass after collision is
- (A) $\frac{v}{2}$ backwards (B) 0
(C) $\frac{v}{3}$ in same direction (D) $\frac{v}{7}$ backwards
9. The angular speed of L frame immediately after collision is
- (A) $\frac{4v}{7l}$ (B) $\frac{v}{3l}$ (C) $\frac{4v}{2l}$ (D) $\frac{v}{7l}$
10. How soon will the frame come to the orientation shown in figure (B) after collision?
- (A) $\frac{\pi l}{4v}$ (B) $\frac{7\pi l}{2v}$ (C) $\frac{\pi l}{v}$ (D) $\frac{7\pi l}{4v}$

Paragraph for Q. no. (11 and 12)

According to the principle of conservation of linear momentum, if the external force acting on the system is zero, the linear momentum of the system will remain conserved. It means if the centre of mass of a system is initially at rest, it will remain at rest in the absence of external force, i.e. the displacement of centre of mass will be zero.



A plank of mass M is placed on a smooth horizontal surface. Two light identical springs each of stiffness k are rigidly connected to struts at the end of the plank as shown in the figure. When springs are in its natural length,

their free ends are at a distance of 3ℓ from y axis. A block of mass m is placed on the plank and pressed against left spring so that it is compressed by 2ℓ . To keep the block at rest it is connected to the strut by means of a light string. Initially the system is at rest. The right spring is in its natural length. Now the string is burnt.

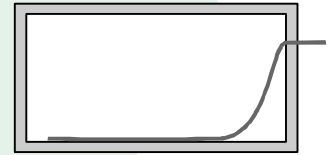
11. The maximum displacement of plank is

(A) $\frac{10m\ell}{M}$ (B) $\frac{10m\ell}{M+m}$ (C) $\frac{4m\ell}{M+m}$ (D) $\frac{8m\ell}{M+m}$

12. Maximum kinetic energy of block m is

(A) $\frac{km^2}{2M(M+m)}$ (B) $\frac{2km^2}{M(M+m)}$ (C) $\frac{kM\ell^2}{2(M+m)}$ (D) $\frac{2kM\ell^2}{(M+m)}$

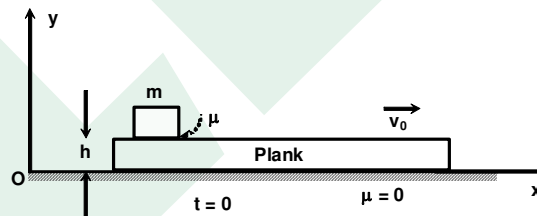
13. An elastic and light cord of length $l_0 = 5.0$ m and stiffness $k = 50$ N/m is inside a fixed box. A small portion of length $x_0 = 20$ cm is left out of a hole made one of the walls. The cord fits tightly in the hole therefore when pulled frictional forces act on the cord. Total magnitude of these frictional forces is $F = 25$ N. How much work has to be done in slowly pulling the complete length of the cord out of the box ?



(A) $F(l_0 - x_0) + \frac{F^2}{k} \left(1 - \frac{x_0}{2l_0}\right)$ (B) $F(l_0 - x_0) + \frac{F^2}{k} \left(1 - \frac{2x_0}{l_0}\right)$
 (C) $F(l_0 - x_0) + \frac{2F^2}{k} \left(1 - \frac{2x_0}{l_0}\right)$ (D) $F(l_0 - x_0) - \frac{2F^2}{k} \left(1 + \frac{2x_0}{l_0}\right)$

Multiple Correct Type

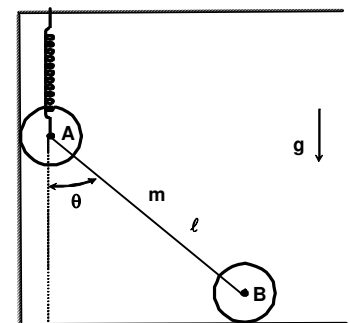
14. A very large plank of mass m is moving along horizontal smooth surface with speed v_0 along positive x -axis. At $t = 0$, a small block of mass m is gently put on the plank as shown in the figure. The coefficient of friction between plank and block is μ . The height of plank is h and width of plank is negligible.



(Block can be considered as point mass). Choose the **INCORRECT** statement(s).

- (A) The kinetic energy of system (Plank + Block) is conserved during motion
 (B) The linear momentum of system (Plank + Block) is conserved during motion
 (C) The angular momentum about origin O of the system (Plank + Block) is conserved during motion.
 (D) The work done by friction on the system (Plank + Block) is zero during the motion.

15. Two identical discs, each of negligible mass and radius $r \ll \ell$ are connected by a uniform rod of mass m and length ℓ as shown in the figure. The disc A is connected with an ideal spring of spring constant k at its centre. The spring is un-stretched when $\theta = 0^\circ$. There is no slipping at any contact at any point of time during the motion. Choose the correct statement [Spring always obeys Hook's Law and $mg = k\ell$]. During the motion, each disc remains in contact with the contact surface.



- (A) The value of θ , when the system is in stable equilibrium is 60°
 (B) The value of θ , when the system is in stable equilibrium is 30°
 (C) Spring potential energy of the system in the state of stable equilibrium is $\frac{k\ell^2}{8}$
 (D) The angular speed of the rod when it passes through the position of the stable equilibrium is $\sqrt{\frac{3g}{4\ell}}$

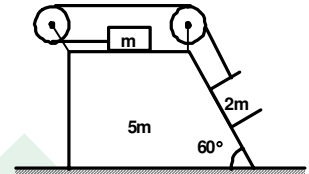
16. A system is shown in the figure, which is released from rest. There is no friction between ground and mass 5m. The coefficient of friction between mass 5m and mass 2m is μ . The coefficient of friction between mass 5m and mass m is also μ . Choose the correct statement(s)

(A) The acceleration of centre of mass of system is zero in horizontal direction.

(B) The magnitude of acceleration of mass 5m is zero if $\mu = 0$

(C) The tension in the string is $\frac{4mg}{3\sqrt{3}}$ if $\mu = \frac{1}{\sqrt{3}}$

(D) The magnitude of acceleration of mass 5m is zero if $\mu = \frac{1}{\sqrt{3}}$



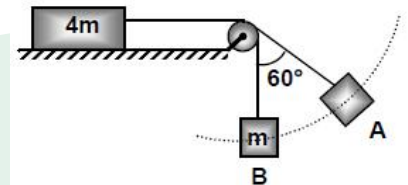
17. In the system shown in the figure the mass m moves in a circular path of angular amplitude 60° and mass 4m is stationary. Then choose the correct statement(s)

(A) The minimum value of coefficient of friction between the mass 4m and the surface of table is 0.5

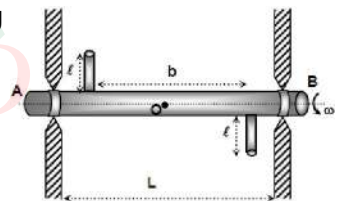
(B) The work done by gravitational force on the block m is positive when it moves from A to B

(C) The power delivered by the tension when m moves from A to B is non-zero

(D) The kinetic energy of m in position B equals the work done by gravitational force on the block when the moves from position A to position B.



18. Figure shows a horizontal rod AB which is free to rotate about two smooth bearing system. Two identical uniform rods each of mass m are attached to rod AB symmetrically about the centre of mass O of the rod AB. All the dimensions are given in the figure. The system is rotating with constant angular velocity ω in such a way that the upper rod is coming outward from the plane of the paper in the position shown. Gravity can be assumed to be absent in the experiment. Then choose the correct option(s).



(A) The hinge reaction at A on the rod AB is downward.

(B) The hinge reaction at B on the rod AB is upward

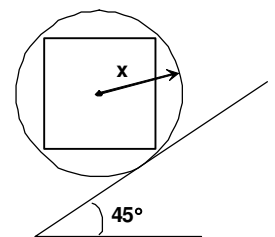
(C) The hinge reaction at B on the rod AB is downward

(D) The angular momentum of the system about point O is NOT along the rod AB.

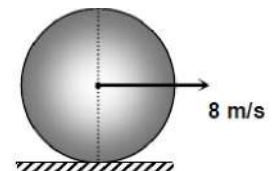
Integer Type

19. For identical rods, each of mass m are welded at their ends to form a square, and the corners are then welded to a light metal hoop of radius r. If the rigid assembly of rods and hoop is allowed to roll down the inclined rough surface. If the minimum value of the coefficient of static friction which will prevent

slipping is $\frac{k}{10}$. Find the value of k.



20. A 10 kg solid sphere of radius $r = 0.8$ m is rolling without slipping on a horizontal rough surface with 8 m/s. The force applied by the right half of the sphere on the left half is $30x$ Newton. Find the value of x.



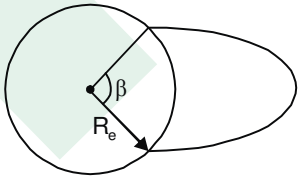
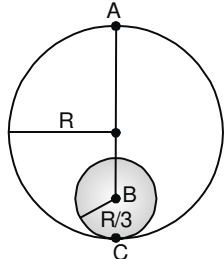
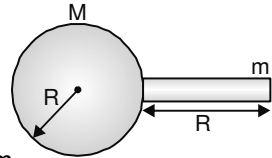
ANSWER KEY

1. (C)	2. (A)	3. (C)	4. (A)	5. (C)	6. (B)	7. (B)	8. (D)
9. (A)	10. (D)	11. (B)	12. (D)	13. (A)	14. (ACD)	15. (ACD)	16. (ABCD)
17. (ABD)	18. (ABD)	19. (4)	20. (5)				

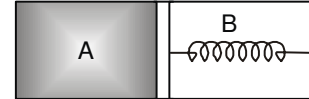
ADVANCE REVISION ASSIGNMENT-08

PHYSICS

One or More Correct Type

- 1 mole of an ideal gas undergoes a process such that $P \propto \frac{1}{T}$. The molar heat capacity of this process is 33.24 J/mol K.
 (A) The work done by the gas is $2R\Delta T$.
 (B) Degree of freedom of the gas is 4.
 (C) Degree of freedom of the gas is 3.
 (D) $\gamma = \left(\frac{C_P}{C_V}\right)$ for the gas is 1.5.
- A particle is projected from the surface of the earth (of radius R_e and mass M_e) with speed equal to $\sqrt{\frac{GM_e}{R_e}}$, at certain angle from local horizontal as shown in the figure such that the angle subtended by arc between launching & landing site at earth's centre is $\beta = 90^\circ$. Calculate the maximum separation of the particle from the centre of the earth. Consider earth to be uniformly dense and air resistance to be absent.
 (A) $R_e\sqrt{2}$
 (B) $R_e(\sqrt{2} + 1)$
 (C) $R_e \frac{(\sqrt{2} + 1)}{\sqrt{2}}$
 (D) $R_e \frac{(\sqrt{2} + 1)}{(\sqrt{2} - 1)}$

- Inside a uniform sphere of mass M and radius R , a cavity of radius $R/3$ is made in the sphere as shown.
 (A) Gravitational field inside the cavity is uniform.
 (B) Gravitational field inside the cavity is non-uniform.
 (C) The escape velocity of a particle of very small mass projected from point A is $\sqrt{\frac{88GM}{45R}}$
 (D) The escape velocity of a particle of very small mass projected from any point on the surface of the sphere is $\sqrt{\frac{88GM}{45R}}$

- An ideal monoatomic gas undergoes a process in which $P = 3V^3$. Choose the correct statements.
 (A) If volume increases then temperature will also increase.
 (B) If temperature of the gas decreases, then pressure must also decrease.
 (C) For expansion, heat will have to be supplied to the gas.
 (D) If temperature increases then work done by gas is positive.
- A satellite is in a circular orbit very close to the surface of a planet with speed v_0 . At some point it is given an impulse along its direction of motion causing its velocity to increase α times (that is, its speed becomes αv_0). It now goes into an elliptical orbit. The maximum possible value of α for satellite to move in the elliptical orbit is.
 (A) 2
 (B) $\sqrt{2}$
 (C) $\sqrt{2} + 1$
 (D) $\frac{1}{\sqrt{2} - 1}$
- A uniform thin rod of mass m and length R is placed normally on surface of earth as shown. The mass of earth is M and its radius is R . Then the magnitude of gravitational force exerted by earth on the rod is:
 (A) $\frac{GMm}{2R^2}$
 (B) $\frac{GMm}{4R^2}$
 (C) $\frac{4GMm}{9R^2}$
 (D) $\frac{GMm}{8R^2}$


7. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant K is connected with the piston and the wall of the container as shown. Initially the spring is unstretched. The ideal gas in chamber A is allowed to expand slowly. Let in equilibrium the spring be compressed by a length x_0 , then



- (A) Change in pressure of the gas is Kx_0/S
 (B) Work done by the gas is $1/2 Kx_0^2$
 (C) Magnitude of change in internal energy of the gas is $1/2 Kx_0^2$
 (D) None of these
8. A certain quaternary star system consist of three star, each of mass ' m ' moving in same circular orbit of radius ' r ' about a central star of mass ' M '. The stars orbit in same sense and are positioned one-third of circumference apart from one another. The period of revolution of each star will be

(A) $2\pi \sqrt{\frac{r^3}{G(M + \frac{m}{\sqrt{3}})}}$ (B) $2\pi \sqrt{\frac{r^3}{G(M + m)}}$ (C) $2\pi \sqrt{\frac{r^3}{G(M + 3m)}}$ (D) $2\pi \sqrt{\frac{r^3}{GM}}$

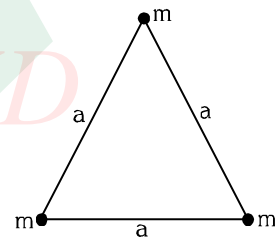
9. If the three point masses are released simultaneously to move under influence of mutual gravitational attraction, speed of each particle when each one of them has moved a distance $\left(\frac{a}{2\sqrt{3}}\right)$

(A) $\sqrt{\frac{6\sqrt{3} Gm}{(2 - \sqrt{3})a}}$

(B) $\sqrt{\frac{2Gm}{a}}$

(C) $\sqrt{\frac{Gm}{a}}$

(D) None of these



10. Force of attraction on a particle of mass m placed at the centre of a semicircular wire of length L and mass M is

(A) $\frac{GMm}{r^2}$

(B) $\frac{GMm}{2r^2}$

(C) Zero

(D) $\frac{2GMm\pi}{L^2}$

11. Starting with the same initial conditions if an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 in the process is purely isothermal, W_2 is purely isobaric and W_3 is purely adiabatic, then,
- (A) $W_2 > W_1 > W_3$ (B) $W_2 > W_3 > W_1$ (C) $W_1 > W_2 > W_3$ (D) $W_1 > W_3 > W_2$

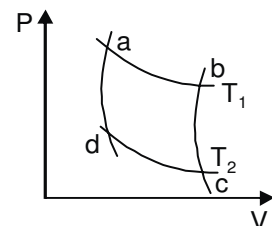
12. Two different adiabatic paths for the same gas intersect two isothermal T_1 and T_2 as shown in the P-V diagram. Then V_a/V_d is equal to

(A) $\frac{V_c}{V_b}$

(B) $V_c V_b$

(C) $\frac{V_b}{V_c}$

(D) $\frac{V_c V_b}{V_c - V_b}$



13. Inside a large isolated shell of mass M having inner and outer radius r and R respectively, a particle is projected with a velocity v tangentially along the inner surface of the shell. Find the time taken for the particle to complete one revolution.

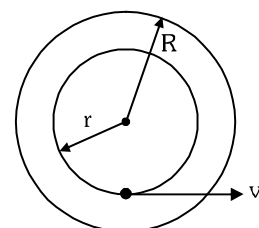
The friction is absent.

(A) $\frac{2\pi r}{\sqrt{GM/R}}$

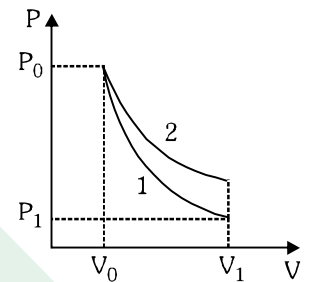
(B) $\frac{2\pi r^{3/2}}{\sqrt{GM}}$

(C) $\frac{2\pi r}{v}$

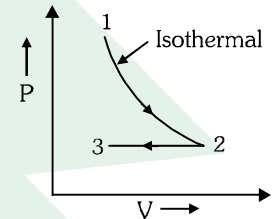
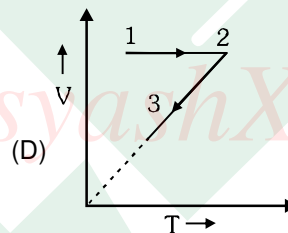
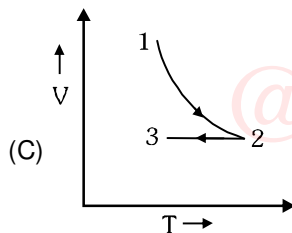
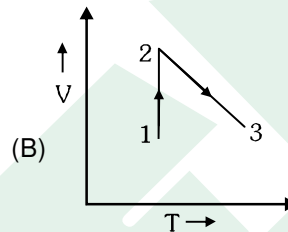
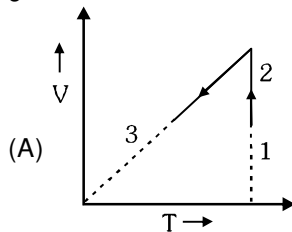
(D) The particle will not be able to complete a revolution, as it will fall off toward the centre.



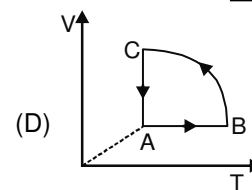
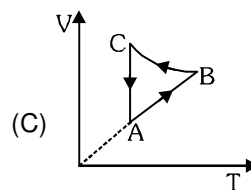
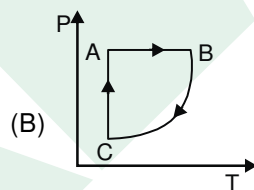
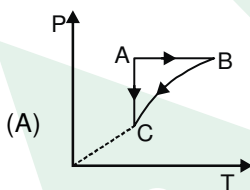
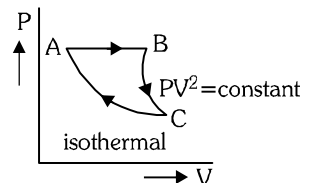
14. An ideal monoatomic gas expansion process is shown by process 1 in the P-V diagram. The process 2 represents an adiabatic expansion of the gas. Then :
- (A) The temperature of the gas decreases in process 1
(B) The work done by the gas in process 1 = heat supplied to the gas in process 1
(C) The heat supplied to the gas in process 1 is negative
(D) The work done in process 1 is less than $\frac{3}{2}(P_0V_0 - P_1V_1)$



15. One mole of a monoatomic gas undergoes the processes 1-2 and 2-3 as shown the corresponding graph for the above process for 2 moles of the gas is

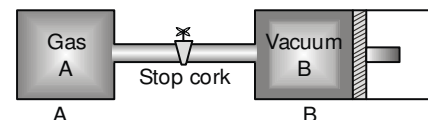


16. An ideal gas is taken through a cyclic process as shown in P-V diagram. Then which of the following option(s) correctly represent the above cyclic process.



Paragraph for Q.Nos. 17 to 19

An ideal diatomic gas is confined in a diathermic cylinder A of volume V_0 . Then, cylinder is connected to another cylinder B with the help of tube of a negligible volume. The cylinder B is fitted with a movable piston, which can be adjusted from outside. Initially, piston is adjusted so that the volume of B is same as volume of A, i.e., V_0 . B is evacuated and stop cork is opened so that gas expands and occupies the volume $2V_0$.



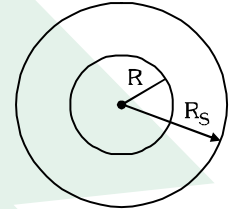
17. During this free expansion, the internal energy of this systems
- (A) Increases (B) Decreases (C) Remains constant (D) Nothing can be said
18. With open stop cork, the piston is slowly moved to compress the gas back to cylinder A at temperature T. Then, for n moles of gas, work done on the gas is
- (A) $nRT \ln 2$ (B) $-nRT \ln 2$ (C) nRT (D) $-nRT$
19. The heat absorbed by the gas in the above process is
- (A) $nRT \ln 2$ (B) $-nRT \ln 2$ (C) nRT (D) $-nRT$

Paragraph for Q. no. (20 to 22)

Supernova refers to the explosion of a massive star. The material in the central case of such a star continues to collapse under its own gravitational pull. If mass of the core is less than 1.4 times the mass of sun, its collapse finally results in a white dwarf star. However, if the core has a mass greater than this, it could end up soon as a neutron star and if its mass is more than about three solar masses, the collapse may still continue till the star becomes a very small object with an extremely high value of density called a 'Black hole'. Escape speed for a black hole is very large. The figure shows a black hole of radius R and another concentric sphere of radius R_s , called the 'Schwarzschild Radius'. It is the critical radius at which escape speed equals the speed of light c . Nothing even the light, can escape from within the sphere of Radius R_s . So light from a black hole cannot escape and hence the terminology 'black hole'.

There has been astronomical evidence of a small and massive object at the centre of our galaxy the 'Milky way'. Suppose that there is a particle at a distance about 6 light years that orbits this massive object with an orbital speed of about 2×10^5 m/s. Use the given data wherever necessary and answer the questions that follow.

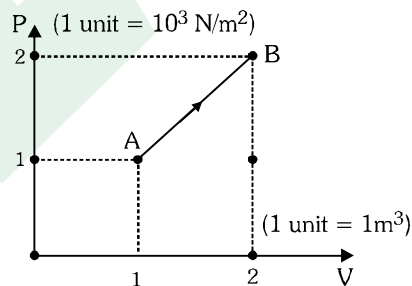
 $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$. Solar mass $M = 2 \times 10^{30} \text{ kg}$, $C = 3 \times 10^8 \text{ m/s}$,

 $1 \text{ light year} = 9.5 \times 10^{15} \text{ m}$


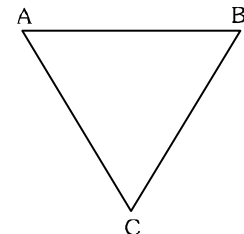
20. Mass (in kg) of the massive object at the centre of the milky way galaxy is of the order :
 (A) 10^{32} (B) 10^{37} (C) 10^{43} (D) 10^{29}
21. Theories suggest that it is not possible for a single star to have a mass of more than 50 solar masses. The massive object at the centre of milky way galaxy is most likely to be a
 (A) white dwarf (B) neutron star (C) black hole (D) single ordinary star
22. If mass of earth $M_e \approx 6 \times 10^{24} \text{ kg}$ and its radius $R_e = 6400 \text{ km}$ to what fraction of its presents radius does the earth need to be compressed in order to become a black hole ? (Give only the order of your answer)
 (A) 10^{-4} (B) 10^{-9} (C) 10^{-7} (D) 10^{-14}

Integer type

23. A certain amount of a mono-atomic ideal gas undergoes a process $pu^\eta = C$, where p is the density of the gas and u is the internal energy of the gas. It was found that the ratio $r = \frac{\Delta W}{\Delta Q}$ for the process was $r = 2/3$. What is the value of η ?
24. For P-V diagram shown for a diatomic ideal gas, heat given to the gas from A to B is (in kJ).



25. Three rods of equal length are joined to form an equilateral $\triangle ABC$. The temperature coefficient of linear expansion is α_1 for AB, α_2 for AC and BC. For $\alpha_1 = K\alpha_2$, $\angle ABC$ remains constant for small changes in temperature. Find K.



ANSWER KEY

1. (ABD) 2. (C) 3. (AC) 4. (ABCD) 5. (B) 6. (A) 7. (A) 8. (A)
 9. (B) 10. (D) 11. (A) 12. (C) 13. (C) 14. (ACD) 15. (A) 16. (BC)
 17. (C) 18. (A) 19. (B) 20. (B) 21. (C) 22. (B) 23. (3) 24. (9)
 25. (1)

ADVANCE REVISION ASSIGNMENT-09

PHYSICS

SINGLE CORRECT TYPE (+3, -1)

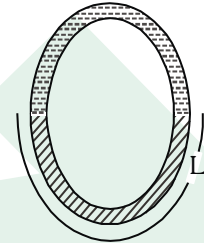
1. A tube of given shape has cross-sectional area S and total length $2L$. Its bottom and upper halves are filled with two non-viscous, non-compressible liquids of densities 3ρ , ρ respectively. If the liquid interface is slightly displaced then find its angular frequency.

(A) It is not SHM

(B) $\sqrt{\frac{3g}{2L}}$

(C) $\sqrt{\frac{2g}{L}}$

(D) $\sqrt{\frac{g}{L}}$



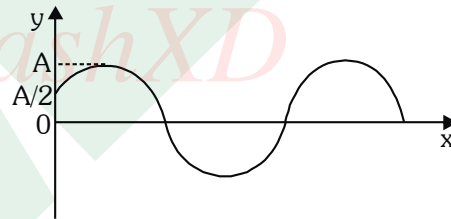
2. Here snap shot of a progressive wave at $t = \frac{T}{6}$ is given with time period = T . Then the equation of the wave will be if wave is going in positive x -direction and if wave is going in negative x -direction respectively. (Here, $T = \frac{2\pi}{\Omega}$) (symbol have their usual meaning)

(A) $y_1 = A \sin\left(kx - \omega t + \frac{\pi}{2}\right)$, $y_2 = A \sin\left(kx + \omega t - \frac{\pi}{6}\right)$

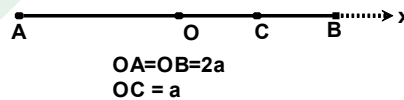
(B) $y_1 = A \cos(kx - \omega t)$, $y_2 = A \sin(kx + \omega t)$

(C) $y_1 = A \sin(kx - \omega t)$, $y_2 = A \sin\left(kx + \omega t - \frac{\pi}{6}\right)$

(D) $y_1 = A \sin\left(kx - \omega t + \frac{\pi}{2}\right)$, $y_2 = A \sin\left(kx + \omega t + \frac{\pi}{6}\right)$



3. A particle of mass m is performing simple harmonic motion along line AB with amplitude $2a$ with centre of oscillation as O . At $t = 0$ particle is at point C ($OC = a$) and is moving towards B with velocity $v = a\sqrt{3}$ m/s. The equation of motion can be given by



(A) $x = 2a (\sin t + \sqrt{3} \cos t)$

(B) $x = 2a (\sqrt{3} \sin t + \cos t)$

(C) $x = a (\sin t + \sqrt{3} \cos t)$

(D) $x = a (\sqrt{3} \sin t + \cos t)$

4. A wire is stretched between two fixed points separated by a distance of $2m$ such that tension in it is equal to 30.25 kgwt. The wire vibrates in its first overtone. A closed organ pipe of length $\frac{5}{3} m$ is brought near the wire. The temperature of the gas in the pipe is $27^\circ C$. When the organ pipe is made to vibrate in second overtone, five beats are heard every second. If the tension in the string is reduced slightly then the number of beats heard per second is reduced to three. Find the linear (mass) density of the wire in gram/metre. Given c_p/c_v of gas in the organ pipe = 1.44 and its mean molar mass is 27.7 gm.

(A) 10^{-2} kg/m

(B) 10^{-3} kg/m

(C) 10^{-5} kg/m

(D) 10^{-7} kg/m

5. Density of a liquid varies with depth as $\rho = \alpha h$. A small ball of density ρ_0 is released from the surface of the liquid

- (A) the ball will execute SHM of amplitude $\frac{\rho_0}{\alpha}$ (B) the mean position will be at a depth $\frac{\rho_0}{\alpha}$
 (C) the ball will sink to a maximum depth of $\frac{2\rho_0}{\alpha}$ (D) all the above

Paragraph for Q. Nos. (6 to 8)

A tube of length ℓ and radius R carries a steady flow of a fluid whose density is ρ and viscosity η . The velocity v of flow is given by $v = v_0[1 - (r^2/R^2)]$, where r is the distance of flowing fluid from the axis. Then

6. Volume of fluid flowing across the section of tube in unit time is

- (A) $\frac{1}{4}\pi v_0 R^2$ (B) $\frac{1}{2}\pi v_0 R^2$ (C) $\pi v_0 R^2$ (D) $\frac{1}{8}\pi v_0 R^2$

7. Kinetic energy of the fluid within the volume of the tube is

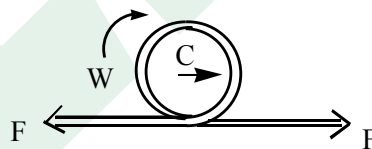
- (A) $\frac{1}{6}\pi \ell R^2 \rho v_0^2$ (B) $\frac{1}{8}\pi \ell R^2 \rho v_0^2$ (C) $\frac{1}{4}\pi \ell R^2 \rho v_0^2$ (D) $\frac{1}{3}\pi \ell R^2 \rho v_0^2$

8. The difference of pressure at the end of the tube is

- (A) $\frac{4\eta \ell v_0}{R^2}$ (B) $\frac{8\eta \ell v_0}{R^2}$ (C) $\frac{2\eta \ell v_0}{R^2}$ (D) $\frac{\eta \ell v_0}{R^2}$

Paragraph for Q. Nos. (9 to 11)

Along, heavy, flexible rope with mass ρ per unit length is stretched by a constant force F . A sudden movement caused a circular loop to form at one end of the loop. In a manner similar to that in which transverse waves propagate, the loop runs (rolls) along the rope with speed C as shown in the figure.



9. The linear speed with which the loop moves along the rope is

- (A) $< \sqrt{\frac{F}{\rho}}$ (B) $\sqrt{\frac{F}{\rho}}$
 (C) a function of radius R of the loop (D) none of these

10. If R is the radius of the loop, the energy carried by the loop is

- (A) $4\pi FR$ (B) $3\pi FR$ (C) $2\pi FR$ (D) πFR

11. In the frame of reference which moves along with the loop with speed C , the acceleration of any small part of the loop will be

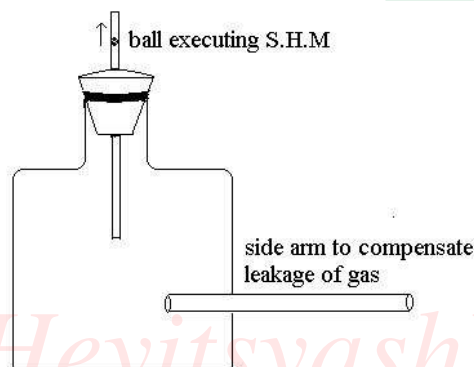
- (A) $\frac{F}{2\pi R\rho}$ (B) $\frac{C^2}{R} + \frac{F}{2\pi RC}$
 (C) $\frac{C^2}{R} - \frac{F}{2\pi RC}$ (D) $\frac{C^2}{R}$

Paragraph for Q. Nos. (12 to 14)

A large bottle is fitted at its neck with one – hole stopper through which passes a precision bore glass tube. A

steel ball bearing of mass m , radius r just fits inside the tube. The bottle is filled with the gas for which $\gamma \left(= \frac{C_p}{C_v} \right)$

is to be calculated. Inside the tube gas leaks through the narrow space between the ball and the tube, gas forms a gas cushion that keeps the ball from touching the wall. The ball can slide with little friction. A very slow flow of gas into the bottle through the side arm compensates the leakage of gas. When the ball is lowered into the tube, it would compress the gas to an equilibrium pressure p_0 , volume V_0 . A small displacement from the equilibrium position will cause oscillations of the ball with frequency ν .



12. Assuming the process to be adiabatic find γ .

(A) $\frac{\nu^2 m V_0}{p_0 r^4}$ (B) $\frac{2\nu^2 m V_0}{p_0 r^4}$ (C) $\frac{3\nu^2 m V_0}{p_0 r^4}$ (D) $\frac{4\nu^2 m V_0}{p_0 r^4}$

13. The frequency of oscillation if the process is isothermal

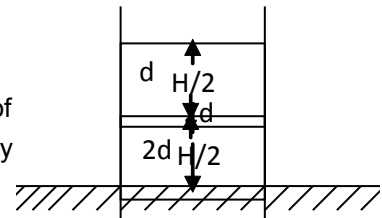
(A) $\frac{1}{2\pi} \sqrt{\frac{p_0 (\pi r^2)^2}{m V_0}}$ (B) $\frac{2}{2\pi} \sqrt{\frac{p_0 (\pi r^2)^2}{m V_0}}$ (C) $\frac{1}{2\pi} \sqrt{\frac{2p_0 (\pi r^2)^2}{m V_0}}$ (D) $\frac{1}{2\pi} \sqrt{\frac{p_0 (\pi r^2)^2}{2m V_0}}$

14. The amplitude of oscillation if the process is adiabatic (maximum acceleration of particle is g)

(A) $\frac{mgV_0}{\gamma p_0 (\pi r^2)^2}$ (B) $\frac{2mgV_0}{\gamma p_0 (\pi r^2)^2}$ (C) $\frac{3mgV_0}{\gamma p_0 (\pi r^2)^2}$ (D) $\frac{4mgV_0}{\gamma p_0 (\pi r^2)^2}$

Paragraph for Q.Nos. (15 to 17)

A container of large uniform cross-section area A resting on a horizontal surface, holds two immiscible, non-viscous and incompressible liquids of densities d and $2d$ each of height $H/2$ as shown in figure. The lower density liquid is open to the atmosphere having pressure P_0 .

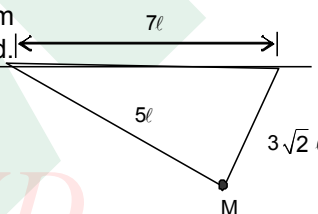


15. A homogeneous solid cylinder of length l ($l < H/2$) cross sectional area $A/5$ is immersed such that it floats with its axis vertical at the interface of two liquids with length $\frac{l}{4}$ in the denser liquid. Find the density of solid cylinder.

(A) $\frac{3d}{4}$ (B) $\frac{3}{2}d$ (C) $\frac{5d}{4}$ (D) $\frac{7d}{4}$

16. The cylinder is removed and original arrangement is restored. A tiny hole of area ($S \ll A$) is punched on the vertical side of the container at a height h ($h < \frac{H}{2}$) from the bottom of the container, find the initial speed of efflux of the liquid at the hole.
- (A) $\sqrt{\frac{g}{2}(3H - 4h)}$ (B) $\sqrt{g(3H - 4h)}$ (C) $\sqrt{g(4H - 3h)}$ (D) $\sqrt{\frac{g}{2}(H - h)}$
17. In Q.No: 42 find the height from the bottom of the container, at which the hole should be punched so that the liquid travels the maximum h-horizontal distance
- (A) $\frac{3}{4}H$ (B) $\frac{3}{8}H$ (C) $\frac{H}{4}$ (D) $\frac{H}{3}$

MULTIPLE CORRECT TYPE (+4, -1)

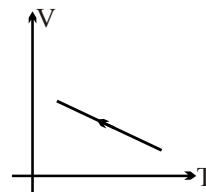
18. A particle is suspended by two ideal strings as shown in the figure. Now mass m is given a small displacement perpendicular to the plane of triangle formed. Choose the correct statement(s).
- 
- (A) The period of oscillation of the system is $2\pi\sqrt{\frac{3\sqrt{3}l}{g}}$.
- (B) The period of oscillation of the system is $2\pi\sqrt{\frac{3l}{g}}$.
- (C) The period of oscillation of the system is independent of M .
- (D) If the distance between the suspension points was kept constant and the lengths of the strings were quadrupled then the period of the system will double.
19. The first overtone of an open organ pipe (of length L_o) beats with the first overtone of a closed organ Pipe (of length L_c) with a beat frequency of 10 Hz. The fundamental frequency of the closed pipe is 110 Hz. If the speed of sound is 330 m/s. Then
- (A) $L_c = 0.75$ m (B) $L_o = \left(\frac{33}{34}\right)m$ or $\left(\frac{33}{35}\right)m$
- (C) $L_o = \left(\frac{33}{32}\right)m$ or 1m (D) $L_o = \left(\frac{33}{32}\right)m$ or $\left(\frac{33}{34}\right)m$
20. There are two narrow pipes A and B of equal length l . Pipe A is filled with Helium gas and pipe B is filled with a hypothetical diatomic gas whose mass is $7/6$ times the molar mass of hydrogen. Both gases are at the same temperature.
- (A) Pipe-A can resonate with pipe-B if both pipe-A and pipe-B are open pipes.
- (B) Pipe-A can resonate with pipe-B if pipe-A is closed and pipe-B is open pipe.
- (C) Pipe-A can resonate with pipe-B if pipe-A is open and pipe-B is closed.
- (D) Pipe-A can resonate with pipe-B if both pipes are closed-pipes.

ANSWER KEY

- | | | | | | | | |
|---------|----------|----------|----------|---------|---------|---------|---------|
| 1. (D) | 2. (A) | 3. (D) | 4. (B) | 5. (D) | 6. (B) | 7. (A) | 8. (A) |
| 9. (B) | 10. (C) | 11. (D) | 12. (D) | 13. (A) | 14. (A) | 15. (C) | 16. (A) |
| 17. (B) | 18. (BC) | 19. (AD) | 20. (AC) | | | | |

ADVANCE REVISION ASSIGNMENT-10
PHYSICS
SINGLE CORRECT TYPE (+3, -1)

- A spaceship is sent to investigate a planet of mass M and radius R . While hanging motionless in space at a distance $5R$ from the centre of the planet, the ship fires an instrument package of mass m ($m \ll \text{mass of spaceship}$) with speed v_0 at an angle θ w.r.t. radial line between the centre of planet and spaceship. For what value of θ , the package will just graze the surface of the planet. Given $v_0 = \sqrt{\frac{GM}{5R}}$
 (A) 30° (B) 60° (C) 37° (D) None of these
- Consider two identical particles each of mass m held at separation r_0 in free space. One of them is given a velocity v_0 perpendicular to r_0 and the other one is simultaneously released. For what range of velocity v_0 the masses would be bound in orbital motion under their mutual gravitational forces.
 (A) $v_0 \leq \sqrt{\frac{2Gm}{r_0}}$ (B) $v_0 < \sqrt{\frac{2Gm}{r_0}}$ (C) $v_0 \leq 2\sqrt{\frac{Gm}{r_0}}$ (D) $v_0 < 2\sqrt{\frac{Gm}{r_0}}$
- A satellite revolves in an equatorial plane in west to east direction around the earth with a period of 8 hrs. At an instant it is observed vertically overhead a longitude. How much minimum time after this instant, the satellite will again be overhead the same longitude?
 (A) 8h (B) 12h (C) 24h (D) 36h
- Suppose the Saturn rings consist of football size particles orbiting in circular orbits around the planet Saturn. If speed of fastest particles in the ring is 1.05 times of the speed of the slowest particles. Minimum speed, find the ratio of radial width to the inner radius of the ring.
 (A) 0.01 (B) 0.05 (C) 0.10 (D) 0.25
- An ice box is built of wood 1.75 cm thick is lined inside by a cork of 2 cm thickness. If the temperature of inner surface of cork is 0°C and that of outer surface of wood is 12°C , what is the temperature of interface? (Given K of wood = 0.0006 units and that of cork is 0.00012 units)
 (A) 10.2°C (B) 8°C (C) 6.4°C (D) 9°C
- Let E be the energy required to raise a satellite to height h above earth's surface and E' be the energy required to put the same satellite into orbit at that height. Then $\frac{E}{E'}$ is equal to (R is earth's radius)
 (A) $\frac{2h}{R+2h}$ (B) $\frac{2h}{2R+3h}$ (C) $\frac{R}{R+h}$ (D) $\frac{2R}{2h+R}$
- The V - T graph for an ideal gas is shown in figure
 (A) work done by the gas is positive
 (B) internal energy of the gas is increasing
 (C) heat is supplied to the gas
 (D) heat is rejected from the gas


Paragraph for Q.no. (8 to 10)

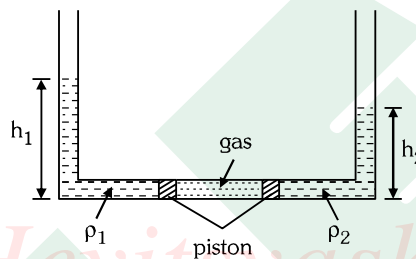
A **diathermic** piston free to move without friction divides an adiabatic cylinder in two equal parts. The two parts contain equal amount of ideal gas. Piston is joined to a string which comes out passing through right boundary. [Assume that no matter and energy flows through hole that passes the string]. Now the string is pulled gently toward right to a certain distance and held.



8. Temperature of gas A :
 (A) increases (B) Decrease
 (C) first decreases then increase (D) first increase then decrease
9. Temperature of gas B :
 (A) Increases (B) Decrease
 (C) first decrease then increase (D) first increase then decrease
10. Work done by gas A on piston is :
 (A) negative (B) positive (C) zero (D) none of the above

Paragraph for Q.no. (11 to 13)

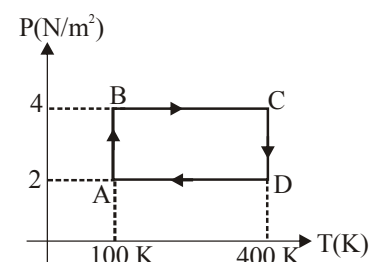
Figure shows a tube whose two limbs are vertical and one is horizontal of sufficient length. An ideal gas is enclosed between two massless non-conducting movable piston of cross-sectional area 10 cm^2 . A liquid of density $\rho_1 = 1000 \text{ kg/m}^3$ and $\rho_2 = 2000 \text{ kg/m}^3$ are poured in the vertical limbs of height of $h_1 = 2\text{m}$ and h_2 . Now gas is slowly heated. Consider the wall of tube is insulated.



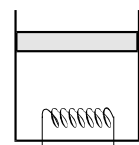
11. Ratio of h_1 and h_2 at any instant will be :
 (A) $\frac{1}{2}$ (B) $\frac{2}{1}$ (C) $\frac{3}{2}$ (D) not constant
12. If due to heating separation between piston is increased by 3m then displacement of piston on left will be :
 (A) 2m (B) 1m (C) 2.5m (D) 1.5m
13. Find the work done by gas for above situation.
 (A) 70 Joule (B) 390 Joule (C) 90 Joule (D) 350 Joule

MULTIPLE CORRECT TYPE (+4, -1)

14. Use the following P-T graph to choose the correct option(s)
 (A) The heat is supplied to the gas during the cycle ABCDA
 (B) During the process AB work is done on the gas.
 (C) The area under the graph gives us the work done by the gas during the cycle.
 (D) The volume of the sample at A is same as the volume of the sample at C.

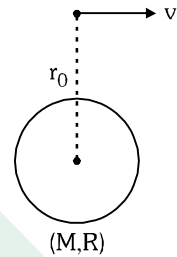


15. An ideal gas has molar heat capacity at constant pressure $C_p = 5R/2$. The gas is kept in a cylindrical vessel fitted with a piston which is free to move. Side walls of the container and the piston are non conducting. Mass of the frictionless piston is 9 kg . Initial volume of the gas is 0.0027 m^3 and cross-section area of the piston is 0.09 m^2 . The initial temperature of the gas is 300 K . Atmospheric pressure $P_0 = 1.05 \times 10^5 \text{ N/m}^2$. An amount of $2.5 \times 10^4 \text{ J}$ of heat energy is supplied to the gas by a heating coil, then
 (A) Initial pressure of the gas is $1.06 \times 10^5 \text{ N/m}^2$
 (B) Final temperature of the gas is 1000 K
 (C) Final pressure of the gas is $1.06 \times 10^5 \text{ N/m}^2$
 (D) Work done by gas is $9.94 \times 10^3 \text{ J}$

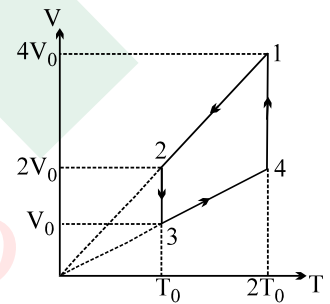


INTEGER TYPE (+3, 0)

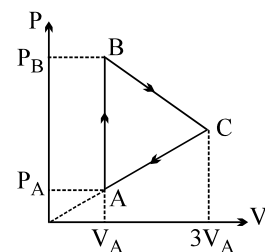
16. A small particle is given a velocity $v = \sqrt{\frac{5GM}{4r_0}}$ at a distance r_0 from center of earth. Here M is mass of earth find the width of band of variation of distance between particle and center of earth. (Take $r_0 = 30000$ km) if your answer is A , fill $\frac{A}{10000}$ in OMR.



17. In a massive metal cylinder of height $h = 1$ m, closed on top by a moving piston is an ideal gas. On top of the piston we gently put weight. The piston immediately dropped by $x = 2.5$ cm, over a long time it was seen that the piston comes down another $x' = 1$ cm. Determine the degree of freedom for the molecules of the gas. Room temperature is constant, gas does not leak out. Use binomial approximation.



19. One mole of an ideal diatomic gas is taken in a cyclic process from initial state P_0, V_0 and T_0 as following :
 (i) Heating at constant volume to a temperature two times.
 (ii) Heating at constant pressure to double volume.
 (iii) Cooling by a process in which pressure is directly proportional to the volume to return to its initial state.
 Calculate the percentage efficiency of the cycle. (Round off the answer to the nearest integer.)
20. A U tube of uniform crosssection is in vertical position as shown. The atmospheric pressure is 76 cm of mercury column. Now end A of tube is closed and gas in the tube is heated so that gas expands and some mercury spills out. During the process, it is seen that the pressure of enclosed gas is directly proportional to the volume of gas. Find ℓ_0 (in cm). Fill $\ell_0 - 68$ in OMR.



* * * * *

ANSWER KEY

- | | | | | | | | | |
|---------|---------|---------|---------|----------|----------|---------|---------|---------|
| 1. (C) | 2. (D) | 3. (B) | 4. (C) | 5. (A) | 6. (A) | 7. (A) | 8. (C) | 9. (D) |
| 10. (B) | 11. (B) | 12. (A) | 13. (B) | 14. (AB) | 15. (AC) | 16. (2) | 17. (5) | 18. (2) |
| 19. (5) | 20. (8) | 21. (3) | | | | | | |

@HeyitsyashXD

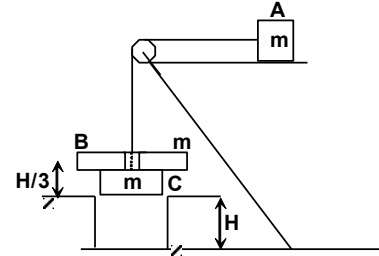
ADVANCE REVISION ASSIGNMENT-11

PHYSICS

SINGLE CORRECT TYPE (+3, -1)

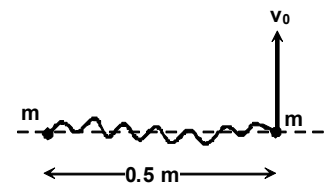
1. The system in figure is released from rest from the position shown. After blocks have moved distance $H/3$, collar B is removed and block A & C continue to move. The speed of C just before it strikes the ground is

- (A) $\frac{4}{3}\sqrt{gH}$ (B) $2\sqrt{\frac{gH}{3}}$
(C) $\frac{\sqrt{13gH}}{3}$ (D) $2\sqrt{2gH}$



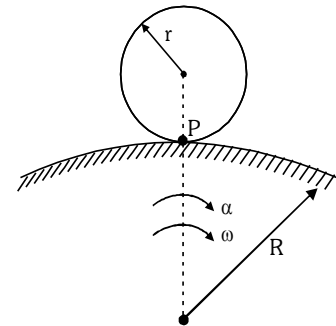
2. Two particles each of mass m are connected by a string of length $1m$. The particles are kept on a smooth horizontal plane. The initial separation between the particles is $0.5 m$. One of the particles is given a velocity v_0 as shown in the figure. The magnitude of angular velocity of one particle with respect to the other particle just after the string becomes taut is

- (A) v_0 (B) $\sqrt{3}v_0$ (C) $\frac{v_0}{2}$ (D) $\frac{\sqrt{3}v_0}{2}$



3. A solid cylinder of mass m and radius r rolls on a fixed cylindrical surface of radius R . at the instant shown line joining both the centre is rotating with angular velocity ω and angular acceleration α . Find the acceleration point P which is the point of the rolling cylinder in contact with the fixed cylinder.

- (A) $\omega^2 \frac{(R+r)R}{r}$
(B) $\omega^2 \frac{(R-r)R}{r}$
(C) $\omega^2 \frac{R^2}{r}$
(D) $\omega^2 \frac{r^2}{R}$



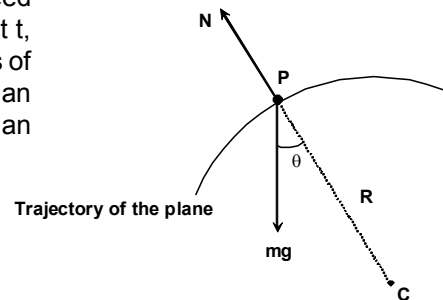
4. A plane of mass M is looping in vertical curve path with constant speed of v . A man of mass " m " is sitting in the plane. At a certain instant t , when plane passes through a point P of the curved path, the radius of curvature of curved path is R and normal force " N " acting on the man exerted by seat of plane is making angle θ with vertical. [mass of man

is 60 kg and $v = \frac{375}{2\sqrt{5}} \text{ m/s}$,

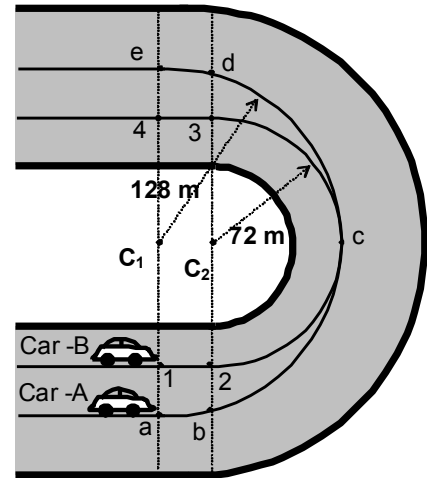
$R = 1250(2 + \cos \theta) \text{ metre}$].

Now choose the CORRECT option.

- (A) The normal force exerted by seat on the man is zero if $\theta = 30^\circ$.
(B) When normal force exerted by seat on the man is zero, then tangential force exerted by seat is $150N$.
(C) The magnitude of force exerted by the seat on the man, when plane is at the bottom of loop is 937.5 N
(D) The magnitude of force exerted by the seat on the man, when plane is at the bottom of loop is 612.5 N

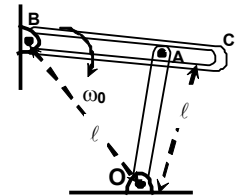


5. Race car A follows path abcde while race car B follows path 12c34 track. If each car has a constant speed corresponding to a normal acceleration of 8 m/s^2 . The tracks abcde and 2c3 are semicircular track while tracks 1-2 and 3-4 are straight track. Point a and point 1 are the starting points of race and point 4 and point e are finishing points of the race. Choose the correct statements.



- (A) car A wins the race with time difference $\frac{14 + 3\pi}{3} \text{ s}$
 (B) car A wins the race with time difference $\frac{14 - 3\pi}{3} \text{ s}$
 (C) car B wins the race with time difference $\frac{14 + 3\pi}{3} \text{ s}$
 (D) car B wins the race with time difference $\frac{14 - 3\pi}{3} \text{ s}$

6. The figure shows two rods BC and OA hinged from B and O respectively. Rod BC has a groove in which point A of rod OA can slide smoothly. At a given instant, the rod BC has an angular velocity ω_0 . The speed of end A of the rod OA at this instant will be

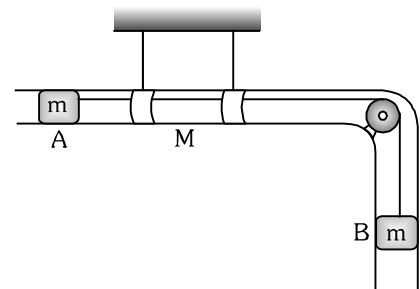


- (A) $\omega_0 \ell$ (B) $2\omega_0 \ell$
 (C) $3\omega_0 \ell$ (D) None of these.

7. A particle A is moving in xy plane with the constant speed of $2\pi \text{ m/s}$ along the path $x^2 + y^2 + 120y = 0$. At time $t = 0$. When A is at the origin, another particle B starts moving from origin with constant acceleration is such a way that at time $t = 5\text{ s}$ velocities of both the particles are found to be equal. If sense of motion of A is clockwise, calculate the average speed of B over first five second.

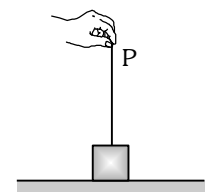
- (A) $2\pi \text{ m/s}$ (B) $\pi \text{ m/s}$ (C) 2 m/s (D) 1 m/s

8. In the figure shown the hollow tube of mass M is free to move without friction in the horizontal direction supported by two fixed vertical lugs attached to the roof. The system is released from rest. Find the velocity of the tube, when the block B has fallen, through a height h. Neglect any friction and mass of the string.



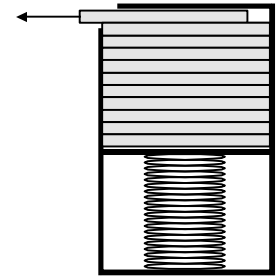
- (A) $m\sqrt{\frac{2gh}{(2M + 3m)(M + m)}}$
 (B) $M\sqrt{\frac{gh}{(2M + 3m)(M + m)}}$
 (C) $M\sqrt{\frac{2gh}{(2M + 3m)(M + m)}}$
 (D) None of these

9. One end of an elastic string is attached to a block of mass m and the other end is held keeping the string relaxed in vertical. Relaxed length of the string is ℓ_0 and there is a mark P at height $0.8\ell_0$ from the lower end. Now the upper end is slowly raised. When the block leaves the ground, the mark P reaches the point where upper end of the relaxed string was. Expression for the work done by the force applied by the hand is



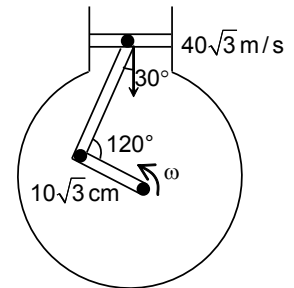
- (A) $\frac{mg\ell_0}{4}$ (B) $\frac{mg\ell_0}{8}$ (C) $\frac{mg\ell_0}{16}$ (D) None of these

10. A special box used to store several plate-like articles contains a light spring loaded piston inside it. When the box is empty the piston touches the top inner surface of the box. From the opening at the top left corner plates can be inserted or dragged out. The box contains $n = 20$ identical square plates each edge length $\ell = 5$ cm. Coefficient of friction between all the surfaces in contact is $\mu = 0.5$. The box is emptied by dragging out all the plates one by one in two different trials, in the first trial the opening was at the top as shown and in the second trial the box was inverted. If difference of work done by the agent pulling the plates in both these trials is $\Delta W = 10$ J, calculate the mass of one plate.



- (A) 3×10^{-2} kg (B) 5×10^{-2} kg (C) 6×10^{-4} kg (D) 7×10^{-5} kg

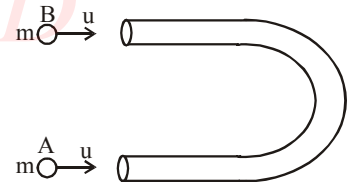
11. A snapshot of a petrol engine is given in which piston is moving downwards with velocity $40\sqrt{3}$ m/s. Find the angular velocity of the shaft.



- (A) 400 rad/s
(B) 300 rad/s
(C) 200 rad/s
(D) 500 rad/s

Paragraph for Q. Nos. 12 to 14

A U shaped tube of mass $2m$ is placed on a smooth horizontal surface. Two identical spherical balls each of mass m and of diameter slightly less than the inner diameter of tube enters into the tube with a velocity u as shown. (Assume no loss of energy anywhere and all collisions to be elastic).



12. Speed of each spherical ball, just before their collision

- (A) $\frac{u}{\sqrt{3}}$ (B) $\frac{2u}{3\sqrt{3}}$ (C) $\frac{\sqrt{3}u}{2}$ (D) $\frac{u}{2}$

13. At the time of collision, angle between direction of motion of spherical ball A and B is, as observed in the ground frame,

- (A) $2 \tan^{-1} \sqrt{2}$ (B) $2 \tan^{-1} \frac{1}{\sqrt{2}}$ (C) $\tan^{-1} \sqrt{2}$ (D) π

14. In the given situation the maximum velocity that the U-tube can gain (choose one of the given options)

- (A) $\frac{u}{2}$ when spheres are about to collide
(B) $2u$ when spheres move in backward direction with speed u .
(C) u , finally when sphere emerges from U-tube
(D) some extra informations are required to calculate it.

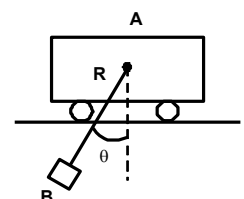
Paragraph for Q. Nos. 15 to 17

A block B of mass M is suspended from a cord of length ℓ attached to a cart A of mass M as shown in the figure. The horizontal surface on which the cart moves is smooth. Initially both the cart and the block are at rest in the position shown. Now B is released. Take $M = m = 2$ kg, $\theta = 45^\circ$ and $g = 10$ m/s².

Answer the following questions.

15. The acceleration of the cart immediately after the system is released from rest is

- (A) $\frac{1}{\sqrt{2}}$ m/s² (B) $\frac{10}{3}$ m/s²
(C) $\frac{10}{3\sqrt{2}}$ m/s² (D) 5 m/s²

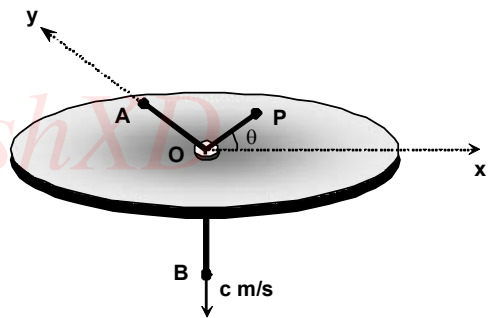


16. The tension in the cord immediately after the system is released from rest is
- (A) 10 N (B) $\frac{10}{3}$ N (C) $\frac{5\sqrt{3}}{2}$ N (D) $\frac{20\sqrt{2}}{3}$ N
17. The magnitude of acceleration of B (with respect to cart) immediately after the system is released from rest is
- (A) $\frac{10}{3}[2 + \sqrt{2}]$ (B) $\frac{10}{3}[\sqrt{2} + 1]$ (C) $\frac{10}{3}[\sqrt{3} + \sqrt{2}]$ (D) $\frac{40}{3\sqrt{2}}$

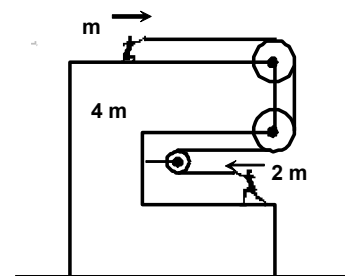
MULTIPLE CORRECT TYPE (+4, -1)

18. A small object is placed on a smooth table at a distance r_0 from a hole in the centre of table. An ideal inextensible chord is attached to the object through this hole. The object is set into motion with initial velocity of magnitude v_0 at right angle to the chord and at the same time, the chord is pulled through the hole at uniform speed c m/s. Initially object is at point A(0, r_0) and at any time t , it is at point P ($r \cos\theta$, $r \sin\theta$). Neglect the dimension of object. Choose CORRECT option(s)

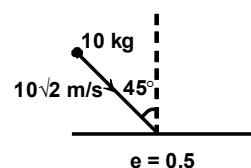
- (A) Tension in the chord at any time t is $\frac{mr_0^2 v_0^2}{(r_0 - ct)^3}$
- (B) Tension in the chord at any time t is $\frac{2mr_0^2 v_0^2}{(r_0 - ct)^3}$
- (C) Angular speed of the object at any time t is $\frac{2r_0 v_0}{(r_0 - ct)^2}$
- (D) Angular speed of the object at any time t is $\frac{r_0 v_0}{(r_0 - ct)^2}$



19. A wedge of mass $4m$ lies on a smooth horizontal surface as shown in the figure. The wedge has two men of masses m and $2m$. The system is initially at rest. The man of mass $2m$ starts pulling the rope so that his acceleration with respect to wedge is $2a$ towards left. The acceleration of m is a towards right (with respect to wedge). The string & pulleys are massless and the coefficient of friction between each pulley & string is μ and not sufficient to stop the slipping.



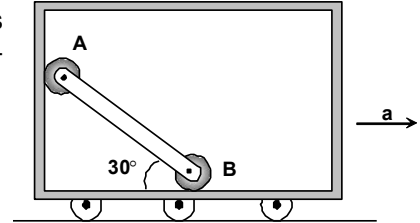
- (A) The acceleration of the wedge is $3a/7$ towards right
- (B) The magnitude of acceleration of m with respect to ground is $10a/7$
- (C) The tension in the string is throughout $10ma/7$
- (D) The force exerted by both the men on the rope is same.
20. A ball of mass 10 kg hits a smooth horizontal surface with a speed of $10\sqrt{2}\text{ m/s}$ at an angle of 45° as shown in the figure. The coefficient of restitution between the ball and the surface is 0.5 and the ball remains in contact with the surface for 0.1 sec . $\vec{F}$ is the instantaneous force exerted by the surface on ball at any time t during the collision.



- (A) The speed with which the ball rebounds is $\sqrt{125}\text{ m/s}$
- (B) During collision $|\vec{F}|$ may be less than or equal to 1800 N
- (C) During collision $|\vec{F}|$ may be greater than or equal to 100 N
- (D) The average force exerted by ground during collision is 1600 N

INTEGER TYPE (+3, -1)

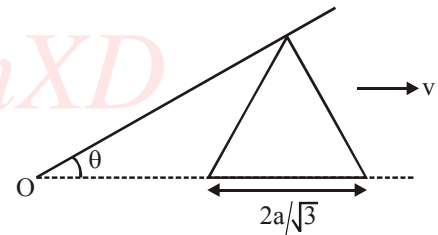
21. If a car is moving rightward with acceleration $a = g\sqrt{k}$ m/s rightward as shown in the figure. Find the value of k so that, rod maintains its orientation as shown in the figure. Neglect the friction and mass of the small rollers at A and B.



22. A sphere of mass m collides elastically with another stationary sphere of mass $m/2$ obliquely. Both the spheres are smooth and there are no external forces acting on them. Maximum angle through which sphere of mass m can be deflected, w. r. t. Its initial direction of motion is $\frac{\pi}{x}$. Value of x is

23. In the arrangement shown the rod is freely pivoted at point O and is in contact with the equilateral triangular block which can move on the horizontal frictionless ground. As the block is given a speed v forward, the rod rotates about point O. Find the angular velocity of rod in rad/s at the instant when $\theta = 30^\circ$.

[Take $v = 20$ m/s, $a = 1$ m]



* * * * *

ANSWER KEY

- | | | | | | | | | |
|----------|------------|---------|---------|---------|---------|---------|---------|----------|
| 1. (C) | 2. (C) | 3. (A) | 4. (C) | 5. (B) | 6. (B) | 7. (B) | 8. (A) | 9. (B) |
| 10. (B) | 11. (A) | 12. (C) | 13. (A) | 14. (C) | 15. (B) | 16. (D) | 17. (D) | 18. (AD) |
| 19. (AB) | 20. (ABCD) | 21. (3) | 22. (6) | 23. (5) | | | | |

ADVANCE REVISION ASSIGNMENT-12

PHYSICS

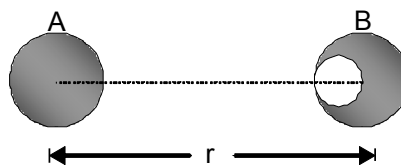
SINGLE CORRECT TYPE (+3, -1)

1. A small sphere of mass $m = 0.5 \text{ kg}$ carrying a positive charge $q = 110 \mu\text{C}$ is connected with a light, flexible and inextensible string of length $r = 60 \text{ cm}$ and whirled in a vertical circle. If a vertically upwards electric field of strength $E = 10^5 \text{ NC}^{-1}$ exists in the space, The minimum velocity of sphere required at highest point so that it may just complete the circle. ($g = 10 \text{ ms}^{-2}$)
 (A) 8 m/s (B) 7m/s (C) 6m/s (D) 9 m/s

2. Two long wires are placed on a smooth horizontal table. Wires have equal but opposite charges. Magnitude of linear charge density on each wire is λ . The work required (for unit length of wires) to increase the separation between the wires from a to $2a$ is $\frac{\lambda^2}{K\pi\epsilon_0} \log_e 2$. The value of K is
 (A) 1 (B) 2 (C) 3 (D) 4

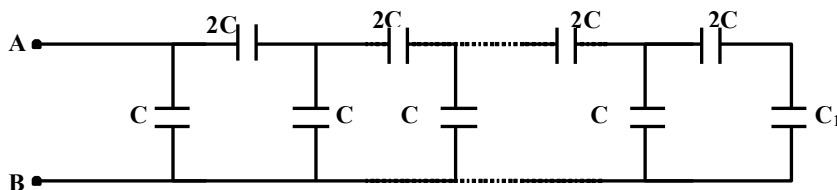
3. Two long wires have uniform charge density λ per unit length each. The wires are non-coplanar and mutually perpendicular. Shortest distance between them is d . The interaction force between them is found to be K times $\frac{\lambda^2}{2\epsilon_0}$. The value of K is
 (A) 5 (B) 2 (C) 3 (D) 1

4. Distance between centers of two spheres A and B, each of radius R as shown in figure. Sphere B has a spherical cavity of radius $R/2$ such that distance of centre of cavity is $(r - R/2)$ from the centre of sphere A and $R/2$ from the centre of sphere B. Di-electric constant of material of each sphere is $K = 1$ and material of each sphere has a uniform charge density ρ per unit volume. The interaction potential energy of the two spheres is found to be $\frac{\pi\rho^2R^6(7r-4R)}{K\epsilon_0r(2r-R)}$. The value of K is



- (A) 11 (B) 7 (C) 5 (D) 9

5. The figure shows a network of capacitance consisting of several repetitive units of capacitance C_1 connected at end. The value of C_1 such that the equivalent capacitance between A and B is independent of number of units.



- (A) $C_1 = 4C$ (B) $C_1 = 3C$ (C) $C_1 = 2C$ (D) $C_1 = C$

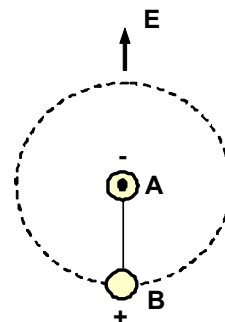
ONE OR MORE THAN ONE CORRECT TYPE (+4, -1)

6. Two small particles A and B having masses $m = 0.5 \text{ kg}$ each and charge

$$q_1 = \left(-\frac{155}{18} \mu\text{C} \right) \text{ and } q_2 = (+100 \mu\text{C}) \text{ respectively, are connected at the ends of a}$$

non-conducting, flexible and inextensible string of length $r = 0.5 \text{ m}$. Particle A is fixed and B is whirled along a vertical circle with centre at A. If a vertically upward electric field of strength $E = 1.1 \times 10^5 \text{ NC}^{-1}$ exists in the space. ($g = 10 \text{ ms}^{-2}$)

Choose the correct statement(s)



- (A) The minimum velocity of the particle B at the lowest position is $\sqrt{37} \text{ m/s}$ so that it may just complete the circle.
(B) The minimum velocity of the particle B at the lowest position is $\sqrt{31} \text{ m/s}$ so that it may just complete the circle.
(C) The minimum velocity of particle B, required at highest point so that it may just complete the circle is $\sqrt{61} \text{ ms}^{-1}$.
(D) The minimum velocity of particle B, required at highest point so that it may just complete the circle is $\sqrt{67} \text{ m/s}$.

7. A non-conducting cylinder of radius a is infinitely long. Its volume charge density ρ varies linearly as the distance from the axis of the cylinder. If ρ is zero at the axis and is ρ_s on the surface, the electric intensity due to it is :

- (A) $\frac{\rho_s a^2}{3\epsilon_0 r}$ at a point distant ' r ' outside it
(B) $\frac{\rho_s r^2}{3\epsilon_0 a}$ at a point distant ' r ' inside it
(C) $\frac{\rho_s a}{3\epsilon_0}$ at a point on its surface
(D) The electric field at axis is zero.

8. An iron wire of diameter 1 mm and length 10 cm is placed in an evacuated chamber. Assume that all heat transfer is by radiation and that the surface of wire radiates according to Stefan's law. Take the temperature of the chamber to be 27°C . Specific resistance of the wire is $10 \times 10^{-8} \Omega\text{-m}$. (Take the emissivity of iron wire is 1)

Choose the correct statement(s)

- (A) The equilibrium temperature of the wire is 301.7 K if it carries a current of 10 A .
(B) The equilibrium temperature of the wire is 457.3 K if it carries a current of 10 A .
(C) If the temperature of chamber is increased to 50°C then the equilibrium temperature of the wire carrying same current will also increase.
(D) If the temperature of chamber is increased to 50°C then the equilibrium temperature of the wire carrying same will decrease.

9. A point charge q is kept at the centre of the cylinder of length L and radius a .

Choose the correct statement(s)

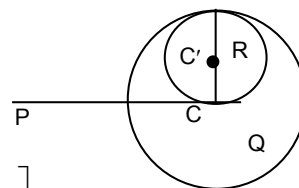
(A) The flux through the curved surface of the cylinder is $\phi_{\text{curved}} = \frac{q\ell}{2\epsilon_0(a^2 + \ell^2/4)^{1/2}}$.

(B) The flux through the curved surface of the cylinder is $\phi_{\text{curved}} = \frac{q\ell}{\epsilon_0(a^2 + \ell^2/4)^{1/2}}$.

(C) The total flux through one base of the cylinder is $\phi = \frac{q}{2\epsilon_0} \left[1 - \frac{\ell/2}{(a^2 + \ell^2/4)^{1/2}} \right]$.

- (D) If the charge q undergoes S.H.M. through a very small amplitude about the centre, then the total flux through the surface of cylinder will remain same.

10. As shown a solid spherical region having a spherical cavity whose diameter R is equal to the radius of spherical region has a total charge $+Q$.
Choose the correct statement(s)

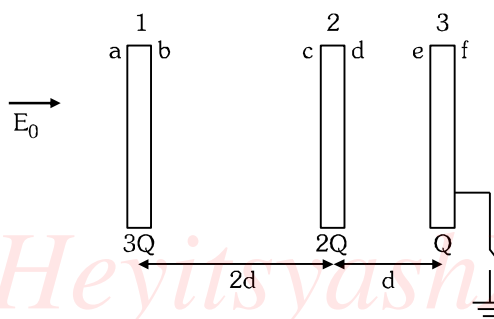


- (A) The potential at a point P at a distance 'x' from C is $\frac{Q}{7\pi\epsilon_0} \left[\frac{2}{x} - \frac{1}{2\sqrt{R^2 + 4x^2}} \right]$.
- (B) The direction of electric field at the centre C of the spherical charged body is along the line joining C and C' and towards the C'.
- (C) The electric field inside the cavity is uniform and perpendicular to the line joining the two centres C and C'.
- (D) The electric field inside the cavity is uniform and along the line joining the two centres C and C'.

Paragraph – 1 for Q.no. (11 to 12)

Consider a system of 3 parallel large conducting plates. A uniform electric field of intensity E_0 is also present as shown through out the region. The charges on plate 1, 2 and 3 are $3Q$, $2Q$ and Q respectively.

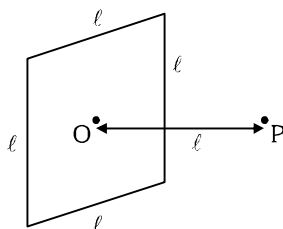
(Given $Q > A\epsilon_0 E_0$)



11. Choose the correct statement(s)
- (A) The charge on surface b is equal to $A\epsilon_0 E_0$.
- (B) The charge on surface b is not equal to zero.
- (C) The potential difference between plate 1 and plate 3 ($V_1 - V_3$) is $3E_0d + \left(\frac{2Qd}{A\epsilon_0} \right)$
- (D) The potential difference between plate 1 and plate 3 ($V_1 - V_3$) is $\left(3E_0d + \frac{Qd}{\epsilon_0} \right)$
12. Now the plate 3 is grounded by closing the switch. Choose the correct statement(s)
- (A) The potential difference between plate 1 and plate 3 is zero.
- (B) The potential difference between plate 1 and plate 3 is $6E_0d + \frac{11Qd}{\epsilon_0}$.
- (C) The potential of Plate 2 is higher than potential of plate 3.
- (D) The amount of charge flow through the switch after the plate 3 is grounded is $-6Q - 2A\epsilon_0 E$

Paragraph – 2 for Q.no. (13 to 14)

Consider a square frame made of uniformly charged non conducting wires of length ℓ and linear charge density λ C/m.



P is a point at a perpendicular distance of ℓ from centre of a square.

13. Choose the correct statement(s)
- (A) The net electric field at P is perpendicular to plane of the square and away from it .
- (B) The net electric field at P is $\left[\frac{32K\lambda}{\sqrt{30}\ell} \right] \text{N/C}$.
- (C) The net electric field at O is zero.
- (D) The net electric field at P is $\left[\frac{32K\lambda}{\sqrt{150}\ell} \right] \text{N/C}$.

14. Choose the correct statement(s) :

- (A) The potential at point P is $V = 4K\lambda \log_e \left[\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} \right] \text{J/C}$, where θ is $\sin^{-1} \left[\frac{1}{\sqrt{6}} \right]$.
- (B) The potential at point P is $V = 4K\lambda \log_e \left[\frac{\cos \theta + \tan \theta}{\cos \theta - \tan \theta} \right] \text{J/C}$, where θ is $\sin^{-1} \left[\frac{1}{\sqrt{6}} \right]$.
- (C) The potential at centre O is lower than potential at P .
- (D) The potential at centre O is higher than potential at P .

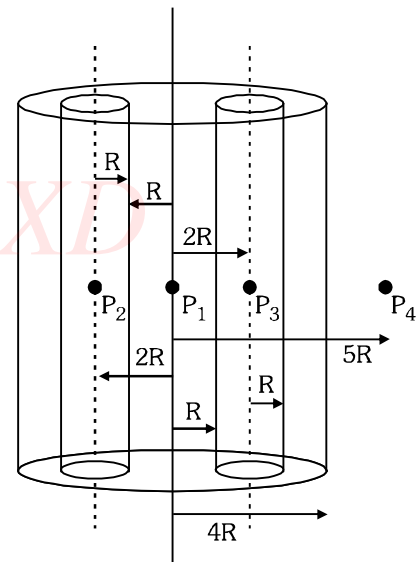
Paragraph – 3 for Q.no. (15 to 16)

Consider a very long uniformly charged non conducting cylinder of radius $4R$.

The volume charge density is $\rho \text{ C/m}^3$.

The cylinder has two symmetrically located cylindrical cavities of radius R as shown in the figure.

The dielectric constant of material of Cylinder is 1.



15. Choose the correct statement(s)
- (A) The magnitude of electric field intensity inside right cylindrical cavity is $\frac{\rho R}{\epsilon_0}$.

- (B) The magnitude of electric field intensity inside right cylindrical cavity is $\frac{2\rho R}{\epsilon_0}$.
- (C) The magnitude of electric field intensity at point P_1 lying on axis of cylinder is zero . .
- (D) The magnitude of electric field intensity at point P_1 lying on axis of cylinder is $\frac{\rho R}{4\epsilon_0}$.

16. Choose the correct statement(s)

- (A) The potential difference between points P_2 and P_3 (lying on axis of each cylindrical cavity) is zero.
- (B) The potential difference between points P_2 and P_3 (lying on axis of each cylindrical cavity) is $\left(\frac{\rho R^2}{4\epsilon_0} \right)$.
- (C) The magnitude of electric field intensity at P_2 is $\left(\frac{7\rho R}{8\epsilon_0} \right)$.
- (D) The electric field intensity at point P_4 is $\left(\frac{7\rho R}{5\epsilon_0} \right)$.

Paragraph – 4 for Q.no. (17 to 18)

Consider a uniformly charged non conducting circular disc . Surface charge density of disc = $+\sigma$, and radius of disc = a .

17. Choose the correct statement(s)

(A) The electric potential at centre is $\frac{\sigma a}{2\epsilon_0}$.

(B) The electric potential at centre is $\frac{\sigma a}{\epsilon_0}$.

(C) The electric potential at any point on the circumference of disc is $V_1 = \frac{\sigma a}{2\pi\epsilon_0}$.

(D) The electric potential at any point on the circumference of disc is $V_1 = \frac{\sigma a}{\pi\epsilon_0}$.

18. Choose the correct statement(s)

(A) The electric potential at centre is higher than electric potential at a point on circumference.

(B) The electric field intensity at any point on the axis of the disc is along the axis and away from it .

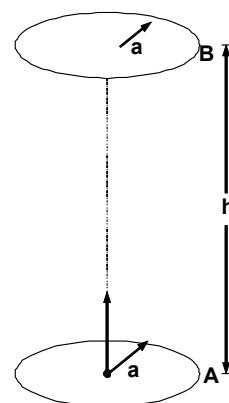
(C) A charge 'q' is brought to the centre of the uniformly charged disc from its circumference. The work done

required is $W = \frac{q\sigma a}{\epsilon_0} \left\{ \frac{1}{\pi} - \frac{1}{2} \right\}$.

(D) If the radius of the circular disc is doubled but the surface charge density remains the same then the electric potential at the centre of disc will increase.

Paragraph – 5 for Q.no. (19 to 20)

Two non conducting circular rings A and B, each of radius $a = 30$ cm, are placed coaxially with their axes vertical as shown in figure. Distance between centres of these rings is $h = 40$ cm. Lower ring A has a positive charge of $10\mu\text{C}$, while upper ring B has a negative charge of $20\mu\text{C}$. A particle of mass $m = 100$ gm and carrying a positive charge of $q = 10\mu\text{C}$ is released from rest at the centre of the ring A. ($g = 10 \text{ ms}^{-2}$)



19. Choose the correct statement(s)

(A) The initial acceleration of the particle is approximately 25 m/s^2 .

(B) The initial acceleration of the particle is approximately 47 m/s^2 .

(C) A null point exists on the common axis between the rings.

(D) There is no null point exists on the common axis between the rings.

20. Choose the correct statement(s)

(A) The velocity of particle when it reaches at the centre of upper ring B is 8 ms^{-1} .

(B) The velocity of particle when it reaches at the centre of upper ring B is 12 ms^{-1} .

(C) During the motion of the charge particle between the the rings ,the electric potential energy decreases.

(D) The motion of the charged particle is oscillatory.

ANSWER KEY

- | | | | | | | | |
|----------|-----------|-----------|-----------|-----------|----------|-----------|----------|
| 1. (C) | 2. (B) | 3. (D) | 4. (D) | 5. (C) | 6. (AC) | 7. (ABCD) | 8. (AC) |
| 9. (ACD) | 10. (ABD) | 11. (ABC) | 12. (BCD) | 13. (ACD) | 14. (AD) | 15. (C) | 16. (AC) |
| 17. (AD) | 18. (ABD) | 19. (BD) | 20. (ACD) | | | | |

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ADVANCE REVISION ASSIGNMENT-13

PHYSICS

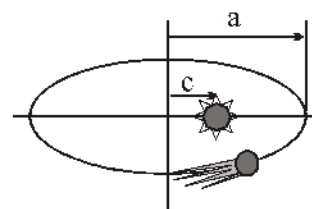
PHYSICS

Time : 1 Hr.

ONE OR MORE THAN ONE CORRECT TYPE (+3, -1)

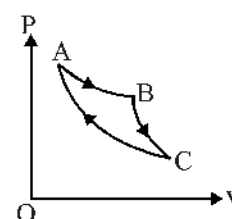
1. A certain comet orbits the sun in an elliptical orbit of eccentricity equal to $1/2$. In this case, the semimajor axis, 'a', is exactly equal to twice 'c', the distance from the sun to the center of the ellipse, as shown below. The speed of the comet when it is closest to the sun (the perihelion) compared to the speed of the comet when it is farthest from the sun (the aphelion) is

(A) $1/3$ as large (B) the same
(C) twice as large. (D) three times as large



2. An ideal gas undergoes a cyclic process ABCA as shown in P-V diagram. The gas expands isothermally along AB and adiabatically along BC and the respective work done are 600 J and 250 J. If the efficiency of the cycle is $1/3$ then

(A) during the compression, the gas gives up 400J of heat to the surrounding
(B) the amount of work done on the gas along CA is 300 J
(C) the net work done is 200 J
(D) the change in the internal energy along CA is 250 J



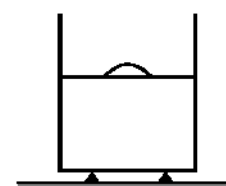
3. The sun orbits the center of the galaxy (milk-way) in almost circular path of radius R in period T and the earth also orbits the sun in almost circular path of radius r in period t. Assume whole mass of the galaxy centered on its center and find an expression for the ratio of the mass of the galaxy to that of the sun.

(A) $\left(\frac{R}{r}\right)^3 \left(\frac{t}{T}\right)^2$ (B) $\left(\frac{R}{r}\right)^3 \left(\frac{T}{t}\right)^2$ (C) $\left(\frac{R}{r}\right)^2 \left(\frac{t}{T}\right)^3$ (D) $\left(\frac{R}{r}\right)^2 \left(\frac{T}{t}\right)^3$

4. The peripheral velocity of a point on the equator of a planet is v_0 due to rotation of the planet about its own axis. The effect of rotation of the planet makes the weight of a body at the equator half of the weight at the pole. What is the escape velocity of polar particle in terms of v_0 ?

(A) v_0 (B) $2v_0$ (C) $3v_0$ (D) $4v_0$

5. A highly conducting cylindrical vessel placed on elevated small supports contains an ideal gas under a piston of mass m. Area of the base of the cylinder is A. The piston is in equilibrium in the mid of the cylinder. Friction force between the cylinder and the piston can be ignored. Total mass of the cylinder and the gas is M. The atmospheric pressure is p_0 . Now the piston is slowly pulled upwards, find the maximum value of M such that the cylinder can be lifted off the supports.

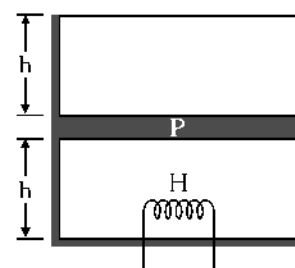


(A) $\frac{p_0 A - mg}{g}$ (B) $\frac{p_0 A - mg}{2g}$ (C) $\frac{2p_0 A - mg}{2g}$

(D) it is impossible in this way

Paragraph for Q.no. (6 & 7)

A cylindrical vessel made of thermally insulating material is fitted with a horizontal piston P of mass m and area A. The piston is also made by thermally insulating material. There is no friction between inner surface of the cylinder and the piston. Above the piston, a liquid of density ρ is filled up to brim of the cylinder. The piston is supported at the position shown due to pressure of an ideal gas filled in the lower portion of the cylinder. When the gas is heated by the heater H, it expands and pushes the piston up, spilling the liquid out of the cylinder. Number of moles of the gas is n and atmospheric pressure is p_0 and initial temperature is T_0 .



6. Work done by the gas is :

(A) $\rho gh^2 A + mgh$

(B) $p_0 hA + \rho gh^2 A + mgh$

(C) $p_0 hA + 0.5\rho gh^2 A + mgh$

(D) $0.5 (p_0 hA + \rho gh^2 A) + mgh$

7. Final temperature of the gas is :

(A) $2T_0 \left(\frac{p_0 A + \rho ghA}{p_0 A} \right)$

(B) $2T_0 \left(\frac{p_0 A}{p_0 A + \rho ghA} \right)$

(C) $2T_0 \left(\frac{p_0 A + \rho ghA + mg}{p_0 A + mg} \right)$

(D) $2T_0 \left(\frac{p_0 A + mg}{p_0 A + \rho ghA + mg} \right)$

8. A vessel is divided into two parts A and B by a fixed partition. The walls of the vessel and the partition are made of a perfect heat insulating material. There is an hole in the partition, which can be opened and closed. Initially the hole is kept closed, an ideal gas is filled in the part A and the part B is evacuated. Now the hole is opened for a short time and then closed, as a result some of the gas enters the part B. Now the hole is again opened, and gas is allowed to flow through the hole until net flow of the gas through the hole ceases. How does the temperature of the gas change during the first and the second opening of the hole ?



(A) During both the first and the second openings, the temperature decreases.

(B) During the first opening, the temperature remains constant but during the second opening, the temperature decreases.

(C) During the first opening, the temperature remains constant but during the second opening, the temperature increases.

(D) During both the first and the second openings, the temperature remains constant.

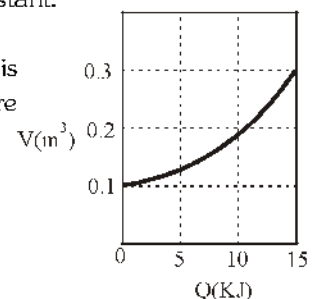
9. Suppose 5 mole of an ideal gas undergoes an isothermal expansion as energy is added to it as heat Q . Graph shows the final volume v_f versus Q . The temperature of gas is (use $\ln 3 = 1.09$ and $R = 25/3 \text{ J/mol-K}$)

(A) 330 K

(B) 260 K

(C) 390 K

(D) None



Paragraph for question nos. 10 to 12

1 mole of a gas at temperature of 127°C having a volume of 10^{-3} m^3 confined under a piston is suddenly expanded. Its temperature is seen fall to 27°C . The work done by the gas during the process is known to be

2500 J. The molar mass of the gas enclosed under the piston is 40gm. Take $R = \frac{25}{3} \text{ J/mol K}$.

10. What is the ratio of $\frac{C_p}{C_v}$ for the gas

(A) $\frac{5}{3}$

(B) $\frac{7}{5}$

(C) $\frac{4}{3}$

(D) 1.5

11. What is the speed of sound in the gas (before expansion)

(A) 340 m/s

(B) 333.33 m/s

(C) 325 m/s

(D) 318.66 m/s

12. What is the volume of gas after expansion?

(A) $\frac{8}{3375} \text{ m}^3$

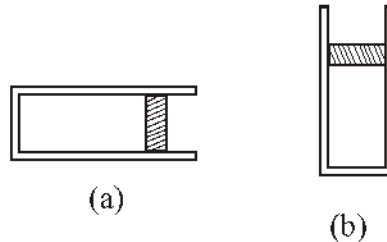
(B) $\frac{16}{3375} \text{ m}^3$

(C) $\frac{4}{3375} \text{ m}^3$

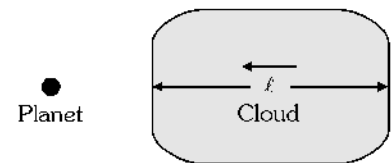
(D) None of these

INTEGER TYPE (+3, 0)

13. Many stars collapse under gravitational pull into a neutron star when most of the hydrogen is used up. The entire star is then a tightly packed ball of neutrons with the density of nuclear matter. Assume that a star of mass 2×10^{32} kg, radius 5.5×10^8 m collapses into a neutron star. It has a time period of rotation of 10^8 sec. about its axis initially. What is its rotation period (in sec.) after collapse. (Take $R_0 = 1.1$ fm, $1 \text{ amu} = 1.6 \times 10^{-27}$ kg)
14. Compute the amount of work done by a gas which is being isobarically heated from 27°C to 127°C if it is in a vessel closed by a movable piston with a cross-sectional area of 20 cm^2 and weighing 5 kg . Consider two cases: (1) the vessel is arranged horizontally (Figure-a), and (2) the vessel is arranged vertically (Figure-b). The initial volume of the gas is 6 litres. Assume standard atmospheric pressure = 100 kPa . Neglect friction. Fill $(w_1 + w_2)/50$ (in J) in OMR.

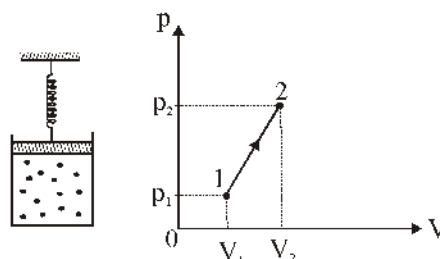


15. Calculate the density (d) of the substance (in kg/m^3) of a spherical planet where the daily period of rotation equals 10^5 sec, if it is known that bodies are weightless at the equator of the planet. Round off to nearest integer. (Take $G = 6.6 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$). Fill $d/7$ in OMR
16. The distance of the Earth from the Sun is 4 times that of the planet Mercury from the Sun. The temperature of the Earth in radiative equilibrium with the Sun is 290 K . Find the radiative equilibrium temperature (T) of the Mercury. Assume all three bodies to be black body. Fill $T/290$ in OMR
17. A uniform cloud of dust particles approaches a planet from a great distance with a velocity v_0 relative to the planet. Radius of the planet is R and escape velocity from the surface of the planet is v_e . Length of the cloud along its direction of motion is l , cross section of the cloud perpendicular to its direction of motion is very large and its density is ρ . How much mass of the dust particles will the planet collect during its passage through the cloud. Mass of the cloud is negligible as compared to the mass of the planet.



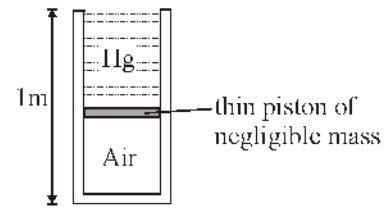
Your answer is $\frac{\pi \rho l R^A}{B} \left[1 + \left(\frac{v_e}{v_0} \right)^C \right]$. Find $A + B + C$.

18. The ideal gas is in a cylinder with base area S under weightless piston, which is held in balance by a spring, the other end of which is secured in position (see figure). Outside the cylinder is vacuum. This gas follows process of 1-2, as shown in the pV -diagram. What is the spring constant K (in N/m) of the spring? Fill $K/625$ in OMR. (Take : $S = 0.1 \text{ m}^2$, $p_1 = 100 \text{ kPa}$, $p_2 = 200 \text{ kPa}$, $V_1 = 1.2 \text{ m}^3$, $V_2 = 2.0 \text{ m}^3$)

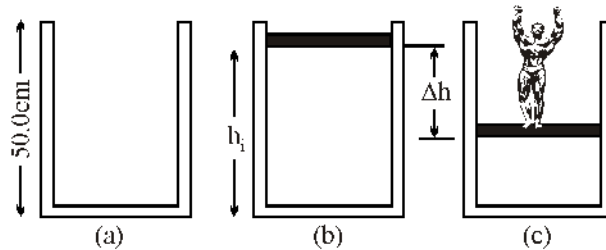


19. One mole helium in a vessel gets the heat from outside and starts expanding to make its volume 2 times the original volume. The heat capacity of the gas in this process is constant and is $\frac{R}{2}$. What is the final temperature (T) of gas (in K)? Initial temperature is 200 K and initial pressure is 40 kPa . Fill $T/20$ in OMR.

- 20.** The 1 m tall conducting cylinder in figure contains air at a pressure of 76 cm of Hg. A very thin piston of negligible mass is placed at the top of the cylinder to prevent any air from escaping, then mercury is very slowly poured into the cylinder until no more can be added without the cylinder overflowing. What is the height h (in cm) of the compressed air? Fill $h/19$ in OMR



- 21.** A highly conducting cylinder that has a cross-section area of 100cm^2 and is 50cm deep is filled with air at 21.0°C and 1.00 atm (figure-a). A 20.0 kg piston is now lowered into the cylinder, compressing the air trapped inside (figure-b). Finally, a 80.0 kg man stands on the piston, further compressing the air, which remains at 21°C (figure-c). What temperature T (in $^\circ\text{C}$) should the gas be maintained to raise the piston and the man back to h_i ? ($g = 10\text{ m/s}^2$). Fill $T/31$ in OMR



- 22.** Three insulated vessels of equal volumes V are connected by thin tubes that can transfer gas but do not transfer heat. Initially all vessels are filled with the same type of gas at a temperature T_0 and pressure P_0 . Then the temperature in the first vessel is doubled (and maintained) and the temperature in the second vessel is tripled (and maintained). The temperature in the third vessel is maintained unchanged. If initial pressure is 77 cm of Hg. Find the final pressure P in the system in cm of Hg. Fill $P/21$ in OMR.

* * * * *

ANSWER KEY

1. (D)	2. (ABCD)	3. (A)	4. (B)	5. (B)	6. (C)	7. (D)	8. (D)
9. (A)	10. (C)	11. (B)	12. (A)	13. (1)	14. (9)	15. (2)	16. (2)
17. (5)	18. (2)	19. (5)	20. (4)	21. (7)	22. 6		

ADVANCE REVISION ASSIGNMENT-14

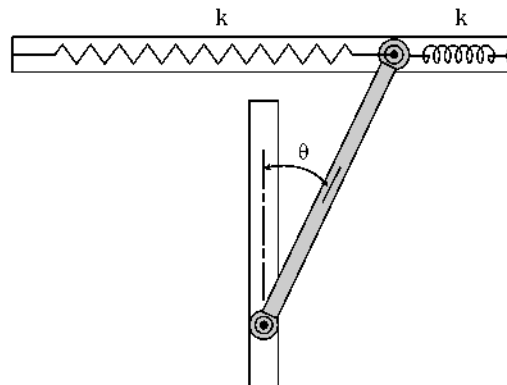
PHYSICS

PHYSICS

Time : 1 Hr.

SINGLE CORRECT TYPE (+3, -1)

1. The ends of the uniform slender bar of mass m and length L move freely in the vertical and horizontal slots under the action of the two precompressed springs each of stiffness k as shown. There is no elongation and compression in spring at $\theta = 0$. Determine the natural frequency f of small oscillations.

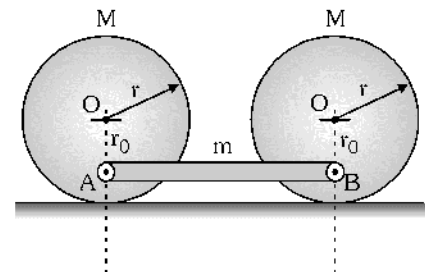


(A) $f = \frac{1}{2\pi} \sqrt{\frac{6K}{m} + \frac{g}{2L}}$ (B) $f = \frac{1}{2\pi} \sqrt{\frac{K}{m} + \frac{3g}{2L}}$ (C) $f = \frac{1}{2\pi} \sqrt{\frac{6K}{m} + \frac{3g}{2L}}$ (D) $f = \frac{1}{2\pi} \sqrt{\frac{2K}{m} + \frac{3g}{2L}}$

2. Derive the natural frequency f_n of the system composed of two homogeneous circular cylinders, each of mass M , and the connecting link AB of mass m . Assume small oscillations. There is no slipping between cylinder and ground.

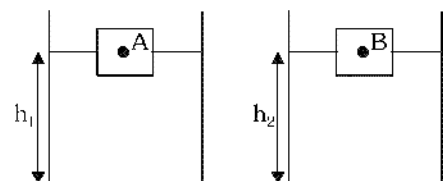
(A) $f = \frac{1}{2\pi} \sqrt{\frac{mgr_0}{3Mr^2 + mr_0^2}}$
 (B) $f = \frac{1}{2\pi} \sqrt{\frac{mgr_0}{3Mr^2 + m(r - r_0)^2}}$
 (C) $f = \frac{1}{2\pi} \sqrt{\frac{Mgr_0}{3Mr^2 + m(r - r_0)^2}}$

(D) None of these



Paragraph for question no. (3 to 4)

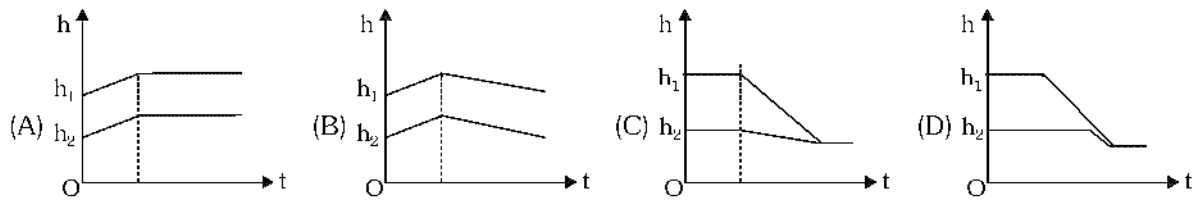
Two ice cubes of side 10 cm, having cavity of volume 20 cm^3 at centre of cube but filled with different materials A and B respectively. The specific gravity of material A is 4.9 and specific gravity of material B is 1.9. Now these cubes are placed in two different vessels of same base area as shown in figure. The water level before putting blocks in vessels are same. Assume that ice melts uniformly from all sides and with same constant rate in both the vessels. (specific gravity of ice = 0.9)



3. Find the ratio of initial submerged volumes of the blocks containing A and B respectively

(A) $\frac{49}{46}$ (B) 1 : 1 (C) $\frac{49}{19}$ (D) 4 : 1

4. Choose the correct graph showing the variation of heights of water-level in two vessels with time.



Paragraph for Q.no. (5 to 6)

An elastic string of natural length $(a + b)$ where $a > b$ and modulus of elasticity Y has a particle of mass m attached to it at a distance 'a' from one end, which is fixed to a point A of a smooth horizontal plane. The other end of the string is fixed to a point B so that the string is just unstretched. If the particle be held at B (by stretching the length a of the string) and then released. Take the cross-sectional area of the string as unity.

5. Find time period of oscillation.

(A) $\pi\sqrt{\frac{ma}{Y}}$ (B) $\pi(\sqrt{a} + \sqrt{b})\sqrt{\frac{m}{Y}}$ (C) $\pi[\sqrt{a} + \sqrt{b}]\sqrt{\frac{2m}{Y}}$ (D) None of these

6. In the above problem, the particle oscillate to and fro through a distance .

(A) $\pi[\sqrt{a} + \sqrt{b}]\sqrt{\frac{2m}{Y}}$ (B) $\frac{b}{2}\left(\frac{\sqrt{a} + \sqrt{b}}{\sqrt{a}}\right)$ (C) $\frac{a+b}{2}$ (D) None of these

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Paragraph for Q.no. (7 to 8)

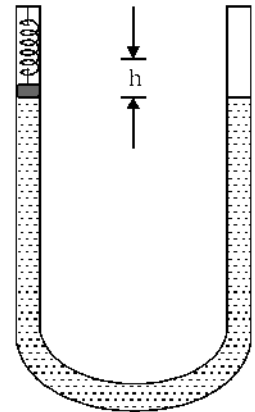
Both the limbs of a U-tube are vertical. One end of a light spring of force constant $K = 78 \text{ Nm}^{-1}$ is fixed with top of left limb and a piston of mass $m = 50 \text{ gm}$ is attached with lower end of the spring as shown in figure. Cross-sectional area of tube is $S = 1 \text{ cm}^2$. Water (density $\rho = 1000 \text{ ms}^{-3}$) is poured into right limb till elongation of spring reduces to $a = 6 \text{ mm}$.

7. Calculate difference h between level of water in right limb and level of lower face of the piston

(A) 8 mm (B) 12 mm
(C) 20 mm (D) 32 mm

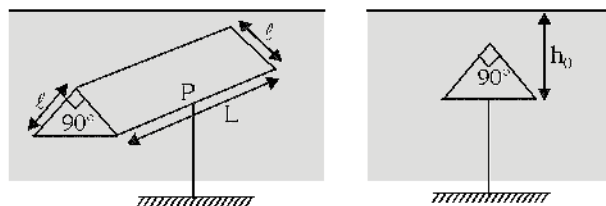
8. If mass of whole in the tube is $M = 150 \text{ gm}$, calculate angular frequency of small oscillations. (Neglect atmospheric pressure).

(A) 5 rad/sec. (B) 20 rad/sec. (C) 30 rad/sec. (D) 50 rad/sec.



Paragraph for Q.no. (9 to 11)

A prism shaped styrofoam of density $\rho_{\text{styrofoam}} (< \rho_{\text{water}})$ is held completely submerged in water. It lies with its base horizontal. The base of foam is at a depth h_0 below water surface and atmospheric pressure is P_0 . Surface of water is open to atmosphere. Styrofoam prism is held in equilibrium by the string attached symmetrically as shown. (Take : $\rho_{\text{styrofoam}} = \rho_f$; $\rho_{\text{water}} = \rho_w$).



9. Net force exerted by liquid on the styrofoam is

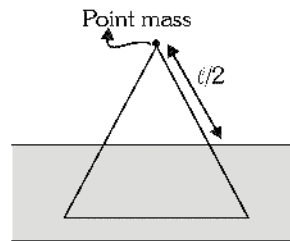
- (A) $\sqrt{2}\rho_w g \ell^2 L$ (B) $2\rho_w g \ell^2 L$ (C) $\rho_w g \frac{\ell^2 L}{2}$ (D) $\rho_w g \ell^2 L$

10. Magnitude of force on any one of the slant face of styrofoam is-

- (A) $(P_0 + \rho_w g(h_0 - \sqrt{2}\ell))L\ell$ (B) $(P_0 + \rho_w g(h_0 - \frac{\ell}{\sqrt{2}}))L\ell$
 (C) $(P_0 - \rho_w g(h_0 - \frac{\ell}{\sqrt{2}}))L\ell$ (D) $(P_0 - \rho_w g(h_0 - \frac{\ell}{2\sqrt{2}}))L\ell$

11. Now string is cut and styrofoam is allowed to come to surface. A point mass is to be placed symmetrically on the upper surface of styrofoam such that it is in equilibrium with its base in horizontal plane. In equilibrium position styrofoam has half of its slant length submerged. Surface tension of water is T , contact angle is 0° . Determine mass m to achieve equilibrium.

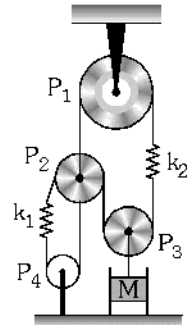
- (A) $\frac{\rho_r g L \ell^2}{2} + \frac{3}{4} \rho_w g L \ell^2 - 2T[L + \ell]$
 (B) $\frac{-\rho_r g L \ell^2}{2} + \frac{3}{8} \rho_w g L \ell^2 - \sqrt{2}T[L + \ell]$
 (C) $\frac{-\rho_r g L \ell^2}{2} + \frac{3}{8} \rho_w g L \ell^2 - \frac{T[L + \ell]}{\sqrt{2}}$
 (D) None of these



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12. An ideal gas having adiabatic constant γ is enclosed in a adiabatic cylindrical container having area of cross-section A . A piston of mass M can move freely in the cylinder is attached with ideal springs k_1 and k_2 and light pulleys P_1 , P_2 and P_3 as shown in figure. When the piston is in equilibrium the volume of the gas in the cylinder is V_0 and its pressure is P_0 . The piston is slightly displaced from the equilibrium position and released. Find the time period of oscillation. Pulley P_2 is fixed.

- (A) $2\pi \sqrt{\frac{M}{\frac{k_1 k_2}{k_1 + k_2} + \frac{\gamma P_0 A^2}{V_0}}}$ (B) $2\pi \sqrt{\frac{M}{\frac{k_1 k_2}{2k_1 + k_2} + \frac{\gamma P_0 A^2}{V_0}}}$
 (C) $2\pi \sqrt{\frac{M}{\frac{3k_1 k_2}{k_1 + k_2} + \frac{\gamma P_0 A^2}{V_0}}}$ (D) $2\pi \sqrt{\frac{M}{\frac{4k_1 k_2}{k_1 + k_2} + \frac{\gamma P_0 A^2}{V_0}}}$



Paragraph for Q.no. (13 & 14)

A string 120 cm in length sustains a standing wave with the points of the string at which the displacement amplitude is equal to 3.5 mm being separated by 15.0 cm. Find

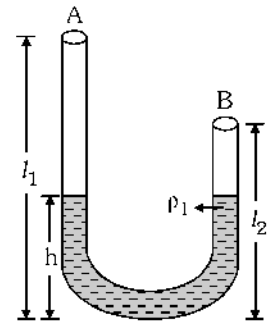
13. The maximum displacement amplitude

- (A) $\sqrt{2}$ mm (B) 2.95 mm (C) 4.95 mm (D) 6.95 mm

14. To which overtone do these oscillations correspond ?

- (A) 4 (B) 3 (C) 5 (D) 6

15. A U-tube having uniform cross-section but unequal arm lengths l_1 and l_2 ($l_2 < l_1$) has same liquid of density ρ_1 filled in it upto a height h as shown in figure. Another liquid of density $\rho_2 = \rho_1/2$ is poured in arm A. Both liquids are immiscible. What length of the second liquid should be poured in A so that first overtone of A is in unison with fundamental tone of B.



(A) $Y = 2 \left(\frac{l_1 - 3l_2 + 2h}{3} \right)$

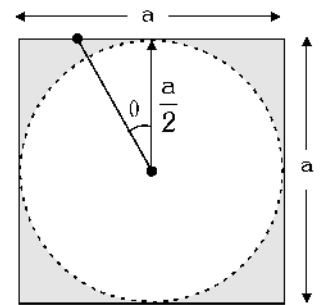
(B) $Y = \left(\frac{l_1 - 3l_2 + 2h}{3} \right)$

(C) $Y = \frac{2(3l_1 - l_2 + h)}{3}$

(D) Situation not possible

16. A square ground of side $a = \frac{10}{\sqrt{2}}\text{m}$ has a circular running track of radius $a/2$ with

its centre coinciding the centre of the ground. A man is running on the track with an angular velocity, $\omega = 22 \text{ rad/s}$ while a car is moving on a road adjacent to ground as shown in the figure. The car moves in such a way that the car, the man and the centre of the ground always lie on the same straight line. If a source of sound of frequency $f = 300 \text{ Hz}$ is being placed at the centre of the ground find the minimum frequency received by the man in the car. Assume velocity of sound in air is $v = 330 \text{ m/s}$.



(A) 200 Hz

(B) 100 Hz

(C) 300 Hz

(D) 150 Hz

MORE THAN ONE CORRECT TYPE (+4, -1)

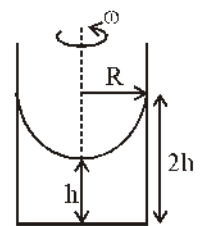
17. A fluid of density ρ is rotated with a certain angular velocity in a cylindrical vessel as shown :-

(A) The angular velocity is $\omega = \sqrt{\frac{2gh}{R^2}}$

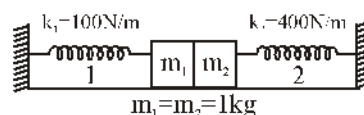
(B) The angular velocity is $\omega = \sqrt{\frac{4gh}{R^2}}$

(C) The kinetic energy of fluid is $\frac{16\pi\rho gh^2 R^2}{15}$

(D) The kinetic energy of fluid is $\frac{5\pi\rho gh^2 R^2}{6}$



18. In the given figure m_1 & m_2 are attached to springs on left & right respectively and kept on smooth level ground. The left block is pushed to the left to compress spring 1 by 10 cm and right block is pushed to right to compress spring 2 by 10 cm. Both the blocks are released simultaneously. Springs are initially relaxed as shown in the diagram:



(A) They will collide at time when both springs come to the natural length.

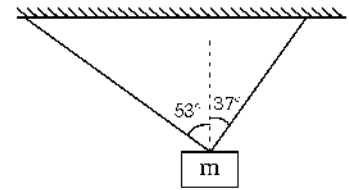
(B) They will collide 5cm to left of position shown.

(C) They will first collide at $t = \frac{\pi}{30} \text{ sec.}$

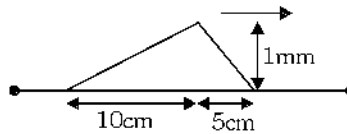
(D) They will first collide at $t = \frac{\pi}{10} \text{ sec.}$

INTEGER TYPE (+3, 0)

19. A heavy mass is hanging from ceiling as shown. Both wires are identical. A pulse is produced in both the strings simultaneously at the point of contact with mass. If it reaches the ceiling in $3\sqrt{3} \times 10^{-3}$ sec. for string on the right, in what time will it reach the ceiling for the left string (in 10^{-1} sec.)



20. The kinetic energy of pulse travelling in a taut string is $(0.3 \times n)$ mJ. Find the value of n .
(Given Tension $(T) = 10$ N and $\mu = 0.1$ kg/m.)



* * * * *

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ANSWER KEY

1.(C)	2. (B)	3. (A)	4. (C)	5. (B)	6. (A)	7. (D)	8. (B)
9. (C)	10. (D)	11. (D)	12. (D)	13. (C)	14. (B)	15. (D)	16. (A)
17. (AD)	18. (BC)	19. (8)	20. (5)				

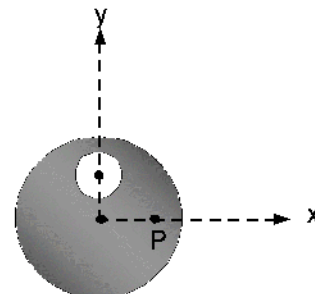
ADVANCE REVISION ASSIGNMENT-15
PHYSICS
PHYSICS
SECTION A (Q.no. 1 to 5) has only One Answer Correct (+3, -1)

1. A non-conducting sphere of radius $R = 5$ cm has its center at origin O of co-ordinate system, shown in figure. It has a spherical cavity of radius $r = 1$ cm, whose centre is at $(0, 3)$ cm. Solid material of sphere has uniform positive

charge density $\rho = \frac{10^{-6}}{\pi}$ coul m^{-3} . Assume dielectric constant of material

to be 1. The potential at point P (4 cm, 0) is approximately

- (A) 35 volt (B) 30 volt
 (C) 25 volt (D) 40 volt

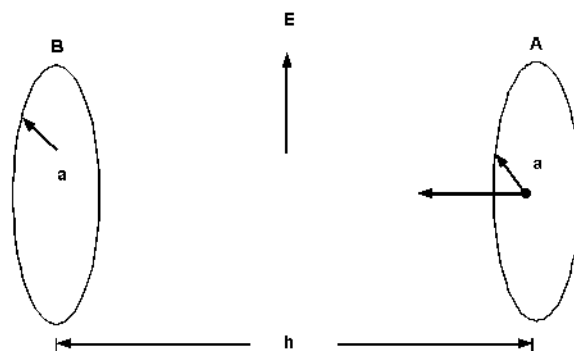


2. A solid non-conducting hemisphere of radius R has a uniformly distributed positive charge of density ρ per unit volume. A negatively charge particle having charge q is transferred from centre of its base to infinity. Dielectric constant of material of hemisphere is unity.

The work performed in the process is found to be $\frac{q\rho R^2}{K\epsilon_0}$. The value of K is

- (A) 3 (B) 5 (C) 6 (D) 4

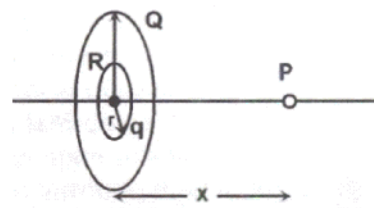
3. Two non conducting circular rings A and B, each of radius $a = 30$ cm are placed coaxially with their axes horizontal in a uniform electric field $E = 10^5$ NC $^{-1}$ directed vertically upwards as shown in figure. Distance between centres of these rings A and B is $h = 40$ cm. Ring A has a positive charge $q_1 = 10\mu$ C while ring B has a negative charge of magnitude $q_2 = 20\mu$ C. A particle of mass $m = 100$ gm and carrying a positive charge $q = 10\mu$ C is released from rest at the centre of the ring A. The velocity when it has moved a distance of 40 cm is



- (A) 6 m/s (B) $6\sqrt{2}$ m/s (C) $6\sqrt{5}$ m/s (D) $6\sqrt{7}$ m/s

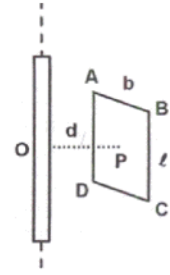
4. Two concentric rings of radius $R = 8$ cm and $r = 2$ cm having charge $Q = 8\mu$ C and $q = -1\mu$ C are placed as shown in diagram. Then electric field is zero at axis of rings

- (A) $x = \pm 4\sqrt{3}$ cms (B) $x = \pm 4$ cms
 (C) $x = \pm 2$ cms (D) $x = \pm 2\sqrt{3}$ cms



5. The electric flux crossing the wire frame ABCD of length ℓ width b and whose centre is at a distance $OP = d$ from an infinite line of charge with linear charge density λ is (consider that the plane of the frame is perpendicular to the line OP)

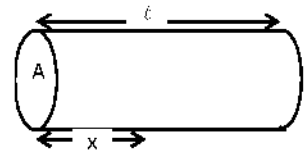
- (A) $\frac{\lambda \ell}{\pi \epsilon_0} \tan^{-1} \left(\frac{b}{2d} \right)$ (B) $\frac{\pi \epsilon_0}{2\lambda \ell} \tan^{-1} \left(\frac{b}{2d} \right)$
 (C) $\frac{\lambda \ell}{\pi \epsilon_0} \tan^{-1} \left(\frac{2d}{b} \right)$ (D) zero



SECTION B (Q.no. 6 to 20) have One or More than One Correct Answers (+4, -1)

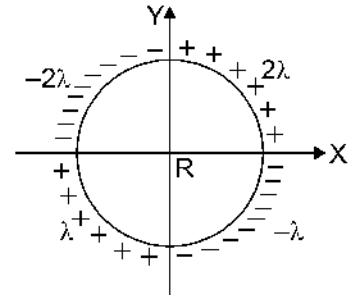
6. The cross-sectional area and length of cylindrical conductor are A and ℓ respectively. The specific conductivity varies as $\sigma(x) = \sigma_0 \ell / \sqrt{x}$ where x is the distance along the axis of the cylinder from one of its ends. A potential difference V_0 is applied across the cylinder. Choose the correct statement(s)

- (A) The resistance of the system along the cylindrical axis is $\frac{2\sqrt{\ell}}{3A\sigma_0}$.
 (B) The current density is constant through out the length of the cylinder.
 (C) The electric field $E(x)$ at each point in the cylinder is $E(x) = \frac{5V_0 \sqrt{x}}{2\ell^{3/2}}$
 (D) The electric field $E(x)$ at each point in the cylinder is $E(x) = \frac{3V_0 \sqrt{x}}{2\ell^{3/2}}$.



7. The charge per unit length of the four quadrant of the ring of radius R 2λ , -2λ , λ and λ respectively as shown in the figure.

- (A) The electric field at the centre is $\frac{\lambda}{2\pi \epsilon_0 R} (-\hat{i})$
 (B) The electric field at the centre is $\frac{\lambda}{2\pi \epsilon_0 R} (-\hat{j})$
 (C) The electric dipole moment of given charge distribution is $4\lambda R^2 (\hat{i})$
 (D) Electric potential at the centre of ring is zero



8. The electric potential V in volts in a region of space is given by

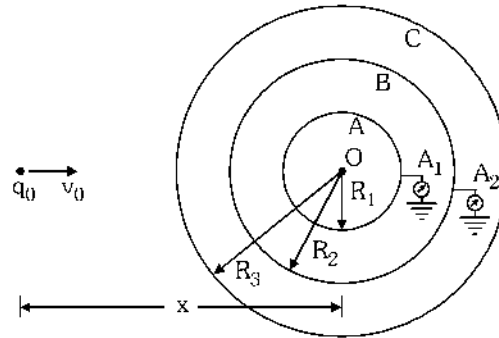
$$V = ax^2 + ay^2 + 2az^2.$$

Where a is a constant of proper dimensions.

Choose the correct statement(s)

- (A) The work done by field when a $2\mu\text{C}$ test charge moves from point $(0, 0, 0.1 \text{ m})$ to origin is $5 \times 10^{-5} \text{ J}$.
The numeric value of a is 1.25×10^3
 (B) The work done by field when a $2\mu\text{C}$ test charge moves from point $(0, 0, 0.1 \text{ m})$ to origin is $5 \times 10^{-5} \text{ J}$.
The value of a is 2.5×10^3
 (C) The radius of the circle of equipotential line in xy -plane corresponding to $V = 6250 \text{ V}$ and $Z = \sqrt{2} \text{ m}$ is 1 m .
 (D) The radius of the circle of equipotential line in xy -plane corresponding to $V = 6250 \text{ V}$ and $Z = \sqrt{2} \text{ m}$ is 0.1 m .

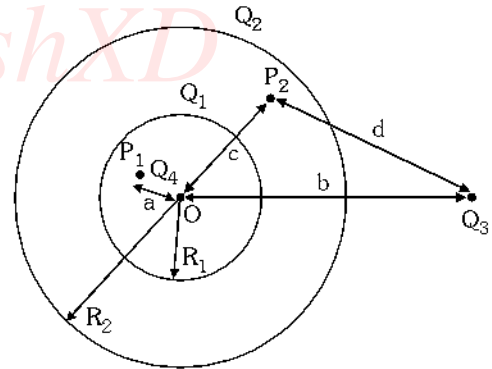
9. A, B, C are three concentric conducting shell of radius R_1, R_2, R_3 respectively. A point charge q_0 is approaching the system of shells (as shown in figure) with a small constant velocity V_0 . The shells A and B are grounded. A_1 and A_2 are ammeters attached to shell A and B.



Choose the correct statement(s)

- (A) The value of current in ammeter A_1 when charge q_0 is at distance of x from centre O is $\frac{R_2 V_0 q_0}{2x^2}$
- (B) The value of current in ammeter A_2 when charge q_0 is at distance of x from centre O is $\frac{R_2 V_0 q_0}{x^2}$
- (C) The value of current in ammeter A_2 when charge q_0 is at distance of x from centre O is $\frac{2V_0 q_0 R_2}{x^2}$
- (D) The Electric field intensity in the region between the shells B and C will increase as charge q_0 approaches towards the system of concentric shells.

10. Consider a system of two conducting spherical shells having common centre at O. The charge given to inner shell and outer shell is Q_1 and Q_2 respectively. A point charge Q_4 kept at distance a from centre O. A point charge Q_3 is kept at distance b from centre O. $\left[K = \left(\frac{1}{4\pi\epsilon_0} \right) \right]$.

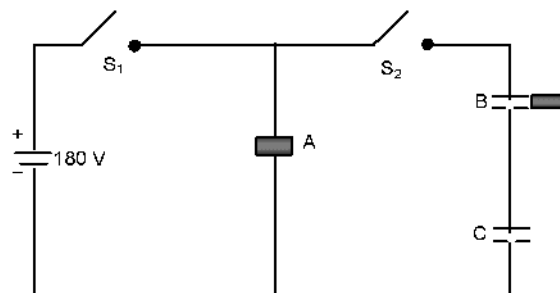


Choose the correct statement(s)

- (A) The potential at centre O is $\left[\frac{KQ_1}{R_1} + \frac{KQ_2}{R_2} + \frac{KQ_3}{b} + \frac{KQ_4}{a} \right]$.
- (B) The magnitude of electric field intensity at centre O is $\left(\frac{KQ_4}{a^2} \right) (N/C)$.
- (C) The magnitude of electric field intensity at centre O due to charge on outer surface of outer shell is $\left[\frac{KQ_3}{b^2} \right]$.
- (D) The magnitude of electric field intensity at point P_2 is $\frac{K(Q_1 + Q_4)}{c^2}$.

Passage for Q.no. (11 to 12)

In the circuit shown in Figure, capacitor A has capacitance $C_1 = 2\mu F$ when filled with a di-electric slab ($K = 2$). Capacitors B and C are air capacitors and have capacitances $C_2 = 3\mu F$ and $C_3 = 6\mu F$, respectively.

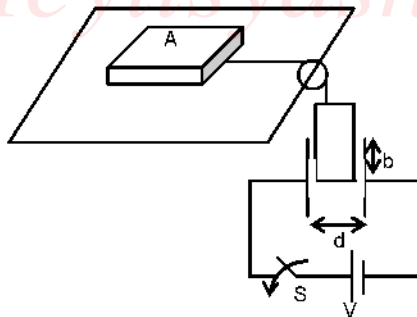


11. Choose the correct statement(s)
- (A) Capacitor A is charged by closing switch S_1 alone, The energy supplied by battery during process of charging is 0.0648 J.
 - (B) Capacitor A is charged by closing switch S_1 alone, The energy supplied by battery during process of charging 0.1296 J.
 - (C) Switch S_1 is now opened and S_2 is closed. The charge on B is $180\mu\text{C}$.
 - (D) Switch S_1 is now opened and S_2 is closed. The energy stored in the system of capacitors is 0.0162 J.
12. Now, switch S_2 is also opened, slab of A is removed. Another di-electric slab of $K = 2$, which can just fill the space in B, is inserted into Capacitor B and then switch S_2 alone is closed. Choose the correct statement(s)
- (A) The electric field intensity in capacitor B is .75 times the initial field intensity.
 - (B) The final charge on Capacitor B and C is equal to $90\mu\text{C}$.
 - (C) The total amount of heat generated in the circuit is $5.4 \times 10^{-3} \text{ J}$ in this process.
 - (D) The final charge on capacitor A is $90\mu\text{C}$.

Passage for Q.no. (13 to 14)

A block A of mass m kept on a rough horizontal surface is connected to a dielectric slab of mass $m/6$ and dielectric constant k by means of a light and inextensible string passing over a fixed pulley as shown in the figure. The dielectric can completely fill the space between the parallel plate capacitor of plate area $\ell \times \ell$ and separation between the plates d kept in vertical position. Initially switch S is open and length of the dielectric inside the capacitor is b . The coefficient of friction between the block A and the surface is $\mu = 1/4$. Ignore any other friction.

13. Choose the correct statement(s)

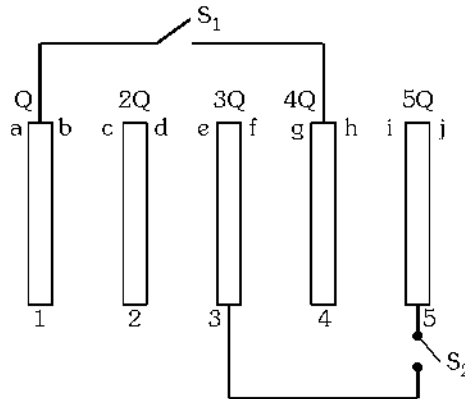


- (A) The minimum value of the $\text{emf} = V$ of the battery so that after closing the switch the block A will move is $\sqrt{\frac{mgd}{6\epsilon_0\ell(k-1)}}$.
 - (B) The work done by electric force on the dielectric will be positive when dielectric starts going inside capacitor.
 - (C) The electric force acting on the dielectric will be constant irrespective of length of dielectric in the capacitor.
 - (D) The potential energy of capacitor will increase with the length of dielectric increases inside capacitor.
14. Choose the correct statement(s)
- (A) If $V = 2V_{\min}$ acceleration of the block A when dielectric completely fills the space between the plates of the capacitor is $\frac{3}{14}g$.
 - (B) The total work done by tension on the system is zero.
 - (C) If $V = 2V_{\min}$ the acceleration of dielectric increases during the motion between the plates of capacitor.
 - (D) If $V = 2V_{\min}$ the acceleration of dielectric is constant during the motion between the plates of capacitor.

Passage for Q.no. (15 to 16)

Consider a system of five large conducting plates of area A. The charge on plate 1, 2, 3, 4, 5 are given as Q, 2Q, 3Q, 4Q and 5Q respectively. Initially the two switches are opened. Area of each plate is A.

The distance between every successive plate is very very small equal to d.



15. Choose the correct statement(s)

(A) The charge on surface b is $\left[-\frac{13Q}{2} \right]$

(B) The potential difference between plate 2 and 3 is $(V_2 - V_3)$ is $-\frac{9}{2} \left(\frac{Qd}{\epsilon_0} \right)$.

(C) The plate 4 is at highest potential among all plates .

(D) The potential of plate 3 is higher than plate 5.

16. Now both the switches S_1 and S_2 are closed simultaneously. Now Consider the steady state.

Choose the correct statement(s)

(A) The charge on surface b is is -1.3 Q .

(B) The plate 2 has highest potential among all plates.

(C) The charge on surface f is is -0.6Q .

(D) The potential of plate 3 is equal to potential of plate 5 .

Passage for Q.no. (17 to 18)

Consider a uniformly charged solid non conducting sphere of radius R. It has two spherical cavities each of radius R' as shown in the figure. Volume charge density for sphere is $\rho \text{ C/m}^3$. Dielectric constant of material

is 1. $\left(K = \frac{1}{4\pi\epsilon_0} \right)$

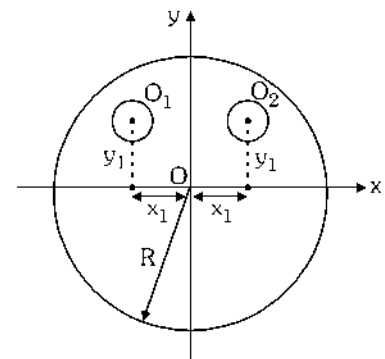
17. Choose the correct statement(s) :

(A) The electric field inside right cavity is uniform .

(B) The electric field at the centre O is $\frac{8K\rho\pi R'^3 y_1 \hat{j}}{3(x_1^2 + y_1^2)^{3/2}}$.

(C) The electric field at the centre O is $\frac{4K\rho\pi R'^3 y_1 \hat{j}}{3(x_1^2 + y_1^2)}$

(D) At least one null point can exist on the Y axis .



18. Choose the correct statement(s) :

(A) The potential at centre O is $2K\rho\pi R^2 - \frac{8}{3} \frac{K\rho\pi R^3}{(\sqrt{x_1^2 + y_1^2})}$

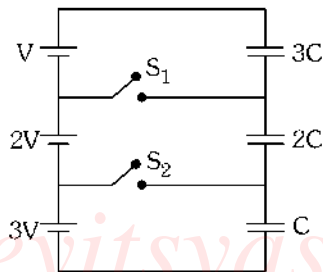
(B) The potential at centre is $2K\rho\pi R^2 + \frac{8}{3} \frac{K\rho\pi R^3}{(\sqrt{x_1^2 + y_1^2})}$

(C) Consider a point P at position (0,2R) ,The electric field at point P is along +Y axis.

(D) If the both the cavities are filled completely with same charge density ρ C/m³. then the magnitude of electric field intensity at every internal point of the sphere will increase in magnitude .

Passage for Q.no. (19 to 20)

At an instant $t=0$ both the switches are closed simultaneously. Consider the circuit when it reaches steady state.



19. Choose the correct statement(s) :

(A) The total amount of charge passing through S_1 after closing the switches is $1.0 CV$ rightwards .

(B) The total amount of charge passing through S_2 after closing the switches is $1.0 CV$ rightwards .

(C) The final potential energy stored in circuit is higher than initial potential energy stored.

(D) The amount of heat energy release after closing of switches is $0.19 CV^2$ approximately .

20. Now a dielectric slab of dielectric constant $K = 2$ is inserted slowly by external agent in the capacitor of $2C$. Then choose the correct statement:

(A) The electric field intensity between the plates of capacitor $2C$ is decreased to half .

(B) The total work done by batteries of emf V and $3V$ are zero during and after the insertion of slab.

(C) The total amount of heat produced in the circuit during insertion of slab is equal to work done by external agent.

(D) The amount of charge stored on capacitor of capacitance $2C$ is $8CV$.

ANSWER KEY

1.(A)	2. (D)	3. (B)	4. (B)	5. (A)	6. (ABD)	7. (ACD)	8. (AC)
9. (BD)	10. (ACD)	11. (ACD)	12. (ACD)	13. (ABCD)	14. (ABD)	15. (ABCD)	16. (ABD)
17. (BD)	18. (ACD)	19. (ACD)	20. (BCD)				

ADVANCE REVISION ASSIGNMENT-16

PHYSICS

PHYSICS

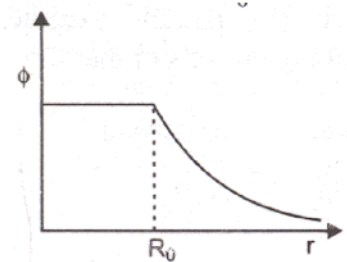
SINGLE CORRECT TYPE (+3, -1) (Q.1 to 5)

- Under the influence of the Coulomb field of a fixed charge $+Q$, a charge $-q$ is moving around it in an elliptical orbit. Find out the correct statement.
(A) The angular momentum of the charge $-q$ w.r.t. $+Q$ is constant.
(B) The linear momentum of the charge $-q$ is constant.
(C) The angular velocity of the charge $-q$ is constant.
(D) The total energy of system of two charges is not constant.

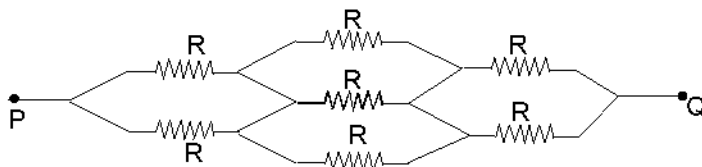
- A spherical symmetric charge system is centred at origin. Given, electric potential

$$\phi = \frac{Q}{4\pi\epsilon_0 R_0} (r \leq R_0), \quad \phi = \frac{Q}{4\pi\epsilon_0 r} (r > R_0)$$

- Within a sphere of radius $r = 2R_0$ enclosed net charge is Q .
- Electric field is discontinuous at $r = R_0$
- Charge is thinly distributed at $r = R_0$
- All of above.



- Find equivalent resistance for the figure given between P and Q



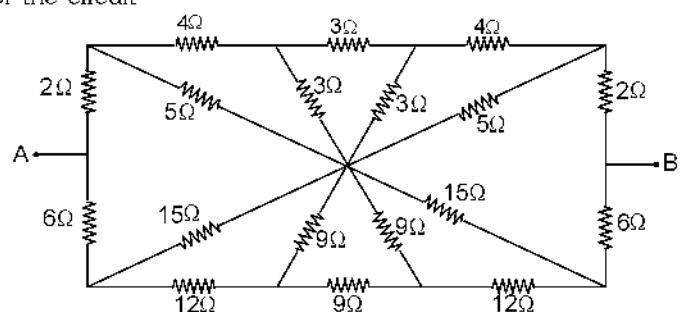
- $\frac{4R}{3}$
- $\frac{10R}{3}$
- $2R$
- $\frac{R}{7}$

- If v_o be the potential at origin in an uniform electric field $\vec{E} = E_x \hat{i} + E_y \hat{j}$. The potential at point P(x, y) is

- $v_o + xE_x + yE_y$
- $v_o + xE_x - yE_y$
- $v_o - xE_x - yE_y$
- $\sqrt{(x^2 + y^2)} \sqrt{E_x^2 + E_y^2} - v_o$

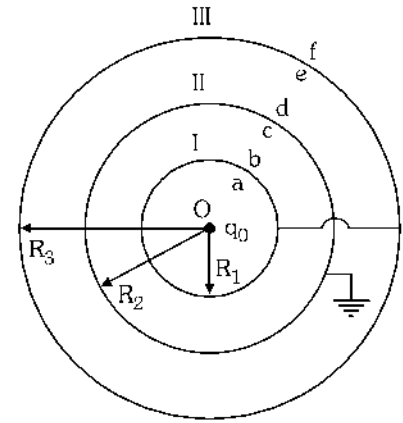
- The equivalent resistance between A and B of the circuit shown in Figure

- 16.5Ω
- 12.50Ω
- 8.75Ω
- 6.75Ω



ONE OR MORE CORRECT TYPE (+4, 0) (Q.no. 6 to 20)

6. Shells I, II, III are conducting spherical neutral shells of radius R , $2R$ and $3R$ respectively. Shell I and shell III are connected. A point charge q_0 is kept at centre O. Now Shell II is also grounded. Choose the correct statement(s)



- (A) Shell I and III are at same potential .
 (B) The charge on surface 'b' (outer surface of Shell I) is $(2/11) q_0$.
 (C) The charge on surface 'b' (outer surface of Shell I) is $(3/11) q_0$.
 (D) The potential energy of system decreases when shell II is grounded.

7. Consider a gravity free region, where a uniform electric field is present as $\vec{E} = [E_0 \hat{i} - E_0 \hat{j}] \text{ N/C}$.

[where x is along horizontally rightwards and y is a vertically upwards]

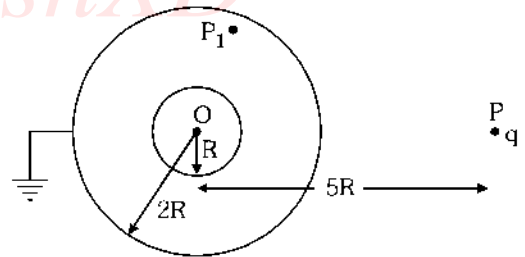
A particle of charge q is projected with velocity u and at angle of θ with horizontal. Finally it strikes the same horizontal level.

Choose the correct statement(s)

- (A) The value of θ should be 67.5° to have maximum horizontal range .
 (B) The value of θ should be 63.5° to have maximum horizontal range .
 (C) The speed of particle at highest position is $u \cos \theta + u \sin \theta$.
 (D) If the particle strikes the same horizontal level at angle of β (the acute angle) then $\beta < \theta$.

8. Consider a solid conducting sphere of radius $2R$. It has a spherical cavity of radius R has its centre at centre of solid sphere. The sphere is grounded. A point charge q is fixed at point P at distance of $5R$.

Choose the correct statement(s) :



- (A) The magnitude of charge on outer surface of solid sphere

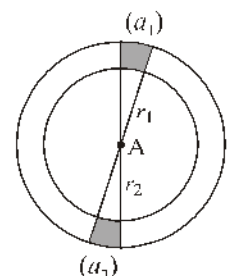
is $\left(\frac{2Q}{5R}\right) C$.

- (B) The electric field energy per unit volume at point P1 is zero.

- (C) The magnitude of net electric force experienced by "q" due to charge on the sphere is $\left[\frac{2Kq}{125R^2}\right] \text{ N/C}$

- (D) If the connection of sphere of ground is now removed then the potential energy of the system will increase.

9. A wire having uniform linear charge density $+\lambda$, is bent in the form of a ring of radius R . Point A as shown in Fig., is in the plane of ring. But not at the centre. Two elements of the ring of lengths a_1 and a_2 subtends small equal angle at the point A. They are at distance r_1 and r_2 from point A respectively ($r_2 > r_1$) :



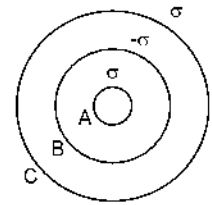
- (A) The ratio of charge of elements a_1 and a_2 is $\frac{r_1}{r_2}$

- (B) The element a_1 produces greater magnitude of Electric Field at A than produced by element a_2

- (C) The elements a_1 and a_2 produce same potential at A.

- (D) If the charge of ring is reversed to negative charge (of linear density $-\lambda$), Then element a_1 produces greater magnitude of Electric Field at A than element a_2

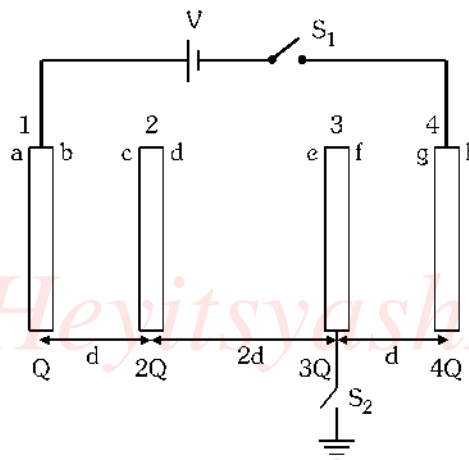
10. Three concentric non conducting shells A, B and C of radii a , b and c ($a < b < c$) have surface charge densities σ , $-\sigma$ and σ respectively. Choose the correct statement(s)



- (A) The potential of shell A is $\frac{\sigma}{\epsilon_0}(a - b + c)$.
- (B) If no work is required to bring a charge 'q' from shell A to shell C then the relation between the radii a , b and c is $c = a + b$.
- (C) The potential of shell B is $\frac{\sigma}{\epsilon_0} \left[\frac{a^2}{b} - b + c \right]$.
- (D) The electric flux through shell B due to charge of shell C is zero.

Paragraph for Q.no. (11 to 12)

Consider a system of 4 large conducting plates. The charges on plates 1, 2, 3, 4 are Q , $2Q$, $3Q$, $4Q$ respectively. The area of each plate is A . Given that $(A\epsilon_0 V = 13Qd)$. The plates are separated from each other as shown in the figure.



11. Now switch S_1 is closed, an ideal battery of emf V is connected to plates 1 and plate 4. Choose the correct statement(s)

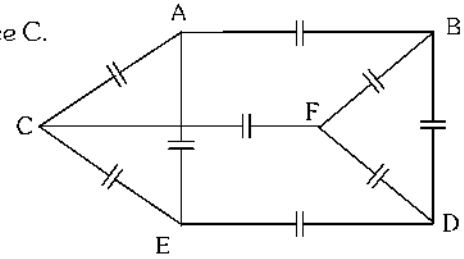
- (A) The charge on surface b of plate 1 after first steady state attained is $\left[\frac{7Qd + A\epsilon_0 V}{4d} \right]$.
- (B) The charge on surface b of plate 1 after first steady state attained is $\left[\frac{-9Qd + A\epsilon_0 V}{4d} \right]$.
- (C) The amount of charge on left surface of plate 1 is $5Q$.
- (D) The plate 3 is at highest potential among all plates.

12. Now the switch S_2 is also closed. Choose the correct statement(s)

- (A) The charge on surface b of plate 1 is $\left[\frac{A\epsilon_0 V + Qd}{4d} \right]$.
- (B) The charge on surface b of plate 1 is $\left[\frac{A\epsilon_0 V + Qd}{2d} \right]$.
- (C) The plate 1 is highest potential among all plates.
- (D) The charge on surface "a" of plate 1 is zero.

Paragraph for Q.no. (13 to 14)

Consider a symmetric circuit of 9 identical capacitors of capacitance C .



13. Choose the correct statement(s) :

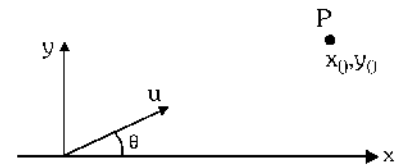
- (A) The equivalent capacitance across AB is $\frac{5C}{3}$.
- (B) The equivalent capacitance across AB is $\frac{5C}{6}$.
- (C) The equivalent capacitance across AC is $\frac{15C}{8}$.
- (D) The equivalent capacitance across AC is $\frac{15C}{16}$.

14. Choose the correct statement(s) :

- (A) The equivalent capacitance across A and D is $\frac{3C}{4}$.
- (B) The equivalent capacitance across A and D is $\frac{4C}{3}$.
- (C) When the network is analyzed across terminals A and D, then B and E are at same potential.
- (D) When the network is analyzed across terminals A and D, the points C and F are not at same potential.

Paragraph for Q.no. (15 to 16)

Consider a gravity free space where a uniform electric field of intensity of E_0 is present vertically upwards. A small particle of mass m and charge $-q$ is projected with velocity of u and at an angle θ with horizontal. The particle is projected from origin. It has to collide point P located at x_0, y_0 with minimum possible velocity u . X Y plane is vertical.



15. Choose the correct statement(s)

- (A) The minimum possible speed with which it has to be projected is $\left[\sqrt{\frac{qE_0}{m} \left[y_0 + \sqrt{x_0^2 + y_0^2} \right]} \right]$.
- (B) The minimum possible speed with which it has to be projected is $\sqrt{\frac{qE_0}{m} \left(\frac{y_0 + \sqrt{x_0^2 + y_0^2}}{2} \right)}$.
- (C) The minimum possible speed with which it has to be projected is $\sqrt{\frac{qE_0}{m} \left(\frac{y_0 + \sqrt{x_0^2 + y_0^2}}{4} \right)}$.
- (D) The particle will move along parabolic path.

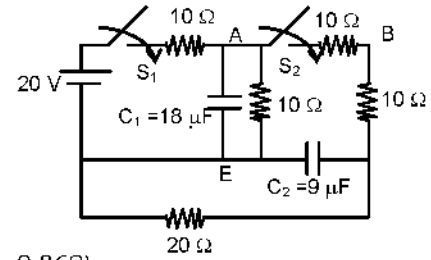
16. Choose the correct statement(s)

- (A) The corresponding angle of projection is $\theta = \tan^{-1} \left[\frac{y_0 + \sqrt{x_0^2 + y_0^2}}{2x_0} \right]$
- (B) The corresponding angle of projection is $\theta = \tan^{-1} \left[\frac{y_0 + \sqrt{x_0^2 + y_0^2}}{x_0} \right]$
- (C) The corresponding angle of projection is $\theta = \tan^{-1} \left[\frac{y_0 + \sqrt{x_0^2 + y_0^2}}{4x_0} \right]$
- (D) During the motion the tangential acceleration of particle is not constant

Paragraph for Q.no. (17 to 18)

For the circuit shown in the figure.

S_1 is closed at $t = 0$ and S_2 is kept open.



17. Choose the correct statement(s)
- (A) The charge on C_1 in steady state is $180\mu\text{C}$.
 - (B) At $t=0$, the power delivered by battery is 40 watt.
 - (C) At $t = 90\mu\text{ sec}$, the current through capacitor C_1 is 0.736A . ($e^{-1} = 0.368$)
 - (D) Under steady state(after long time) , potential difference between point A and B is zero.
18. When the circuit is under steady state. The circuit is again disturbed by closing the switch S_2 . Now S_1 and S_2 are closed. Choose the correct statement(s)
- (A) The charge on C_1 in final steady state is $160\mu\text{C}$
 - (B) The charge on C_2 in final steady state is $40\mu\text{C}$
 - (C) Under final steady state , potential difference between point A and B is $20/9$ volts .
 - (D) Under final steady state, the charge on C_1 is zero.

Paragraph for Q.no. (19 to 20)

A parallel plate capacitor placed in a cylindrical tank is filled with a liquid of dielectric constant k . The area of cross-section of the tank is A and height of the liquid is equal to the length of the square plate of plate area ℓ^2 . The separation between the plates is d . A very small hole of area ' a ' is opened at the bottom of the tank at $t = 0$. If the capacitor in the process remains connected with a battery of emf E . Assume the level of liquid in the capacitor remains same as outside. Assume that bernoulli's equation is applicable in the situation.

19. Choose the correct statement(s)
- (A) The current in the circuit as a function of time is $i = \frac{\epsilon_0 \ell E}{d} (k-1) \frac{a}{A} (\sqrt{2g\ell} - \frac{a}{A}gt)$.
 - (B) The current in the circuit as a function of time is $i = \frac{\epsilon_0 \ell E}{2d} (k-1) \frac{a}{A} (\sqrt{2g\ell} - \frac{a}{A}gt)$.
 - (C) The capacitance of capacitor decreases with time.
 - (D) The potential difference between the plates of capacitor remains constant.
20. Choose the correct statement(s)
- (A) The rate at which the level of liquid decreases is $\frac{dh}{dt} = \frac{a}{2A} (\sqrt{2g\ell} - \frac{a}{A}gt)$.
 - (B) The rate at which the level of liquid decreases is $\frac{dh}{dt} = \frac{a}{A} (\sqrt{2g\ell} - \frac{a}{A}gt)$.
 - (C) The power supplied by battery decreases with time.
 - (D) When the container gets empty the power delivered by the battery become zero.

ANSWER KEY

- | | | | | | | | |
|-----------|------------|-----------|-----------|----------|-----------|----------|----------|
| 1. (A) | 2. (D) | 3. (A) | 4. (C) | 5. (D) | 6. (ABD) | 7. (ACD) | 8. (AB) |
| 9. (ABCD) | 10. (ABCD) | 11. (AC) | 12. (ACD) | 13. (AC) | 14. (BCD) | 15. (AD) | 16. (BD) |
| 17. (ABC) | 18. (ABC) | 19. (ACD) | 20. (BCD) | | | | |

ADVANCE REVISION ASSIGNMENT-17

PHYSICS

PHYSICS

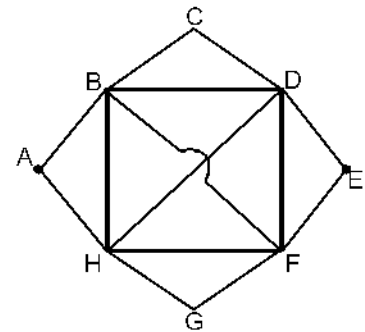
SINGLE CORRECT TYPE (+3, -1) (Q.1 to 8)

1. A particle of mass m having negative charge q move along an ellipse around a fixed positive charge Q so that its maximum and minimum distances from fixed charge are equal to r_1 and r_2 respectively. The angular momentum L of this particle is

(A) $\sqrt{\frac{mr_1r_2Qq}{\pi\epsilon_0(r_1+r_2)}}$ (B) $\sqrt{\frac{mr_1r_2Qq}{2\pi\epsilon_0(r_1+r_2)}}$ (C) $\sqrt{\frac{mr_1r_2Qq}{3\pi\epsilon_0(r_1+r_2)}}$ (D) $\sqrt{\frac{mr_1r_2Qq}{4\pi\epsilon_0(r_1+r_2)}}$

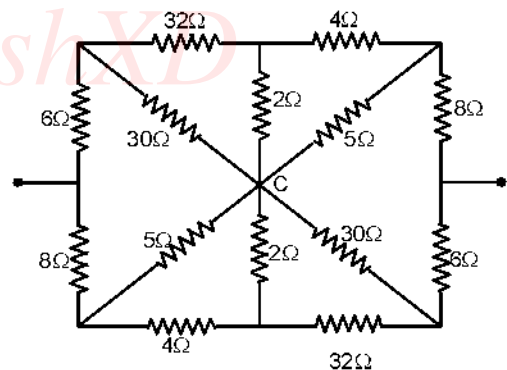
2. Fourteen identical resistors, each of resistance r are connected as shown in figure. The equivalent resistance between A and E is

- (A) r
(B) $1.4r$
(C) $1.2r$
(D) $1.6r$



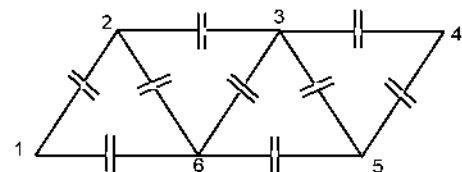
3. The equivalent resistance between A and B of the circuit shown in Figure is

- (A) 12Ω
(B) 18Ω
(C) 16Ω
(D) 14Ω



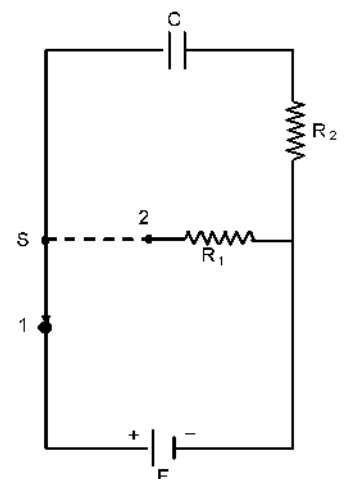
4. Nine identical capacitors, each of capacitance $C = 15\mu\text{F}$ are connected as shown in Figure. The equivalent capacitance between terminals 1 and 4 is

- (A) $10\mu\text{F}$ (B) $5\mu\text{F}$
(C) $15\mu\text{F}$ (D) $11\mu\text{F}$



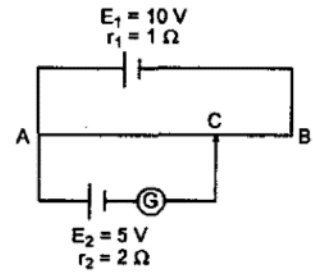
5. Switch S of circuit shown in figure is in position 1 for a long time. At instant $t = 0$, it is thrown from position 1 to 2. The thermal power $P_1(t)$ generated in resistance R_1 .

- (A) $\frac{E^2 R_1}{(R_1 + R_2)^2} \cdot e^{-2t/(R_1 + R_2)C}$
(B) $\frac{E^2 R_1}{(R_1 + R_2)^2} \cdot e^{-t/(R_1 + R_2)C}$
(C) $\frac{E^2 R_1}{(R_1 + R_2)^2} \cdot e^{-t/2(R_1 + R_2)C}$
(D) $\frac{E^2 R_1}{2(R_1 + R_2)^2} \cdot e^{-t/(R_1 + R_2)C}$

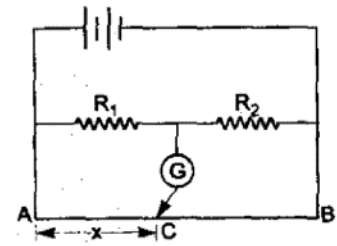


6. A galvanometer of resistance $20\ \Omega$ gives a full scale deflection when a current of $0.04\ \text{A}$ is passed through it. It is desired to convert it into an ammeter of full scale reading $20\ \text{A}$. The only shunt available is $0.050\ \Omega$ resistance. The resistance that must be connected in series with the coil of the galvanometer is:
 (A) $4.95\ \Omega$ (B) $5.94\ \Omega$ (C) $0.01\ \Omega$ (D) $0.04\ \Omega$

7. In the figure the potentiometer wire of length $l = 100\ \text{cm}$ and resistance $9\ \Omega$ is joined to a cell of emf $E_1 = 10\ \text{V}$ and internal resistance $r_1 = 1\ \Omega$. Another cell of emf $E_2 = 5\ \text{V}$ and internal resistance $r_2 = 2\ \Omega$ is connected as shown. The galvanometer G will show no deflection when the length AC is:
 (A) $50\ \text{cm}$ (B) $55.55\ \text{cm}$
 (C) $52.67\ \text{cm}$ (D) $54.33\ \text{cm}$

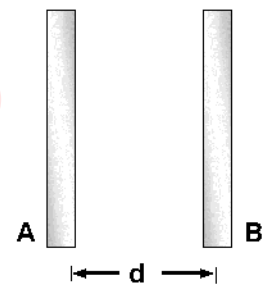


8. In the shown arrangement of the experiment of the meter bridge if AC corresponding to null deflection of galvanometer is x , what would be its value if the radius of the wire AB is doubled?
 (A) x (B) $x/4$
 (C) $4x$ (D) $2x$



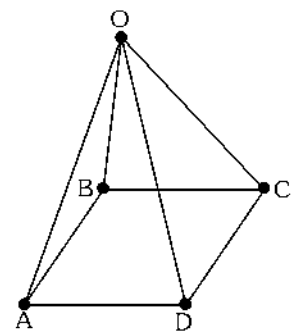
ONE OR MORE CORRECT TYPE (+4, -1) (Q.9 to 20)

9. Each plate of a parallel plate air capacitor has area $A = 5 \times 10^{-3}\ \text{m}^2$ and are $d = 8.85\ \text{mm}$ apart as shown in figure. Plate A has positive charge $q_1 = 10^{-10}$ coulomb and plate B has charge $q_2 = 2 \times 10^{-10}$ coulomb. When its positive terminal of battery of emf $10\ \text{volts}$ is connected with plate A and negative terminal with plate B. Choose the correct statement(s)
 (A) The amount of charge on left surface of plate A will decrease when the battery is connected.
 (B) The amount of charge on right surface of plate B will not change when the battery is connected.
 (C) The amount of charge passes through the battery is $2 \times 10^{-10}\ \text{C}$.
 (D) The amount of charge passes through the battery is $1 \times 10^{-10}\ \text{C}$.



10. Eight identical resistance r each are connected along edges of a pyramid having square base $ABCD$ as shown in Figure. Choose the correct statement(s)

- (A) The equivalent resistance between A and D is $\frac{8r}{15}$.
 (B) The equivalent resistance between A and D is $\frac{r}{5}$
 (C) The equivalent resistance between A and O is $\frac{7}{15}r$
 (D) The equivalent resistance between A and O is $\frac{11r}{15}$

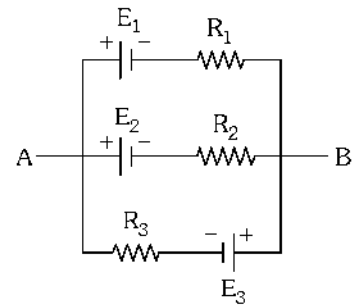


11. A voltmeter of resistance R_v and an ammeter of resistance R_A are connected in series across a battery of emf E and of negligible internal resistance. When a resistance R is connected in parallel to voltmeter, reading of ammeter increases to three times while that of voltmeter reduces to one-third. Choose the correct statement(s)

- (A) The value of R_A is $\frac{8}{3}R$ (B) The value of R_A is $\frac{10}{3}R$
 (C) The value of R_v is $8R$ (D) The voltmeter and ammeter are not ideal

12. Three batteries having emf $E_1 = 12\text{ V}$, $E_2 = 14\text{ V}$ and $E_3 = 2\text{ V}$ and internal resistance $R_1 = 4\Omega$, $R_2 = 8\Omega$ and $R_3 = 12\Omega$ are connected with each other as shown in Figure.

- (A) The equivalent emf across A and B is 10 volt .
 (B) The equivalent emf across A and B is 20 volt.
 (C) The internal resistance of the combination across A and B is $\frac{24}{11}\Omega$.
 (D) The internal resistance of the combination across A and B is $\frac{34}{11}\Omega$.



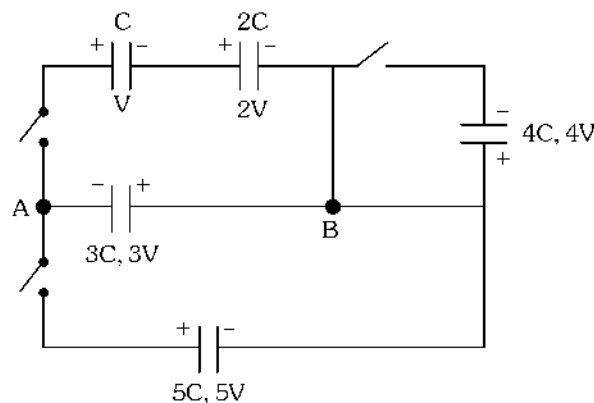
Paragraph for Q.no. (13 to 14)

A steady beam of α -particles travelling with kinetic energy $E = 83.5\text{ keV}$ carries a current of $I = 0.2\text{ }\mu\text{A}$. If this beam strikes a plane surface at an angle $\theta = 30^\circ$ with normal to the surface
 (Mass of α -particles $= 6.68 \times 10^{-27}\text{ kg}$.)

13. Choose the correct statement(s)
 (A) The number of α -particles strike the surface in $t = 4\text{ sec}$ is 2.5×10^{12} .
 (B) The number of α -particles strike the surface in $t = 4\text{ sec}$ is 3.5×10^{12} .
 (C) The number of α -particles striking the surface per second depends on the angle θ .
 (D) The number of α -particles striking the surface per second does not depends on the angle θ .
14. Choose the correct statement(s)
 (A) The number of α -particles are there in length $l = 20\text{ cm}$ of the beam is 6.25×10^4 .
 (B) The number of α -particles are there in length $l = 20\text{ cm}$ of the beam is 3.25×10^4 .
 (C) The power of the source used to accelerate these α -particles from rest is $5 \times 10^{-3}\text{ watts}$.
 (D) The power of the source used to accelerate these α -particles from rest is $8.35 \times 10^{-3}\text{ watts}$.

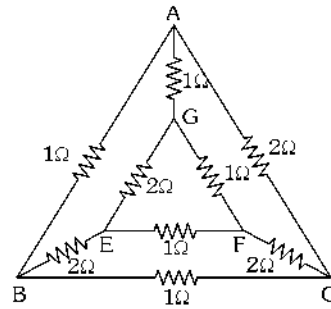
Paragraph for Q.no. (15 to 16)

Five capacitor are shown with their capacitance, potential and polarities. At $t = 0$ all keys are closed.



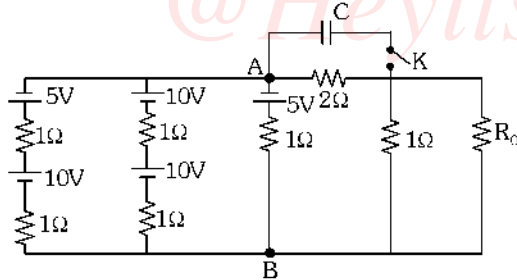
15. Choose the correct statement(s) :
 (A) Under steady state, the charge on capacitor of capacitance $5C$ is 10.4 CV approximately.
 (B) Under steady state the charge on capacitor of capacitance $5C$ is 14.3 CV approximately.
 (C) Under steady state the charge on capacitor of capacitance $4C$ is zero.
 (D) Under steady state the charge on capacitor of capacitance $4C$ is 0.5 CV approximately.
16. Choose the correct statement(s) :
 (A) Under steady state the potential difference $(V_A - V_B)$ is 2.08 V approximately.
 (B) Under steady state the potential difference $(V_A - V_B)$ is 1.1 volt approximately.
 (C) Under steady state, the charge on capacitor of capacitance is 0.385 CV approximately.
 (D) The heat energy will be generated after closing the keys.

Paragraph for Q.no. (17 to 18)



17. Choose the correct statement(s) :
- (A) The equivalent resistance across AB is 0.82Ω approximately.
 - (B) The equivalent resistance across AB is 0.62Ω approximately.
 - (C) The equivalent resistance across AB is 0.42Ω approximately.
 - (D) The current passing through FC branch will not be zero when battery is attached across A and B terminal.
18. Choose the correct statement(s)
- (A) The equivalent resistance across terminal B and C is 0.64Ω approximately
 - (B) The equivalent resistance across terminal B and C is 0.84Ω approximately
 - (C) The equivalent resistance across terminal B and C is 1.2Ω approximately
 - (D) The current through branch AG will not be zero when battery is connected across B and C terminals.

Paragraph for Q.no. (19 to 20)



Consider the circuit shown.

19. At $t = 0$, switch K is closed. Given $R_0 = 1\Omega$.
Choose the correct statements(s).
- (A) The current $i(t)$ passing through capacitor C is $\left(\frac{15}{4}\right)e^{-\frac{3t}{2C}}\text{A}$.
 - (B) The current $i(t)$ passing through capacitor C is $\frac{5}{4}e^{-\frac{3t}{2C}}\text{A}$.
 - (C) The potential difference across R_0 after long time is $\frac{5}{8}\text{V}$.
 - (D) The potential difference across C after long time is not zero.
20. Now the capacitor is removed from the circuit. R_0 is now an unknown resistance.
Choose the correct statement(s)
- (A) For the maximum power dissipation in R_0 the value of R_0 should be $\frac{5}{7}\Omega$.
 - (B) For the maximum power dissipation in R_0 the value of R_0 should be $\frac{3}{7}\Omega$.
 - (C) The potential difference between A and B depends on the value of R_0 .
 - (D) The potential difference between A and B does not depend on the value of R_0 .

* * * * *

Yashpatil TG~ @bohrring_bot

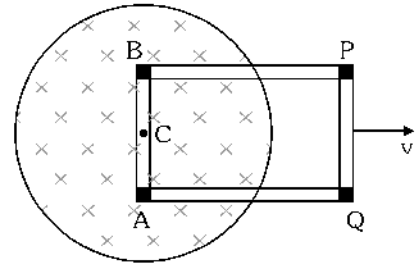
ANSWER KEY

1. (B)	2. (C)	3. (D)	4. (D)	5. (A)	6. (A)	7. (B)	8. (A)
9. (BD)	10. (AC)	11. (ACD)	12. (AC)	13. (AD)	14. (AD)	15. (AC)	16. (ACD)
17. (BD)	18. (AD)	19. (ACD)	20. (AB)				

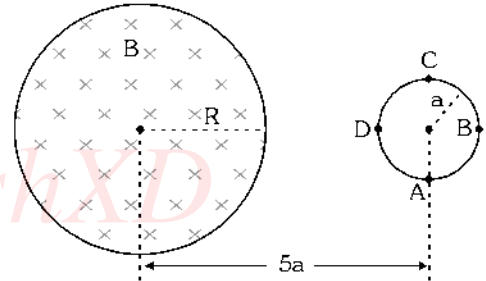
@HeyitsyashXD

ONE OR MORE THAN ONE CORRECT TYPE (+4, -1)

1. Uniform but time varying magnetic field $B = \frac{B_0 t}{T}$ present in the cylindrical region. There is small section at corners which is made with insulating material and other portion of rectangular loop is conducting. Loop is moving with constant velocity v . At $t = T$ situation shown in figure. C is centre of cylindrical region. Choose correct option/s at given instant.
- (A) There is no emf in section PQ
(B) From B to P potential decreases
(C) From A to Q potential decreases
(D) There is no electrostatics field in section PQ

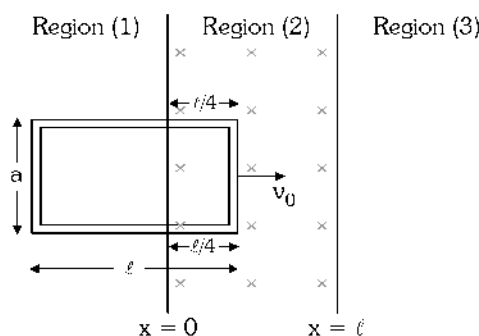


2. Uniform but time varying magnetic field $B = \frac{B_0 t}{T}$ present in cylindrical region of radius R . There is a circular conducting ring of radius a is kept at distance $5a$ (center to center distance). Choose correct options :
- (A) The value of emf in ring from B to C is zero.
(B) The value of emf in ring from B to D is zero.
(C) The maximum value of emf in ring from one point on the ring to other point on the ring is $\frac{R^2 B_0}{T} \sin^{-1} \left(\frac{1}{5} \right)$



- (D) The value of emf in ring from A to C is zero.

3. Uniform magnetic field $B = -B_0 \hat{k}$ present from $X = 0$ to $X = \ell$. A super conducting loop of given side length and self inductance L . Start with speed v_0 with no initial current choose correct options. (mass of loop is m)

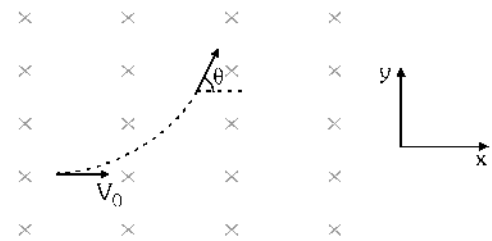


- (A) If v_0 is slightly greater than $\frac{3 B_0 a \ell}{4 \sqrt{mL}}$ then loop never returns to initial position.
(B) If v_0 is slightly less than $\frac{3 B_0 a \ell}{4 \sqrt{mL}}$ then loop returns to initial position only one time.
(C) If v_0 is slightly less than $\frac{B_0 a \ell}{4 \sqrt{mL}}$ then loop oscillates.
(D) If v_0 is slightly greater than $\frac{B_0 a \ell}{4 \sqrt{mL}}$ then loop will oscillates.

4. A super conducting ring of radius 'R' and self inductance L is kept in uniform but time varying magnetic field $B = B_0(1 - t/T)$. The direction of magnetic field perpendicular to plane of ring. At $t = 0$ no current in ring. If cross-section radius of wire of ring is b. Then find minimum breaking stress of material of ring required so that rupture of ring avoid. (Neglect force applied by any section of ring on the other section)

(A) $\frac{B_0^2 R^3 T}{4Lb^2}$ (B) $\frac{B_0^2 R^3 T}{8Lb^2}$ (C) $\frac{B_0^2 R^3 T}{2Lb^2}$ (D) None of these

5. A particle of mass m charge q is projected with velocity v_0 is viscous medium where uniform and constant magnetic field B exists everywhere is the direction perpendicular to the direction of projection of the particle. A viscous force on the particle is given by $\vec{f} = -b\vec{v}$, where b is constant and $\vec{v}$ is velocity of particle. Choose correct options :



(A) $\frac{d\theta}{dt}$ is given by $\frac{qB}{m}$

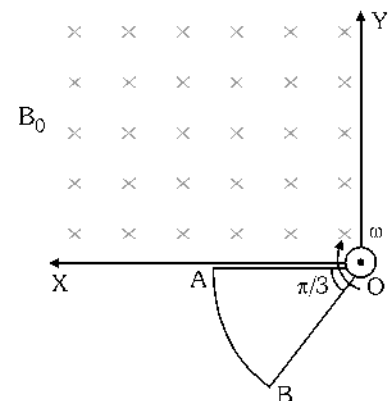
(B) $\frac{d\theta}{dt}$ is time dependent

(C) $\vec{v}$ at any time is $\vec{v} = v_0 e^{-\frac{bt}{m}} \left[\cos\left(\frac{qB}{m}t\right)\hat{i} + \sin\left(\frac{qB}{m}t\right)\hat{j} \right]$

- (D) Distance travelled by the particle during a time interval from the instant of projection until velocity vector

turns by 2π radian is $\frac{mv_0}{b} \left[1 - e^{-\frac{2\pi b}{qB}} \right]$

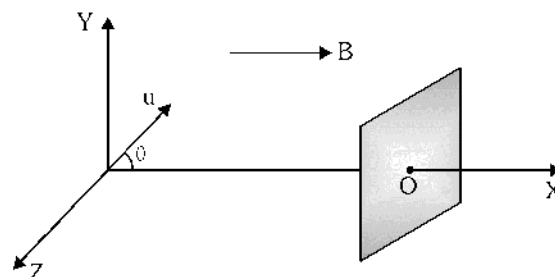
6. Uniform magnetic field B_0 present in the given region shown in figure. A part of circular loop of resistance R, self inductance L and radius ℓ rotating with constant angular speed ω . Situation shown in figure at $t = 0$, and no current in the loop. Choose correct option :



- (A) When side OB coincide with Y-axis first time no current in the loop.
 (B) When side OA coincide with Y-axis first time current in loop is in anticlockwise direction.
 (C) Current in loop when it is at starting position first time (after one complete revolution) is in clockwise direction
 (D) The value of current in the loop when OA coincide with Y-first time

is $\frac{B_0 \ell^2}{2R} \left(1 - e^{-\frac{\pi R}{3\omega L}} \right) \cdot e^{-\frac{\pi R}{6\omega L}}$

7. Uniform magnetic field B is present parallel to x-axis. A charge particle charge q mass m is projected from origin in x-y plane with speed u at an angle θ with x-axis. After some time particle collide with given smooth surface (parallel to yz plane) elastically and then returns to origin. (no gravity and neglect any force applied by losing charge)



- (A) Collision takes place at some special points on the surface.
 (B) Collision takes place at any point on the surface.
 (C) If during collision particle loses half of its charge, then collision must be takes place at O.
 (D) If during collision particle loses half of its charge, the it is not possible in any condition particle returns to origin.

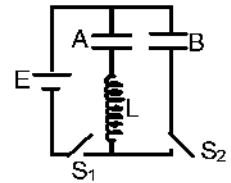
8. Stationary circular loop of radius a is located in a magnetic field which varies with time from $t = 0$ to $t = T$ according law $B = B_0 .t(T - t)$. If plane of loop is normal to the direction of field and resistance of the loop is R . Neglect self inductance of the loop.

- (A) The amount of heat generated in the loop during this interval is $\frac{\pi^2 a^4 B_0^2 T^3}{3R}$.
- (B) The amount of heat generated in the loop during this interval is $\frac{\pi^2 a^4 B_0^2 T^3}{5}$.
- (C) The magnitude of charge flown through the loop from instant $t = 0$ to the instant when current reverse, its direction $\frac{\pi a^2 B_0 T^2}{4R}$.
- (D) The magnitude of charge flown through the loop from instant $t = 0$ to the instant when current reverse, its direction $\frac{\pi a^2 B_0 T^2}{2R}$.

Paragraph for Q. no. 9 to 10

Two uncharged identical capacitors A and B, each of capacitance C , and an inductor of inductance L are arranged as shown in the adjacent figure. At $t = 0$, the

switch S_1 is closed while switch S_2 remains open. At time $t = t_0 = \sqrt{LC} \frac{\pi}{2}$, switch S_2 is closed while switch S_1 is opened.



9. Choose the correct statement(s).

- (A) The charge on capacitor A after time t_0 is $2CE$.
- (B) The charge on capacitor A after time t_0 is CE .
- (C) The current flowing through the inductor at $t = t_0$ is $CE \times \frac{1}{\sqrt{LC}}$.
- (D) The current flowing through the inductor at $t = t_0$ is $\frac{CE}{2} \times \frac{1}{\sqrt{LC}}$.

10. Choose the correct statement(s).

- (A) After switch S_2 is closed and S_1 is opened, the maximum value of current through the inductor is

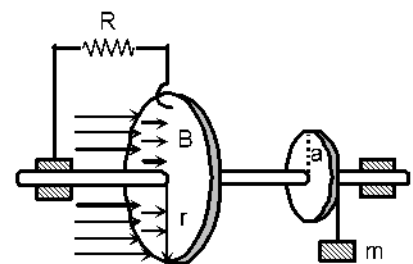
$$\left(\sqrt{\frac{C}{L}} \right) E$$

- (B) After switch S_2 is closed and S_1 is opened, the maximum value of current through the inductor is

$$\left(\sqrt{\frac{3C}{2L}} \right) E$$

- (C) When the current is maximum the amount of charges on each capacitor is equal.
- (D) When the current is maximum the amount of charges on each capacitor is not equal.

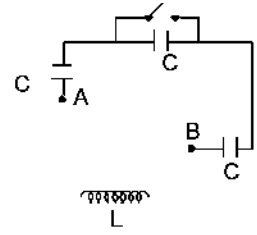
11. A metallic disc of radius r is made of a material of negligible resistance and can rotate about a conducting horizontal shaft. A smaller non conducting disc of radius a is fixed onto the same shaft and has a massless cord wrapped around it, which is attached to a small object of mass m as shown. Two ends of a resistor of resistance R are connected to the perimeter of the disc and to the shaft by sliding contacts. The system is then placed into a uniform horizontal magnetic field B and the mass m is released. Find the terminal angular velocity with which the disc will rotate finally.



(Take $r = 10\text{cm}$, $a = 2\text{cm}$, $R = \frac{1}{100} \Omega$, $B = 0.2 \text{ T}$, $m = 50 \text{ gm}$, $g = 10 \text{ m/s}^2$)

- (A) 200 rad/s (B) 300 rad/s (C) 100 rad/s (D) 10 rad/s

12. Three capacitors of capacitance C each are connected in series as shown in the figure. Initially switch S is open. Now capacitors are charged by a battery of emf V by connecting between terminal A and B . After long time battery is disconnected and inductor of inductance L is connected between A and B at time $t = 0$ so that an oscillatory circuit is formed. Now at an instant t_0 switch S is closed, then find the amplitude of charge oscillations of the remaining capacitors.



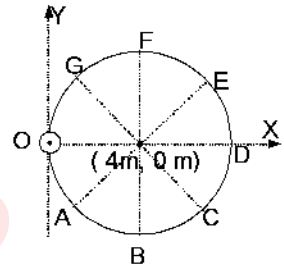
- (A) $CV \sqrt{\frac{1}{6} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$
- (B) $\frac{CV}{3} \sqrt{\frac{1}{5} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$
- (C) $CV \sqrt{\frac{1}{4} \left(1 - \frac{\cos^2 \left(\sqrt{\frac{3}{LC}} t_0 \right)}{3} \right)}$
- (D) $\frac{CV}{3}$

13. An infinite uniform current carrying wire is kept along z -axis, carrying current I_0 in the direction of the positive z -axis. OABCDEFG represents a circle (where all the points are equally spaced), whose centre at point

$(4m, 0m)$ and radius $4m$ as shown in the figure. If $\int_{DEF} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 I_0}{k}$ in

S.I unit, then the value of k is

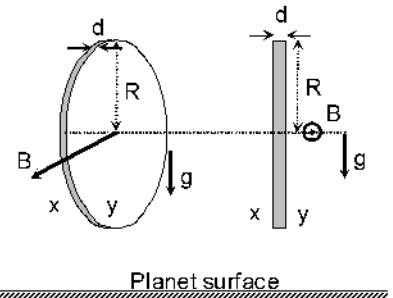
- (A) 4 (B) 8
(C) 3 (D) 6



Comprehension Type

Paragraph for Question Nos. 14 to 16

A space shuttle is designed in such a way that magnetic force is used to reduce its acceleration during the landing of the shuttle on a newly discovered planet. For this planet, it was known that gravitational acceleration above the surface of the planet is constant towards perpendicular downward to the planet surface and is equal to g and the magnetic field produced by the planet is uniformly B and is directed parallel to the planet's surface. There is no atmosphere above the planet surface. The space shuttle is given such a shape, such that it can be idealized like conducting disc of mass m , radius R and thickness d (thickness d is much smaller than R) as shown in the figure. x and y are two flat surfaces of the shuttle. Neglect any type of eddy current appeared during the motion.



14. The amount of charge induced on flat surface (say x) of the shuttle during the falls in the gravitational and magnetic field of the planet in terms of the shuttle speed v is

- (A) $\epsilon_0 \pi R^2 B v$ (B) $2 \epsilon_0 R^2 B v$ (C) $\epsilon_0 \pi R d B v$ (D) $\frac{\epsilon_0 \pi R^3 B^2 v}{d}$

15. The magnetic force acting on the shuttle during its fall in terms of the shuttle acceleration a is

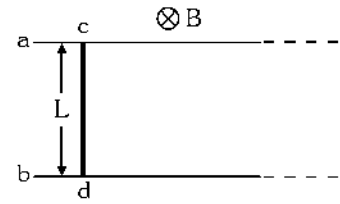
- (A) $\epsilon_0 \pi R^2 B d a$ (B) $\epsilon_0 \pi R^2 B^2 d a$ (C) $2 \epsilon_0 \pi R^2 B^2 d a$ (D) $\epsilon_0 \pi R^3 B^2 a$

16. The acceleration of the space shuttle during the fall is

- (A) $\frac{g}{1 + \frac{B \epsilon_0 \pi R^2 d}{m}}$ (B) $\frac{g}{1 - \frac{B^2 \epsilon_0 \pi R^2 d}{m}}$ (C) $\frac{g}{1 + \frac{B^2 \epsilon_0 \pi R^2 d}{m}}$ (D) $\frac{B^2 \epsilon_0 \pi R^2 d}{m}$

Paragraph-1 for (Q.n 17 to 18)

In the shown figure, there are two long fixed parallel conducting rail (having negligible resistance) and are separated by distance L . A uniform rod of resistance R and mass M is placed at rest on frictionless rails. Now at time $t = 0$, a capacitor having charge Q_0 and capacitance C is connected across rails at ends a and b such that current in rod (cd) is from c towards d and the rod is released. A uniform and constant magnetic field having magnitude B exists normal to plane of paper as shown. (Neglect acceleration due to gravity)



17. When the acceleration of rod is zero, the speed of rod is :

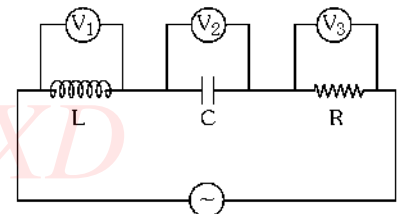
(A) $\frac{Q_0 R^2 B C}{M + B^2 L^2 C^2}$ (B) $\frac{Q_0 L B}{M + B^2 L^2 C}$ (C) $\frac{Q_0 L B}{M + B^2 R^4 C^4}$ (D) $\frac{Q_0 L B}{M + B^2 L^2 C^2}$

18. When the acceleration of rod is zero, the charge on capacitor is :

(A) $\frac{B^2 L^2 C Q_0}{M + B^2 L^2 C}$ (B) $\frac{B^2 R^4 C^3 Q_0}{M + B^2 L^2 C^2}$ (C) $\frac{B^2 R^4 C^3 Q_0}{M + B^2 R^4 C^4}$ (D) $\frac{B^2 L^2 C Q_0}{M + B^2 L^2 C^2}$

Paragraph-2 for (Q.n. 19 to 20)

Reading of AC voltmeter V_1 , V_2 & V_3 shown in figure is 100 V, 80 V & 160 V respectively. Current in circuit is 10A.



19. Value of L, C, and R respectively :

(A) $\frac{1}{10\pi}$ mH, $\frac{1}{800\pi}$ μ F, 16 Ω (B) $\frac{100}{\pi}$ mH, $\frac{1}{800\pi}$ μ F, 16 Ω
 (C) $\frac{1}{10\pi}$ H, $\frac{1}{800\pi}$ F, 16 Ω (D) $\frac{100}{\pi}$ H, $\frac{1}{800\pi}$ F, 16 Ω

20. If voltage in circuit at an instant of time is $20\sqrt{130} \sin 100\pi t$, then current in circuit at the same time is :

(A) $\sqrt{40} \sin\left(100\pi t - \tan^{-1} \frac{1}{8}\right)$ (B) $\sqrt{200} \sin\left(100\pi t - \tan^{-1} \frac{1}{8}\right)$
 (C) $\sqrt{200} \sin\left(100\pi t + \tan^{-1} \left(\frac{1}{8}\right)\right)$ (D) Required current cannot be calculated

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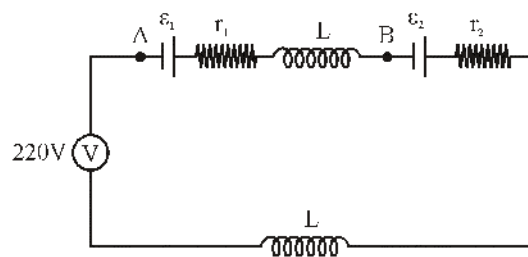
ANSWER KEY

1. (B)	2. (BC)	3. (ABC)	4. (D)	5. (ACD)	6. (BCD)	7. (AC)	8. (AC)
9. (BC)	10. (BC)	11. (C)	12. (A)	13. (B)	14. (A)	15. (B)	16. (C)
17. (B)	18. (A)	19. (C)	20. (B)				

ONE OR MORE THAN ONE CORRECT TYPE (+4, -1)

Paragraph for (Q.no. 1 to 2)

In AC circuits, deflection based galvanometers always fail to record the current as coil of galvanometer experience an alternating torque of fast frequency and hence gives zero reading. A.C. ammeters are based on heating effect and can measure alternating current although an a.c. ammeter measures i^2 but are calibrated to give i_{rms} . In a circuit two coils of equal inductances and two cells of emfs $\varepsilon_1 = 50 \text{ V}$ and $\varepsilon_2 = 100 \text{ V}$ are connected to a 220 V AC source. The internal resistances of the cells are r_1 and r_2 . In the circuit current lags the voltage by 45° . ADC voltmeter connected between points A and B reads zero.



1. Value of $\frac{r_2}{r_1}$ is :-

(A) 2 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

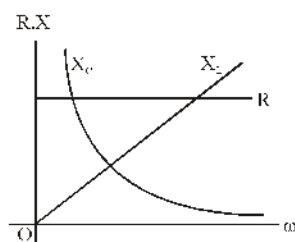
2. When an AC voltmeter is connected across A and B, its reading in steady state is :-

(A) 220 V (B) 153.5 V (C) 107 V (D) 93.5 V

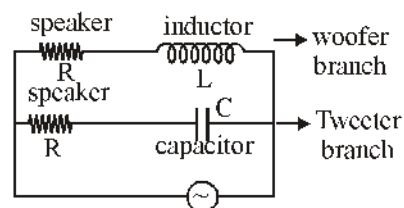
Paragraph for (Q.no. 3 to 4)

Figure shows how the resistance of a resistor and the reactances of an inductor and a capacitor vary with angular frequency ω . Resistance R is independent of frequency, while the reactances X_L and X_C are not. If $\omega = 0$, corresponding to a dc circuit, there is no current through a capacitor because $X_C \rightarrow \infty$, and there is no inductive effect because $X_L = 0$. In the limit $\omega \rightarrow \infty$, X_L also approaches infinity, and the current through an inductor becomes vanishingly small; recall that the self-induced emf opposes rapid changes in current. In this same limit, X_C and the voltage across a capacitor both approach zero.

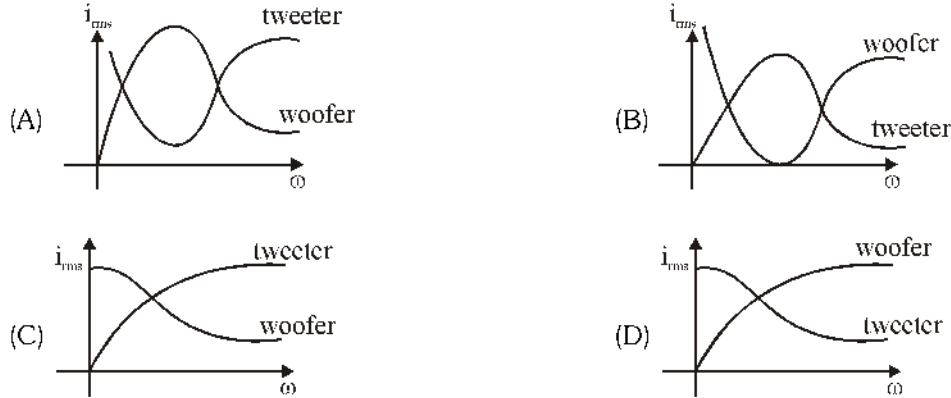
Figure shows an application of the above discussion to a loudspeaker system. Low-frequency sounds are produced by the woofer, which is a speaker with large diameter; the tweeter, a speaker with smaller diameter, produces high frequency sounds. In order to route signals of different frequency to the appropriate speaker, the woofer and tweeter are connected in parallel across the amplifier output. The capacitor in the tweeter branch blocks the low-frequency components of sound but passes the higher frequencies; the inductor in the woofer branch does the opposite.



Graphs of R , X_L , and X_C as functions of angular frequency ω .



3. The correct graph for i_{rms} versus angular frequency ω through tweeter and woofer is shown by :-

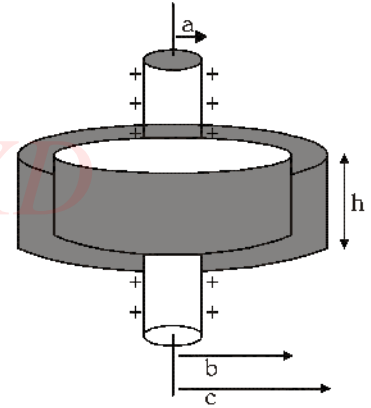


4. The rms current in tweeter and woofer can be equal if :-

(A) $\omega = \frac{1}{\sqrt{LC}}$ (B) $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$ (C) $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$ (D) $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}$

Paragraph for Q.no. (5 to 6)

A uniformly charge long solenoid having charge density σ , radius 'a' and a thick hollow cylinder of inner radius b, outer radius c, and height h kept co-axially. (Resistivity for material of cylinder is ρ). Now Solenoid start rotating with constant angular acceleration α about its axis as shown.



5. Choose correct statements :

- (A) The direction of magnetic dipole moment of cylinder is opposite to the direction of angular velocity ω .
 (B) The direction of magnetic dipole moment of cylinder is opposite to the direction of angular acceleration

(C) The value of current in cylinder is $\frac{\mu_0 \sigma a^3 \alpha h}{2\rho} \ln \frac{c}{b}$

(D) The value of current in cylinder is $\frac{\mu_0 \sigma a^3 \alpha h}{3\rho} \ln \frac{c}{b}$

6. Choose correct statements :

(A) Value of magnetic dipole moment of cylinder $\frac{\mu_0 \sigma a^3 \alpha h \pi (c^2 - b^2)}{4\rho}$

(B) Value of magnetic dipole moment of cylinder $\frac{3\mu_0 \sigma a^3 \alpha h \pi (c^2 - b^2)}{2\rho}$

(C) Rate of heat developed per unit time in cylinder is $\frac{2\pi \mu_0 \sigma^2 a^6 \alpha^2 h}{\rho} \ln \frac{c}{b}$

- (D) None of these

7. Some electrical elements are shown in below figure. All ammeters and voltmeters are ideal and read rms values. An alternating source of emf $e = e_0 \sin \omega t$ is connected in each circuit between P and Q. Now choose the correct option(s) :

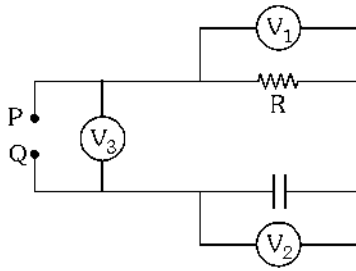


Figure (i)

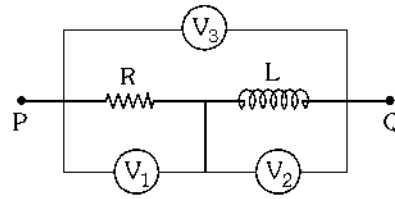


Figure (ii)

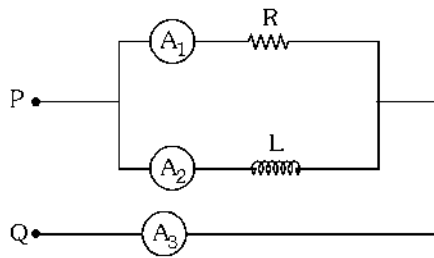


Figure (iii)

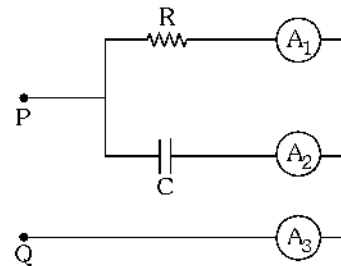
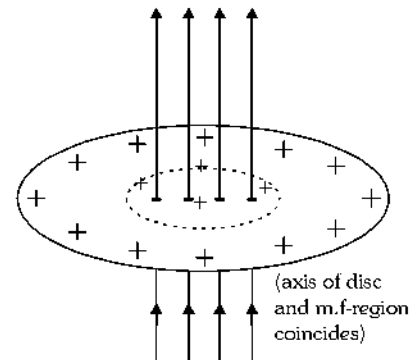


Figure (iv)

- (A) The reading of voltmeter V_3 is more than the sum of the readings of the voltmeters V_1 and V_2 for each case in figure (i) and (ii)
 (B) The reading of ammeter A_3 is more than the sum of the readings of the ammeters A_1 and A_2 for each case in figure (iii) and (iv)
 (C) The reading of voltmeter V_3 is less than the sum of the readings of the voltmeters V_1 and V_2 for each case in figure (i) and (ii)
 (D) The reading of ammeter A_3 is less than the sum of the readings of the ammeters A_1 and A_2 for each case in figure (iii) and (iv)

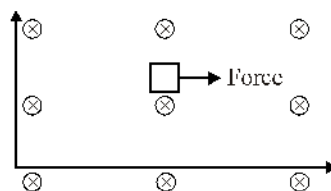
8. Uniform magnetic field $B = 2t^2$ Tesla present in cylindrical region of radius $R/2$. And a uniformly charged disc charge Q , mass m and radius R kept on rough horizontal plane with coefficient of friction μ . Find the time after disc start rotating.

- (A) $\frac{20\mu mg}{3R}$ (B) $\frac{32\mu mg}{27R}$
 (C) $\frac{32\mu mg}{3R}$ (D) None of these



9. A very small square loop of finite resistance at time $t = 0$ starts moving parallel to x-axis with a constant acceleration 1 m/s^2 as shown in a space where magnetic field is changing with position and time according to relations

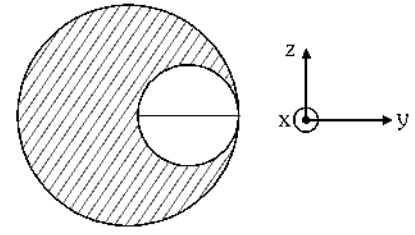
$$\frac{dB}{dt} = 10^{-3} \text{ T/s}, \quad \frac{dB}{dx} = -10^{-3} \text{ T/m}$$



- (A) Initially current in loop is anticlockwise (B) Initially current in loop is clockwise
 (C) At $t = 1\text{s}$, current in loop is zero (D) After $t > 1\text{s}$ current is anticlockwise

Paragraph for (Q.no. 10 to 11)

A long cylindrical conductor of radius R having a cylindrical cavity of radius $R/2$ through out its length is lying with its axis parallel to x -axis. The distance between the axes of the cylinder and the cavity is $R/2$. A constant current I is passed through the cylinder along its length in the positive x -direction. The cross-sectional view of the cylinder is shown in the figure.



10. The magnetic induction at point on the cavity is :

(A) $\frac{\mu_0 I}{3\pi R}(-\hat{k})$ (B) $\frac{\mu_0 I}{3\pi R}(\hat{k})$ (C) $\frac{\mu_0 I}{6\pi R}(\hat{j})$ (D) $\frac{\mu_0 I}{6\pi R}(-\hat{j})$

11. A charged particle having specific charge (charge to mass ratio) is $\frac{18\pi}{\mu_0 I} \times 10^6$ SI unit is projected from a point on the axis of the cavity along x -axis with the speed of 10^6 m/s. Radius of curvature of the path of the particle is equal to :

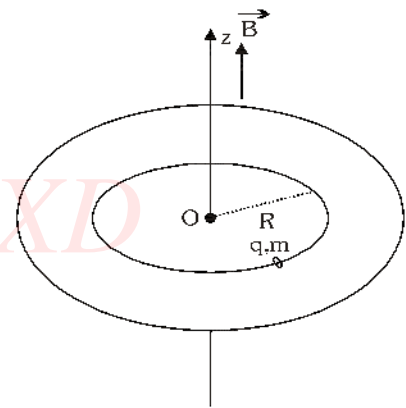
(A) $R/18$ (B) $R/9$ (C) $R/6$ (D) $R/3$

Paragraph for (Q.no. 12 to 13)

A small bead of mass m and positive charge q is threaded on a horizontal ring of radius R made of insulating material. The bead can move on the circular track without friction and is initially at rest. A magnetic field that is cylindrically symmetric about z -axis (axis of the ring) is created, in which magnetic induction that is perpendicular to the plane of the track depends upon only on the distance r , measured

from centre and time t , $\vec{B}(r, t) = \frac{B_0}{r} t(\hat{k})$ where B_0 is a constant.

(In a negligibly small neighbourhood of $r = 0$, the magnetic induction has some small finite value). Neglect gravity.



12. The magnitude of tangential force, acting on the bead depend upon
 (A) depends upon time and radius R
 (B) depends upon time but is independent of radius R
 (C) is independent of time but depends upon radius R
 (D) is independent of time and radius R

13. Determine the speed of bead as function of time

(A) $\frac{B_0 q t}{m}$ (B) $\frac{2B_0 q}{m} t$ (C) $\frac{B_0 q}{2m} t$ (D) $\frac{3B_0 q}{m} t$

14. Determine the normal force between the bead and the track as a function of time.

(A) zero (B) $\frac{2q^2 B_0^2 t^2}{mR}$ (C) $\frac{2q B_0 t}{mR}$ (D) $\frac{q^2 B_0^2 t^2}{mR}$

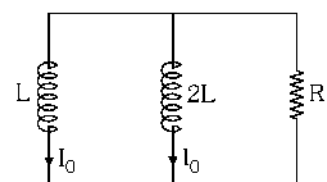
15. A parallel circuit consists two ideal inductor of inductance L and $2L$ and a resistance R . At some instant the same current I_0 is flowing in both the inductor in same direction.

(A) Final current in both inductor is same and equal to $\frac{I_0}{2}$

(B) Total heat produced in resistance is equal to $\frac{2}{3} L I_0^2$

(C) Total charge flow through resistance is $\frac{4 L I_0}{3 R}$

(D) Total heat produced in resistance is equal to $\frac{1}{3} L I_0^2$



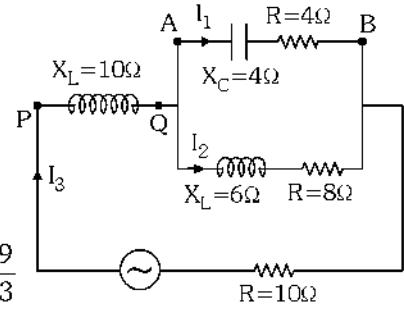
16. In given AC circuit

(A) Phase difference between I_3 and voltage across AB is $\tan^{-1} \frac{13}{41}$

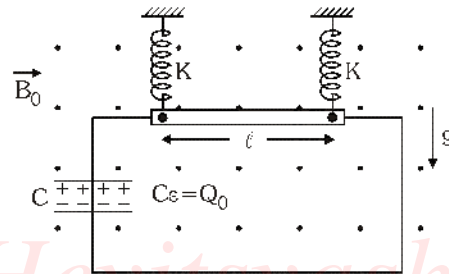
(B) Phase difference between I_1 and I_3 is $\tan^{-1} \frac{13}{41} - \frac{\pi}{4}$

(C) Phase difference between I_1 and voltage across PQ is $\frac{\pi}{4} + \tan^{-1} \frac{19}{33}$

(D) None of these



17. A conducting rod of mass m and length ℓ is suspended horizontally with the springs each of force constant K in the presence of uniform magnetic field B_0 as shown. A completely charged capacitor charge $Q_0 = C\epsilon$ is connected to rod as shown when rod is in equilibrium.



Neglect any resistance, any inductance in the circuit. Consider connecting wires are extremely flexible and having negligible mass. Connecting wire does not exert any force on the rod.

Choose correct statement(s) :

(A) maximum speed of rod is $\frac{B\ell C\epsilon}{m}$

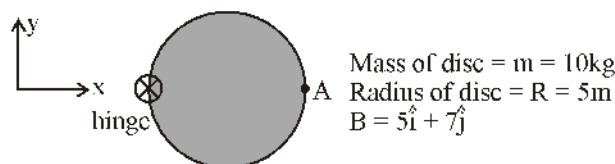
(B) time period of oscillation is $2\pi\sqrt{\frac{m}{2K}}$

(C) amplitude of oscillation is $\frac{B\ell C\epsilon}{\sqrt{2mK}}$

(D) None of these

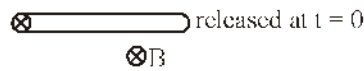
INTEGER TYPE (+3, -1)

18. Diagram shows a disc hanging vertically in presence of an external magnetic field in the plane of motion of the disc. The potential difference between two points one located at hinge & other at A when the disc reaches at lowest position is given by v_0 volt. Fill $\frac{3v_0}{7}$ in OMR sheet.

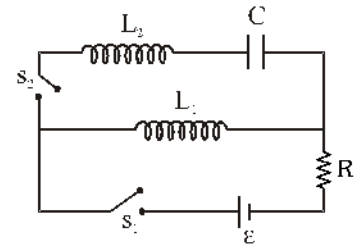


- 19.** A non conducting rod of length 2m is hinged at one end and a charge of $\frac{1}{10}\text{C}$ is distributed uniformly over it.

At $t = 0$, it is released from the position shown. There exist a uniform magnetic field of $\sqrt{15}\text{ T}$ inside the plane of motion of rod. If mass of rod is 100 g, then find the value of hinge force (in N) when rod is rotated by $\frac{\pi}{2}$ due to gravity.



- 20.** Diagram shows a circuit. Switch S_1 was closed for a long time with S_2 open. At $t = 0$ switch S_2 was closed & S_1 was opened. The time period of L-C oscillation after $t = 0$ is given by $2\pi\alpha \times 10^{-4}\text{ s}$. Fill the value of α .
(Given $C = 30\mu\text{F}$, $L_1 = L_2 = 6\text{ mH}$, $R = 10\Omega$, $\varepsilon = 10\text{Volts}$).



* * * * *

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ANSWER KEY

ONE OR MORE THAN ONE CORRECT TYPE (+4, -1)

- | | | | | | |
|---------|---------|---------|---------|-----------|---------|
| 1. (A) | 2. (D) | 3. (C) | 4. (A) | 5. (ABC) | 6. (A) |
| 7. (CD) | 8. (D) | 9. (AC) | 10. (B) | 11. (C) | 12. (D) |
| 13. (A) | 14. (A) | 15. (C) | 16. (A) | 17. (ABC) | |

INTEGER TYPE (+3, -1)

- | | | |
|---------|---------|---------|
| 18. (0) | 19. (4) | 20. (6) |
|---------|---------|---------|



ADVANCE REVISION ASSIGNMENT

**JEE (Main + Advanced) 2024
ENTHUSIAST COURSE**

ADVANCE REVISION ASSIGNMENT-20

PHYSICS

Test Date : 10-04-2015

PHYSICS

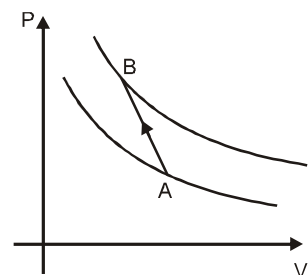
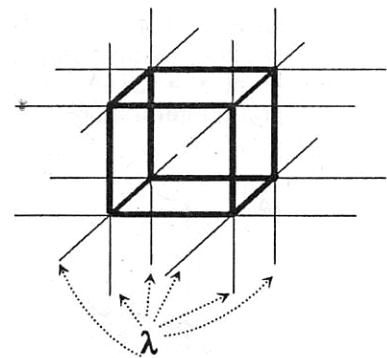
COURSE : SRG

Time Allowed: 1:00 Hrs.

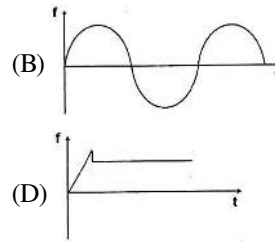
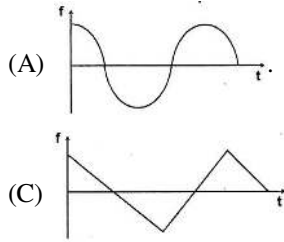
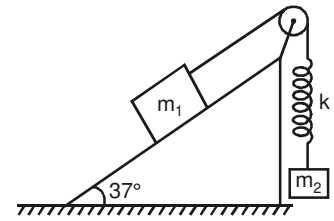
Max. Marks: 69

Single Choice Correct Type Questions.

- The radius of a planet is n times the radius of earth, R . A satellite revolves around it in a circle of radius $4nR$ with angular velocity ω . The acceleration due to gravity on planet's surface is :
 (A) $R\omega^2$ (B) $16 R\omega^2$ (C) $32 nR\omega^2$ (D) $64 nR\omega^2$
- Some successive frequencies of an ideal organ pipe are measured as 150 Hz, 250 Hz, 350 Hz. Velocity of sound is 340 m/s.
 (A) it must be an open organ pipe of length 3.4 m (B) it must be an open organ pipe of length 6.8 m
 (C) it must be a closed organ pipe of length 1.7 m (D) it must be a closed organ pipe of length 3.4 m
- Two identical blocks are kept on frictionless horizontal table and connected by a spring of stiffness ' k ' and of original length l_0 . A total charge Q is distributed on the blocks such that maximum elongation of spring, at equilibrium, is equal to x . Value of Q is :
 (A) $2l_0\sqrt{4\pi\epsilon_0 k(l_0 + x)}$ (B) $2x\sqrt{4\pi\epsilon_0 k(l_0 + x)}$ (C) $2(l_0 + x)\sqrt{4\pi\epsilon_0 kx}$ (D) $(l_0 + x)\sqrt{4\pi\epsilon_0 kx}$
- Two resistances are measured in Ohm.
 $R_1 = 3 \Omega \pm 1\%$
 $R_2 = 6 \Omega \pm 2\%$
 When they are connected in parallel, the percentage error in equivalent resistance is :
 (A) $\frac{14}{3} \%$ (B) 1 % (C) 2 % (D) $\frac{4}{3} \%$
- If $\lambda_{\min}$ is minimum wavelength produced in X-ray tube and $\lambda_{K\alpha}$ is the wavelength of K_{α} line. As the operating voltage is increased :
 (A) $\lambda_{K\alpha}$ increases (B) $\lambda_{K\alpha}$ decreases (C) $(\lambda_{K\alpha} - \lambda_{\min})$ increases (D) $(\lambda_{K\alpha} - \lambda_{\min})$ decreases
- Twelve infinite long wire of uniform linear charge density (λ) are passing along the twelve edges of a cube. Find electric flux through any face of cube.
 (A) $\left(\frac{\lambda l}{2\epsilon_0}\right)$ (B) $\left(\frac{\lambda l}{\epsilon_0}\right)$
 (C) $\left(\frac{\lambda l}{3\epsilon_0}\right)$ (D) $\left(\frac{3\lambda l}{\epsilon_0}\right)$
- Two adiabatic processes involving an ideal gas are plotted on a P-V diagram. A and B are two points on these curves as shown in the diagram. Which of the following statements is true ?
 (A) Heat is given to system in the process AB
 (B) Heat is rejected by system in the process AB
 (C) Whether heat is rejected or absorbed by the system during the process AB depends on the magnitude of the slope of line AB
 (D) Whether heat is rejected or absorbed by the system during the process AB depends on the position of the points A and B on the P-V diagram.



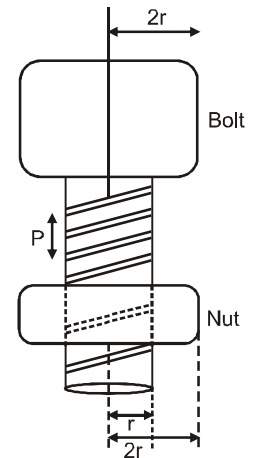
8. A block of mass $m_1 = 5 \text{ kg}$ is placed on rough fixed inclined plane $\mu = 0.8$ and other block of mass $m_2 = 3 \text{ kg}$ is connected by means of a spring as shown in the diagram. Initially spring is relaxed and block m_2 is released from rest. Graph of friction between m_1 and inclined surface as a function of time t is best represented by (Consider friction to be positive if it acts up the plane).



9. The time period of a particle executing linear SHM is T . There is a point P at a distance x from the mean position O . When the particle passes P towards O , it has a speed v . The time in which it return to P again is :

(A) $\frac{T}{\pi} \left[\pi - \tan^{-1} \left(\frac{vT}{2\pi x} \right) \right]$ (B) $\frac{T}{\pi} \tan^{-1} \left(\frac{vT}{2\pi x} \right)$ (C) $T \sin^{-1} \left(\frac{vT}{x} \right)$ (D) $\frac{T}{2\pi} \cot^{-1} \left(\frac{vT}{2\pi x} \right)$

10. A bolt with shaft radius r has pitch of its threading equal to p . The nut has frictionless movement on threads. The bolt is rotated clockwise while looking from top with an angular acceleration α such that the nut moves up by an acceleration g . Find α in terms of other given parameters.



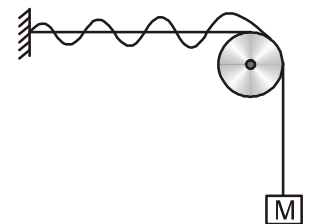
(A) $\frac{gp}{2\pi r^2}$ (B) $\frac{gp}{4\pi r^2}$ (C) $\frac{gp}{\pi r^2}$ (D) $\frac{2gp}{3\pi r^2}$

11. Two uniform rods are welded together to form a letter T as shown in the figure. Each rod is of mass M and length ' l '. If this combination is hinged at ' A ' and kept in vertical plane then time period of small oscillations about A is equal to :

(A) $2\pi \sqrt{\frac{l}{6\sqrt{3}g}}$ (B) $2\pi \sqrt{\frac{3l}{2\sqrt{2}g}}$ (C) $2\pi \sqrt{\frac{l}{2g}}$ (D) $2\pi \sqrt{\frac{11l}{(6\sqrt{5})(g)}}$

12. A string forms ' n ' loops when mass M is hung from it. What total mass should be hung from it to form ' $2n$ ' loops for same frequency ?

(A) $2M$ (B) $M/4$ (C) $4M$ (D) $M/2$



13. One mole of an ideal monoatomic gas is taken from temperature T_0 to $2T_0$ by the process $PT^{-4} = C$. Considering the following statements choose the correct alternative.

I. Molar heat capacity of the gas is $-\frac{3R}{2}$

II. Molar heat capacity of the gas is $\frac{3R}{2}$

III. Work done is $3RT_0$

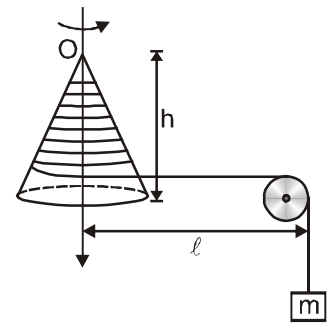
(A) Statements I and IV are correct

(C) Statements II and IV are correct

(B) Statements I and III are correct

(D) Statements II and III are correct

14. Figure shows a cone of moment of inertia I , about its own axis and a block of mass m . They are connected by a massless string as shown in figure. Consider $l \gg h$. If α is the angular acceleration and ω , the angular velocity of the cone at any time t .
- (A) α decreases with time
 (B) α increases with time
 (C) ω decreases with time
 (D) α remain unchanged

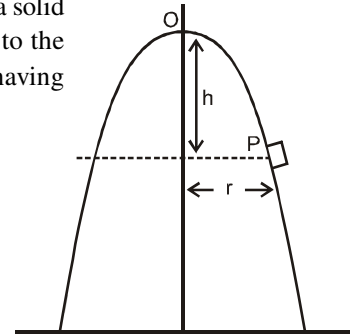


15. A block is held stationary at the position shown in the figure over the surface of a solid paraboloid. What should be the magnitude of the velocity, needed to be given to the block at this point such that it moves along the surface of the paraboloid without having any normal reaction anywhere.

$(g = 10 \text{ m/s}^2) \quad r = \sqrt{2}h, \quad h = (10/3) \text{ m}.$

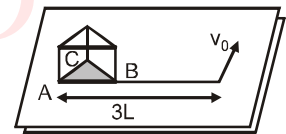
Consider motion only along the plane of paper.

- (A) 20 m/s
 (B) $10\sqrt{\frac{2}{3}}$ m/s
 (C) 10 m/s
 (D) cannot be calculated



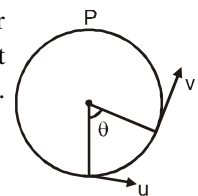
16. A pyramid is placed on a horizontal surface whose base ABC is fixed on a horizontal surface. It is in the shape of an equilateral triangle of side L . One end of a string of length $3L$ is attached at A and other end attached to a small bead. The bead is given a velocity V_0 along the horizontal frictionless plane as shown. The time taken by the string to wound will be :

- (A) $\left(\frac{2L}{V_0}\right)\pi$
 (B) $\left(\frac{3L}{V_0}\right)\pi$
 (C) $\left(\frac{4L}{V_0}\right)\pi$
 (D) None of these



Multiple Choice Correct Type Questions

17. A particle of mass m is tied with an ideal string and is moving in a vertical circle in such a manner that at the top point P of the path, the tension of the string $T = 0$. The particle started at the lowest point of the circle at time $t = 0$. At any amount of time t , it makes an angle θ with vertical as shown. Net magnitude of acceleration of the particle is :
- (A) $5g$ at $\theta = 0^\circ$
 (B) $\sqrt{10}g$ at $\theta = (\pi/2)$
 (C) $4g$ at $\theta = (\pi/3)$
 (D) g at $\theta = \pi$

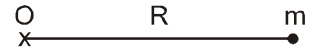


18. There are a number of hydrogen like atoms in free space. Initially all the atoms stationary and in a particular excited level A . They undergo transitions from the initial level A to the ground state. The recoiling atom enters a region of constant magnetic field. If the radii of the paths of the recoiling atoms is observed to have only three different values, then :

- (A) Magnetic moment of the atoms in the initial state A is $\frac{5eh}{4\pi m}$.
 (B) The radii must be in the ratio $5 : 27 : 32$.
 (C) The angular momenta of the electrons in the initial state of the emitting atoms is $\frac{3h}{2\pi}$.
 (D) The principal quantum number of the level A is given by $n = 3$.

19. A Bob of mass m is attached to a massless string, whose other end is connected to the fixed support O . Initially string is horizontal and bob is held at rest. At $t = 0$ bob is released and it takes time ' t ' to reach bottom most position. During the given interval of time ' t '.

(A) Average value of tension in string is $m\sqrt{g^2 + \frac{R}{t^2}}$



(B) Average value of tension in string is mg

(C) Magnitude of angular momentum of bob about point O will increase

(D) Average value of torque due to weight of bob about O will be $m \frac{(\sqrt{2gR})R}{t}$

20. A string fixed at both ends and under tension T vibrates in its 1st overtone with an amplitude A at the antinodes. The total energy of the string is E and the maximum possible speed of a particle of the string is v . If the same string were to vibrate in its fundamental mode under a tension $4T$ and with an amplitude A at the antinode then:

(A) The total energy of the string will be E

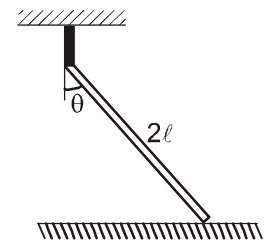
(B) The total energy of the string will be $2E$

(C) The maximum possible speed of a particle on the string is v

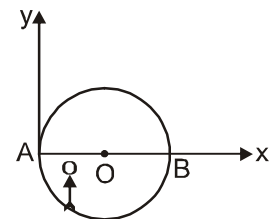
(D) The maximum possible speed of a particle on the string is $2v$

Integer Answer Type Questions

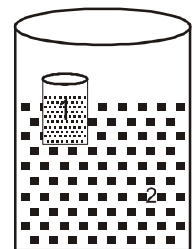
21. A uniform rod of length $2l$ and mass m is suspended from one end by inextensible string and other end lies on smooth ground. The angle made by rod with vertical is $\theta = \sin^{-1}(1/\sqrt{3})$. If N_1 and N_2 represents the contact force from ground on rod just before and just after cutting string then find the ratio of N_1/N_2 .



22. A man is running with constant speed of $v = 10\sqrt{5}$ m/s on a horizontal track of radius $R = 20$ m as shown in figure. At position A man launches a stone in space (without changing his own speed) so that he can catch stone at B (diametrically opposite to A). The speed of launch will be approximately $4K$ m/sec. Then find the value of K . (take $\pi^2 = 10$ and $g = 10$ m/s²)



23. A massless cylindrical container of base area 100 cm^2 contains a solid of volume $V = 500 \text{ cm}^3$ of specific gravity 2. The container is floating in another cylindrical container of base area 1000 cm^2 . If the solid is shifted from container 1 to 2 the height with which the liquid level decreases by $K/2$ cm. Find the value of K .



* * * * *

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ANSWER KEY

Single Choice Correct Type Questions.

- 1. (D) 2. (C) 3. (C) 4. (D) 5. (C) 6. (A) 7. (A) 8. (A) 9. (A)**
10. (A) 11. (D) 12. (B) 13. (B) 14. (B) 15. (C) 16. (A)

Multiple Choice Correct Type Questions

- 17. (A, B, D) 18. (B, C, D) 19. (C, D) 20. (A, C)**

Integer Answer Type Questions

- 21. (1) 22. (5) 23. (1)**

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